VU Medical Zone

ADMIN:

AMAAN KHAN

ECONOMIC ZOOLOGY

ZOO 510 MIDTERM PPT

LECTURE 1 TO 119

ECONOMIC ZOOLOGY

Economic zoology deals with animal world that is associated with the

- Economy
- Health
- Welfare of humans

History of Aquaculture

- Egyptian tombs have bas-relief (sculpture)of fish (tilapia) being removed from ponds – 2500 B.C.
- Carp were farmed in China as early as 2500 B.C.
  - Wen Fang – founder of the Chou Dynasty is called the first fish farmer
  - (during exile he kept records of fish growth and behavior)
  - Fan Li – wrote first book on fish farming 475 B.C.
  - Lee family – Were the first to polyculture carp during the Tang Dynasty 600 to 900 A.D.
- England – 1500 A.D. carp culture was introduced
- U.S. – first fish hatchery was in Oregon 1877

World Aquaculture







Species Selection

- Producer's expertise
- Water supply and climate
- Species biology
- Marketability
- Production methods
- Production economics

Commonly Cultured Fish species

- Food fish
- Many species, Catfish, Tilapia, Rainbow trout, Atlantic Salmon, shrimp, Shellfish, Striped Bass, Others
- Ornamental fish
  - –Aquaria
  - –Backyard ponds
- Bait fish
  - –Minnows
  - –Shiners
  - –Goldfish (carp)
- Natural stock enhancement
  - –Salmon
  - –Trout
  - –Black sea bass
  - –Red Drum
  - –Many others…

Commonly Cultured Crustaceans

- Marine (Penaeid) shrimp
- Freshwater shrimp (prawns)
- Crabs
- Crayfish
- Lobsters
- Brine Shrimp

Commonly Cultured Mollusks

- Clams
- Oysters
- Mussels
- Abalone
- Urchins











Additional Cultured Organisms

- Seaweed
  - –Food for Abalone
  - –Extraction of nutrients for vitamins
- Corals / Sponges / Sea Fans
  - –Extraction of medicines
  - –Aquarium trade
- Live rock
  - –Inhabiting macro and micro life Aquarium trade







Status of Aquaculture

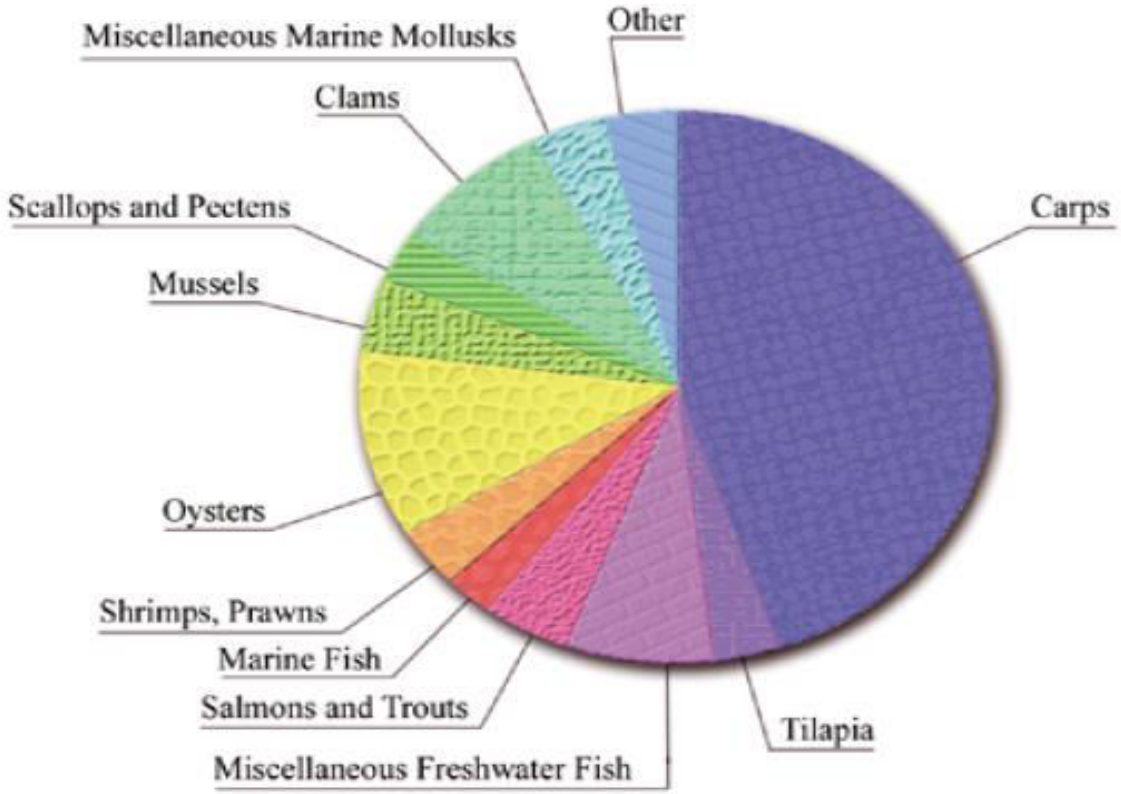
- Aquaculture is the fastest growing sector of agriculture with an approximate annual growth rate of 10%

- Currently aquaculture accounts for 25% of all seafood consumed in the U.S.
- Percentage is very low in Pakistan about 2 kg against world consumption of above 16 kg

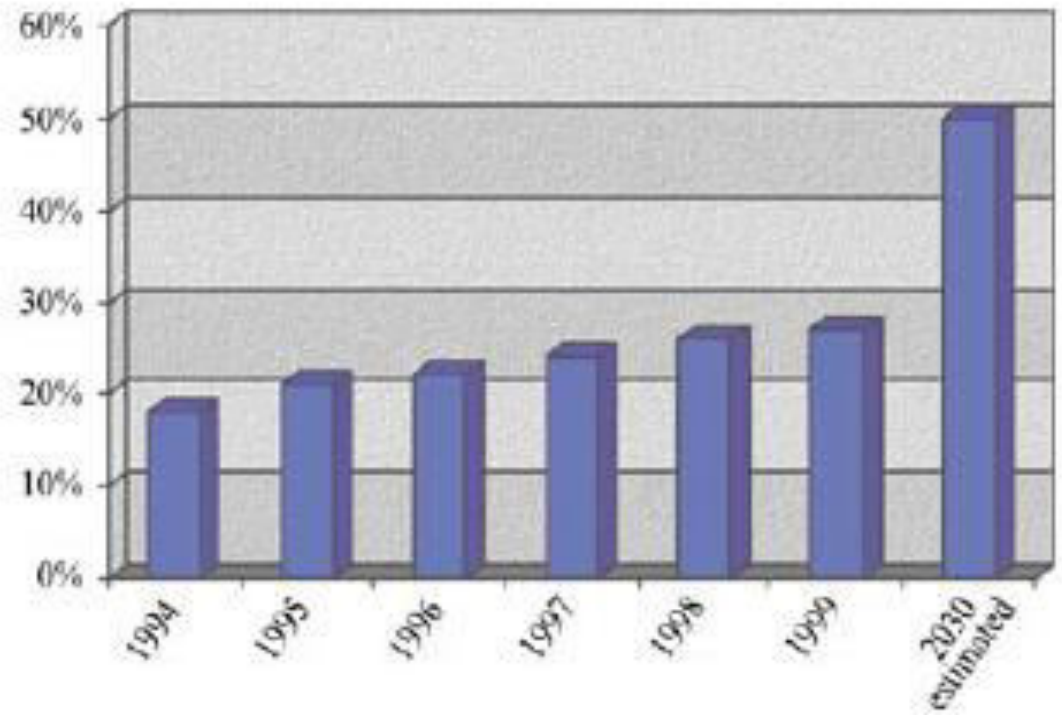
World Aquaculture Production

- In 2000 45.51 million metric tons by weight of aquaculture products
- Equal to US $56.47 billion
- China is the largest aquaculture producing country in the world

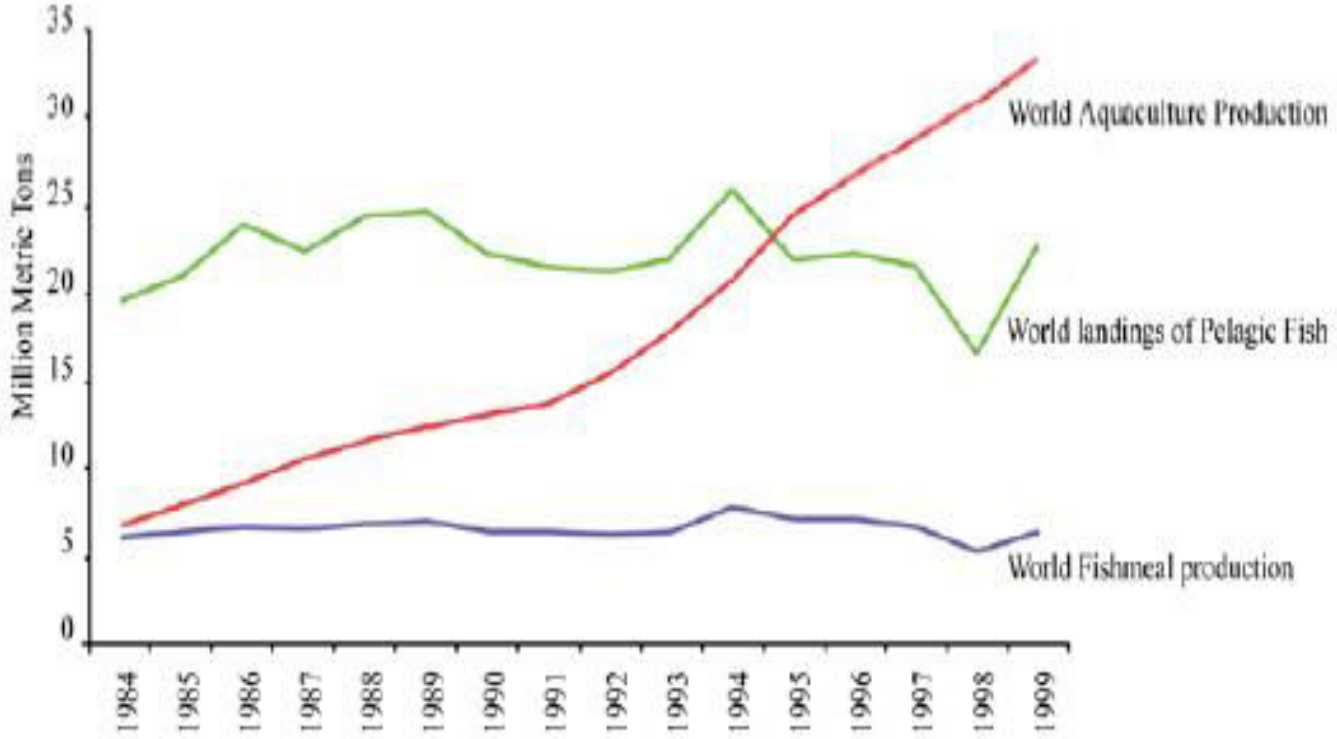
Proportion of Total Aquaculture Production for Different Taxonomic Groups



Percent of Total Food Fish Supplied by Aquaculture



Aquaculture Production, Ocean Fisheries, and Fishmeal Production



Why Aquaculture?

- Control: Food fed, Density, Quality of product
- Sustainable in the face of Finite Resources—overfishing and habitat destruction antagonists
- Diversify farm income
- Proximity—Farms may be closer to local markets.

- Health Consciousness (protein, FA's, micronutrients)

  -2 fish meals/week decreases mortality from heart problems 50%

  -Omega-3 fatty acids decreases occurrence of heart disease (oily marine fish – Salmon)

- American Cancer Association
  - -Regular fish consumption decreases chances of colon cancer 50%

- Efficiency of growth

Feed Conversion (grain/feed:flesh)

- Beef cattle on feedlot 8:1
- Swine 3.3:1
- Poultry 2.25:1
- Rainbow trout 1.5:1
- Tilapia 1.25:1









- Why *are* fish so efficient?

Aquaculture is a Diverse Field

- Biology
- Ecology
- Nutrition
- Handling and hauling
- Water quality
- Disease
- Marketing
- Culture techniques

Employment Opportunities

- Fisheries biology
- Public aquariums
- Research positions
- Education
- Laboratories
  - –Genetic studies
  - –Nutritional studies
  - –Disease studies
  - –Water quality
- State hatcheries
  - –Technicians
  - –Biologists
- Private operations
  - –Biologist
  - –Assistant manager
  - –Manager
- Open your own operation

Aquaculture Journals

- Journal of the World Aquaculture Society
- North American Journal of Aquaculture (PFC)
- Aquaculture
- Journal of Applied Aquaculture
- Aquaculture Nutrition
- Aquaculture Research
- Journal of Aquatic Animal Health
- Transaction of the American Fisheries Society

Types of Aquaculture Systems

- Pond Fish Culture
- Raceway Culture
- Cage Fish Culture
- Recirculating Water Fish Culture

Pond construction and Designing

- Ponds were used as one of the first forms of aquaculture.
- Dates back to ancient China.
- Already had the water...just add fish and feed,
- Pond production has come along way since then!

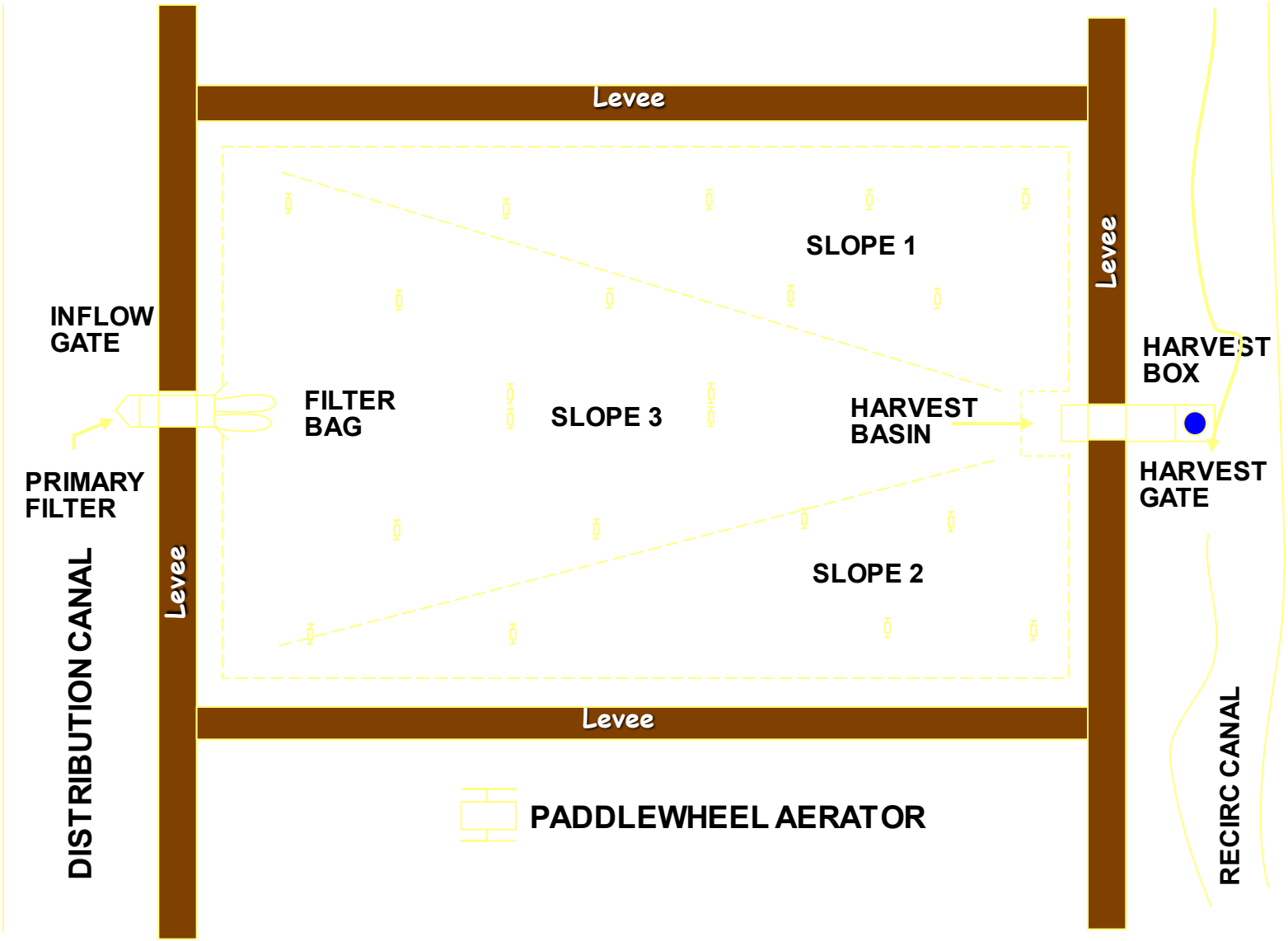
Parameters to Consider

- Site Considerations
  - Water
  - Soil
  - Topography
  - Types of ponds
  - Cost
- Construction of ponds
  - Levee ponds
  - Watershed ponds
  - Lined ponds

Pond Design Criteria

- Screened inflow gates at shallow end of pond
- Screened harvest gates at deep end
- Slope to harvest basin (0.5-1.0%)
- Water depth 1.25 ➔ 2.00 M
- Feeding tray piers(docks)
- Rounded or square corners, steps or ramps for entry
- Primary dikes (levees) wide enough to accommodate vehicles

GENERAL DESIGN, INTENSIVE POND



- Heights determined by pond bottom elevation, tidal amplitude
- Perimeter levee often required for protection in flood areas
- Levees with slopes 1:2 for high clay, 1:3-4 low clay
- Levee crown width varies with use
- Width of crown: 5 m (driving), 3m (walking)
- Crown is sloped to reduce puddles on levee top
- Once formed, levees are sprigged with grass to reduce erosion

- Levees are typically constructed by Caterpillar sized bulldozers
- Construction is first undertaken on ponds nearest the sedimentation basins and pump station
- Bulldozers push earth up to create general form of the levee walls
- Follow stakes set along the length of the pond
- Smaller dozers used to put on finishing touches

- Erosion is the main problem in maintaining levee slopes
- Source: both rainfall and wave action
- Solution: plants and vegetation (local grasses or *Salicornia sp.*) as soon as possible
- Pond sides receiving wind could be reinforced with rocks (contractor services)
- Tops of levees definitely need layer of rocks, especially if high clay content

Water

- Availability
  - 25 gallons per minute per acre
- Source
  - Wells
  - Springs
  - Streams
- Pesticides
  - TEST Water!!
- Alkalinity
  - 80 ppm minimum

Soil

- Clay Content
  - Take samples at various locations.
- Water Table
- Wet soils aren't best!
- Pesticides
  - TEST Soil

Topography

- Slope
  - 2-10% best
- Flooding
  - Do not build on floodplain

Levee Pond Construction

- Freeboard and Depth
  - 1-2 feet freeboard
  - Depth 5 feet in shallow end
  - 7 feet in drain end
- Slope of Levees
  - 3: 1 minimum
  - 4: 1 maximum

Levee Pond Construction

- Size
  - Larger ponds are cheaper per acre
  - 1-10 acres for new farmers
- Shape
  - 10 acre square
    - requires 2,569 linear feet
  - 10 acre rectangular
    - requires 2,729 linear feet
- Levee width
  - 20 ft for main
  - 16 ft for others

Levee Pond Construction





Photos by Cortney Ohs

Levee Pictures





Newly Constructed Pond



Photo by Cortney Ohs

Small Research Pond



Photo by Cortney Ohs

Graveled vs. Not Graveled Levees





Photos by Auburn University Department of Fisheries and Allied Aquaculture

Orientation

- Right angle to prevailing winds reduces levee erosion
- Parallel increases wind action and aeration
- Site Preparation and Construction
  - Remove vegetation and topsoil.
  - Compaction is critical

Pond Drains

- Typically 6 inch pipe is sufficient for ponds less than 10 acres
- If ponds must be drained to harvest (shrimp) then use one 10 inch per acre of water
- PVC pipe is preferred
- Use anti-seep sleeve around pipe deep in levee

Water Inlets

- Water source pipes should have control valves on them
- May be designed to flow into adjacent ponds
- An independent **Electrical source** for each pond if filled by tube well



Lined Ponds





Watershed Ponds

- Watershed is containment (control)free
- Depth is a function of topography
- Requires 5-7 acres of watershed per acre of water
- Must have emergency spillway

Preventing Leaks

- Minimize amount of loss due to seepage
  - Proper compaction
  - Core trenching
- - Optimum clay content
  - Vertical plastic membranes
  - Vegetative coverage
- Remove burrowing animals (turtles, muskrat)

  Optimal clay content
- Construction during dry season

Pond Bottom
CONSTRUCTION CRITERIA

- If detailed pond bottom slopes are required, usually accomplished by scrapers
- Small 4-6 m^3 earthmovers towed by 4X4 tractors, laser-guided
- Bottom slope from upper end to lower end of pond usually 1m:250-500m or 0.4-0.2% for large ponds, pond should be drainable all the time
- In simple ponds, follows natural slope to estuary
- Must insure at least 20 cm height of harvest gate above high tide elevation (varies considerably by site)

Pond Control Structures
Inflow gates

- Used for control of pond water exchange
- Concrete structures with screen/bag filters on both sides of Levee
- Dual primary screens for pre-filtration (1/4"
- to 1/2“)
- Secondary filtration screen bag eliminates Potential predators (250-500 µM)
- Flashboards for controlling flow rate of water entering pond
- Multiple gates in larger ponds

WATER CONTROL STRUCTURES

HARVEST GATE

- Concrete w/harvest basin in pond
- Number/size of gates depends on speed of harvest required
- Screen to retain shrimp, mesh according to size
- Use of flashboards
- Canal side often modified for harvest pump



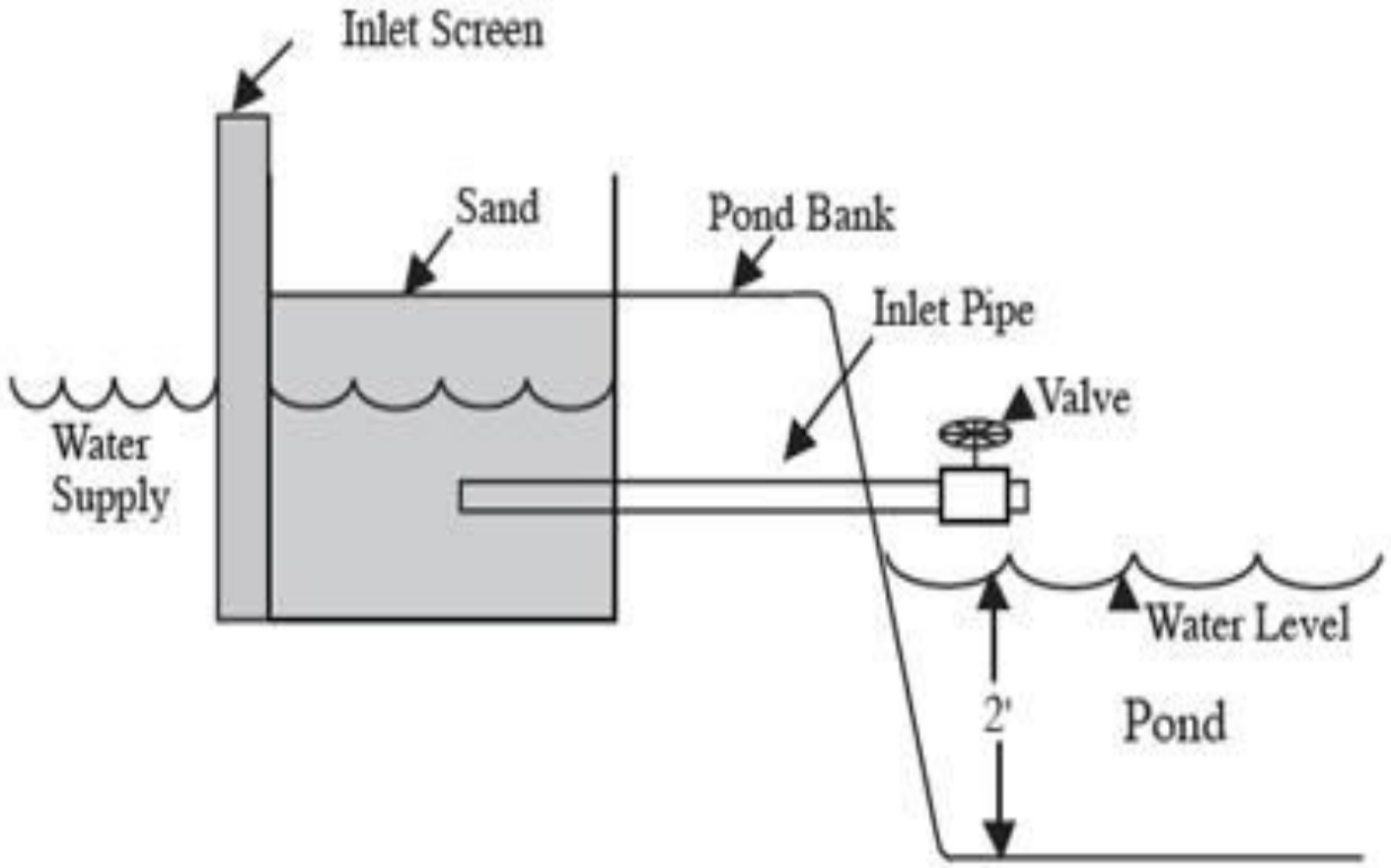
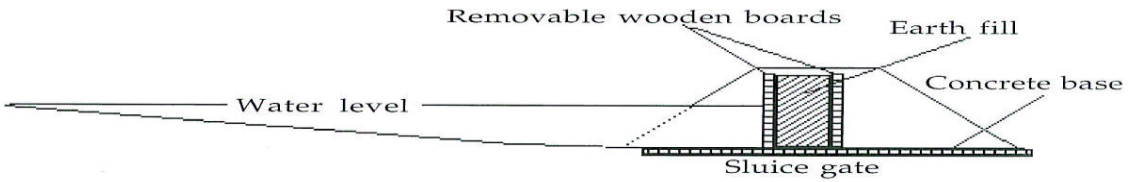
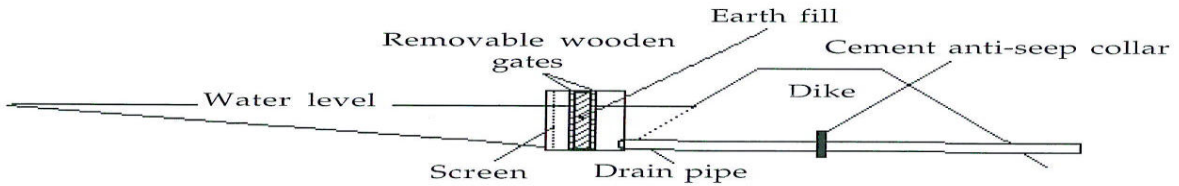

Inlet Screen
Sand
Pond Bank
Inlet Pipe
Valve
Water
Supply
Water Level
2'
Pond

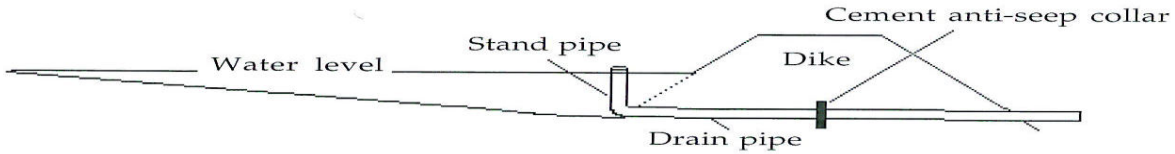
Figure 3



SLUICE GATE STRUCTURE



MONK STRUCTURE



DRAIN WITH STANDPIPE

Figure 15: Cross sections of three structures used for draining ponds.

Harvest Gate: outflow



Harvest Gates: outflow

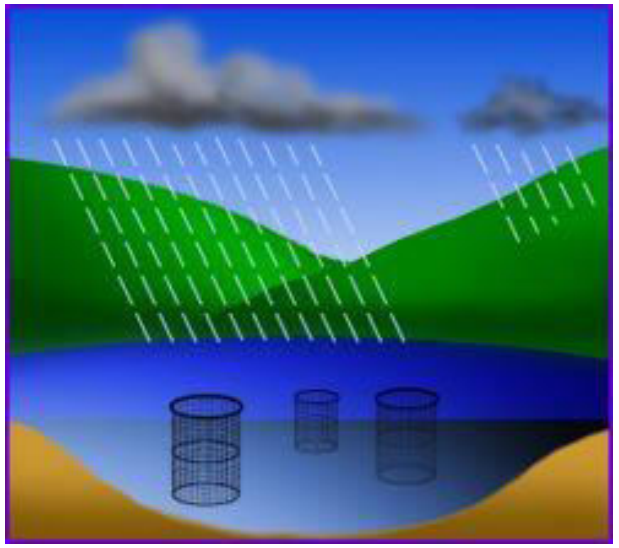


Water Sources

- Rivers, lakes and streams
- Advantages
  - large volumes
  - inexpensive
- Disadvantages
  - excessive nutrients
- Springs
- Advantages
  - few or no predators
  - no pathogens



- Wells
- Advantages
  - no predators
  - no pathogens
- Disadvantages
  - low O_2
- Surface
- Advantages
  - inexpensive
- Disadvantages
  - contaminates
  - 5-7 acre watershed per surface acre of water





Qualities of Cultured Fish species

- Hardy (resistance to diseases)
- Fast growth
- Easy adaptability
- Easy to breed
- Cheaper to culture
- Compatibility with other species
- High market demand
- Ability to tolerate low oxygen levels if it happens

Species cultured in ponds

- Rohu *Labeo rohita*
- Morakhi *Cirrhinus mrigala*
- Catla/Thaila *Catla catla*
- Grass carp *Ctenopharyngodon idella*
- Silver carp *Hypophthalmichthys molitrix*
- Big head carp *Aristichthys nobilis*
- Common carp *Cyprinus carpio*
- Tilapia *Oreochromis niloticus*

Stocking density and ratio

- Stocking density
- 800 to 1000 fish per acre
- 3 species culture
- Rohu 50% Morakhi (Mori)20% Catla 30%
- 5 species culture
- Rohu 35%, Morakhi 20%, Grass carp 15%, Silver carp 30%
- 6 species culture
- Rohu 35%, Morakhi 20%, Grass carp 15%, Catla 20%, Silver carp 10%

Addition of fertilizers

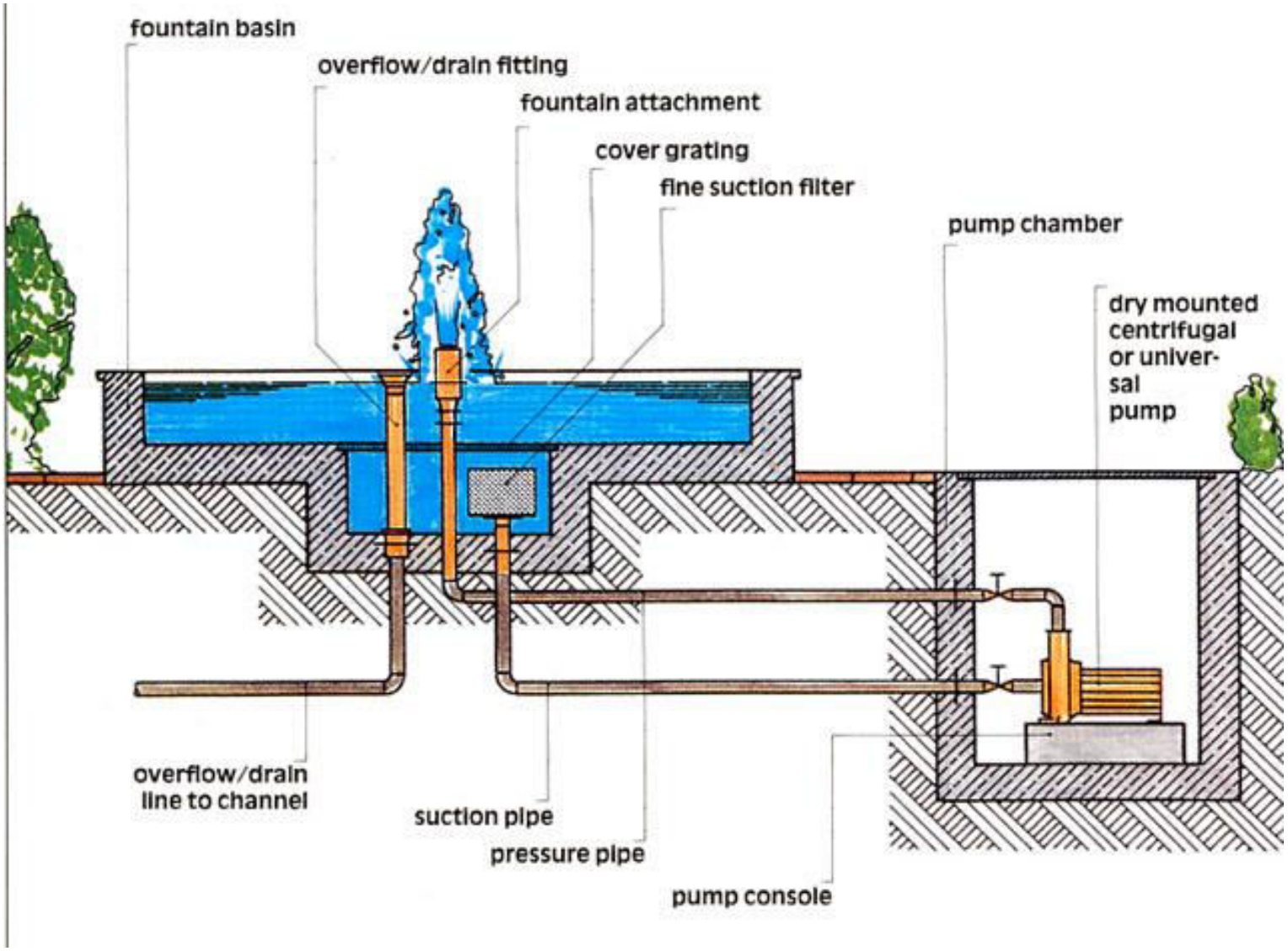
- Natural feeding
- Inorganic fertilizers
- DAP, NH_4SO_4, $NaNO_3$, SSP, Nitrophos, triple super phosphate, Urea
- Organic fertilizers
- Cow dung, poultry waste, duck waste, green manure, compost
- Artificial feeding
- Rice polish, maize gluten meal, sunflower meal, cotton seed meal, soybean meal, wheat bran, canola meal, mustard oil cake

Time and ratio of artificial feeding

- Up to 400gm 360 kg per acre feed 9% protein feed
- Up to 700 gm, 560 kg per acre 22% protein feed
- >700 gm, 560 kg per acre feed >22% protein
- Share of natural feed
- Up to 100 gm fish 100% Natural food
- Up to 400 gm natural food 50%
- Up to 700 gm natural food 32%
- > 700 gm natural food 25%

POND AERATION/OXYGENATION

- level determined by oxygen demand
- pumping vs. artificial aeration
- used for oxygenation and solids mobilization
- efficiency of devices varies
  - paddlewheels: 2.13 kg O_2/kwh
  - propeller/aspirator: 1.58 kg O2/kwh
  - diffusors: 0.97 kg O2/kwh

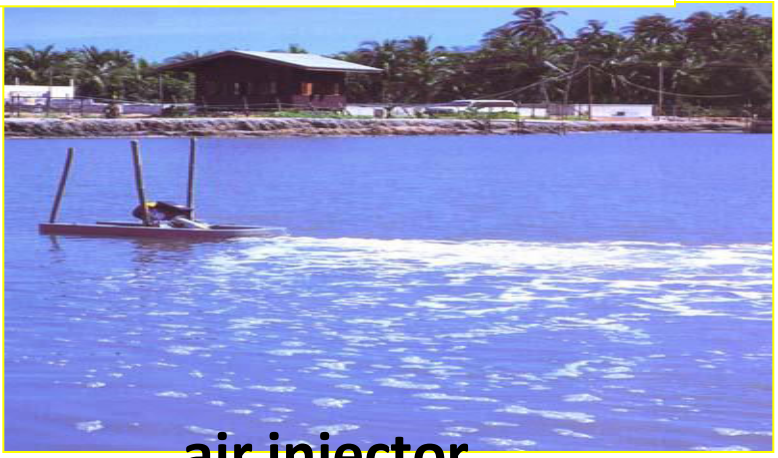

fountain basin
overflow/drain fitting
fountain attachment
cover grating
fine suction filter
pump chamber
dry mounted centrifugal or universal pump
overflow/drain line to channel
suction pipe
pressure pipe
pump console

- If oxygen deficiency occurs;
- Pump water preferably with sprinklers
- Stop feeding and fertilizers forthwith
- Turn on emergency aeration
- If problem becomes chronic then stop feeding and fertilizers for several days

- Typical Aerators



paddlewheel



air injector

Estimating Oxygen Requirement

- During paddlewheel aeration and high density culture O_2 requirement is usually estimated on the basis of feed application to pond e.g.

- 1 kg of feed = 0.2 kg O_2 consumed via respiration
- 300 kg feed = 60 kg O_2 consumed/day

- Caveat(caution): Some O_2 consumed by shrimp/fish, but more by primary productivity

Estimating Paddlewheel Requirements

Biomass density (kg/ha)	Hp (flow-through)	Hp (limited water exchange)
< 1,000	None	None
1,000 – 2,000	2-4	4-8
2,000 – 4,000	4-8	8-16
4,000 – 8,000	8-10	16-20
Above 8,000	Above 10	Above 20


Raceways

Introduction

- Raceways are considered flow-through systems.
- Being simple to construct they are some of the oldest designs in aquaculture.
- Water sources for raceway aquaculture operations are usually streams, springs, reservoirs or deep wells.

Construction material and length

- Raceways are made of concrete!
- Length x width of 6:1 is recommended.
- This prevents the fish stock from swimming in circular movements, which would cause debris to build up in the center.
- If the width is too large this could result in a feeble current speed which is not desirable
- Length is usually constrained by the water quality or stocking density.

Determination of Raceway volume

- If water Flow rate is 20 L/sec X3600 sec/hr ÷ 4 exchanges /hr X m3/1000 = 20 x 3600/4x1000 = 18 m3 /exchange
- Now if flow is 0.020 m3 =W(width) x0.2(depth) x 0.033(velocity meter/second) = 3.0 m then length will be
- L = 18/3.0 x 0.2 = 30 m



Depth and stocking density

- Average depth of a raceway for fin fish, such as rainbow trout, is about three feet.
- So raceway should be about 90 ft long, 6-9 ft wide with slope of 1-2%
- For trout, stocking rates of 30 to 50 kg/m^3 are normal at the end of a rearing cycle!
- While for marine species, such as sea-bass and sea-bream, the achievable load is lower, between 15 and 20 kg/m^3.
- Raceway volume required = total amount of fish in kg/stocking rate in kg per m^3.

- Loading rate = lb fish /gpm
- = (Oa-Ob)(0.0545)/F
- Where Oa and Ob are incoming and outgoing oxygen and F= lb feed/lb fish per day
- Oxygen consumption in mg/kg of fish/hr = Oc ={3xCxΔL/L}9155.23
- Ammonia production and toxicity
- A= 56 P (decimal fraction of protein)
- Oxygen consumption (mg DO/kg of fish per hr = 200 mg in raceway
- In case of feeding O2 consumed per kg of feed is 0.2 kg O2

Recirculating Aquaculture Systems

Recirculating aquaculture systems (RAS) are systems in which aquatic organisms are cultured in water which is serially reconditioned and reused.

Why recirculation

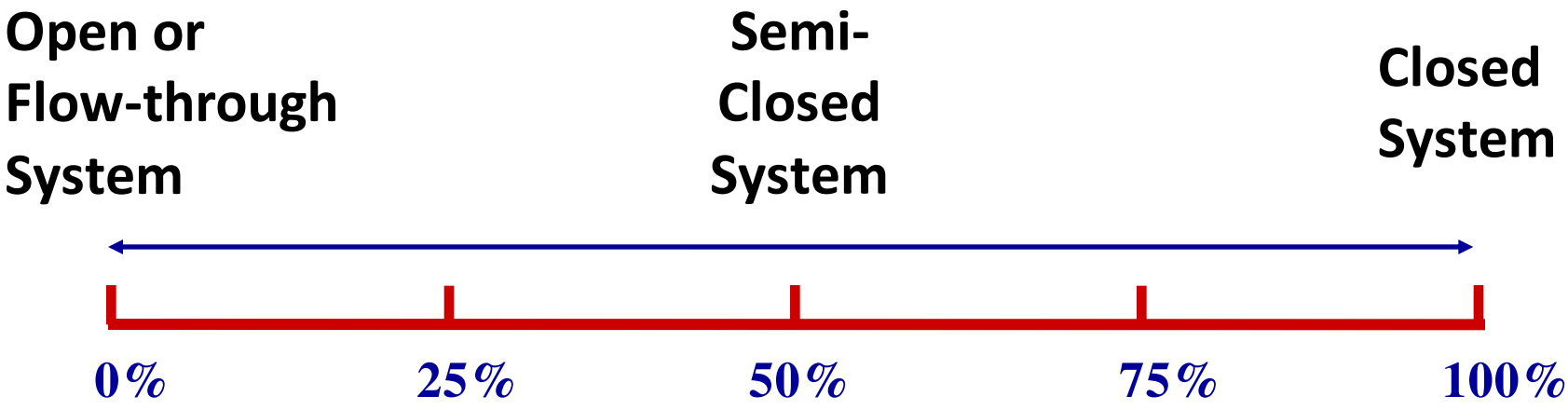
- Conserves water
- Permits high density culture in locations where space and or water are limiting
- Minimizes volume of effluent, facilitating waste recovery
- Allows for increased control over the culture environment, especially indoors
- Improved biosecurity
- Environmentally sustainable

Integrated Treatment

An assembly of components that creates an artificial environment suitable for production, breeding or display of aquatic animals

- Must be reliable
- Must be cost effective
- Must be compatible with the intended user group

Water Reuse Rates



Characteristics of Culture Tank Effluent

- High concentrations of suspended and dissolved solids
- High ammonia levels and high concentration of CO_2
- Low levels of dissolved oxygen
- Application of Recirculation system
- Brood stock maturation
- Nursing and Larval rearing systems
- Nutrition and health research systems
- Short-term holding systems
- Ornamental and display tanks
- High density grow-out of food fish

Factors those dictate Classification of Culture Systems

- Trophic Level
- Temperature
- Salinity

Trophic Level

Distinguishes the level of nutrient enrichment

- **Oligotrophic**
- **Mesotrophic**
- **Eutrophic**

Oligotrophic

- Excellent water quality
- Very Clear
- Used in display aquaria
- Most frequently used for breeding purposes
- Some species are kept in these conditions all of their lives, while others for a period of time



Mesotrophic

Describes the bulk of high-density production systems where risk and economics must be carefully balanced to achieve profitability

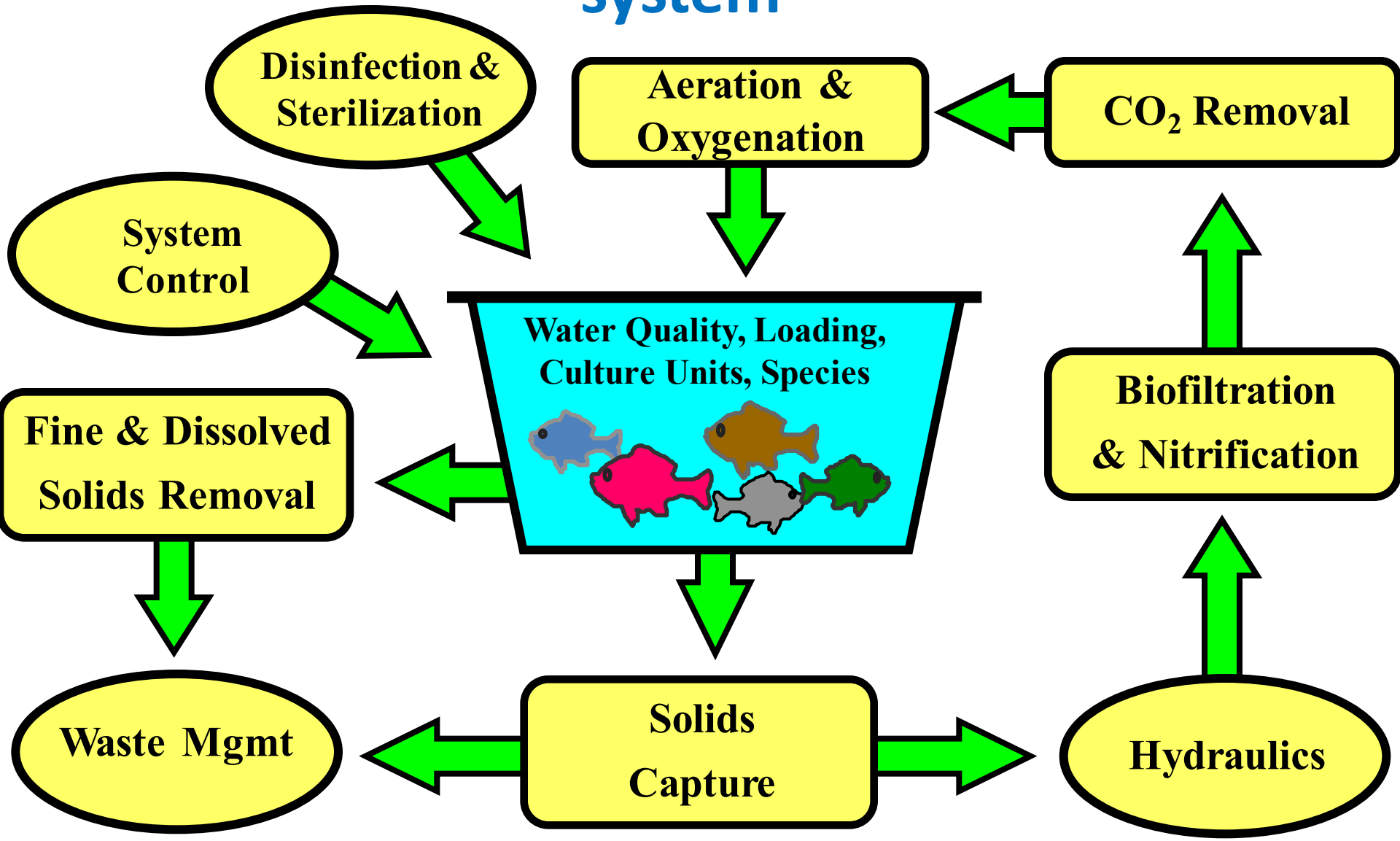
- **Some deterioration in aesthetics**
- **Water quality at safe levels**
- **Dissolved Oxygen- above 5 mg/L**
- **TAN & Nitrite – less than 1mg-N/L**
- **Total suspended solids – less than 15 mg/L**

Eutrophic

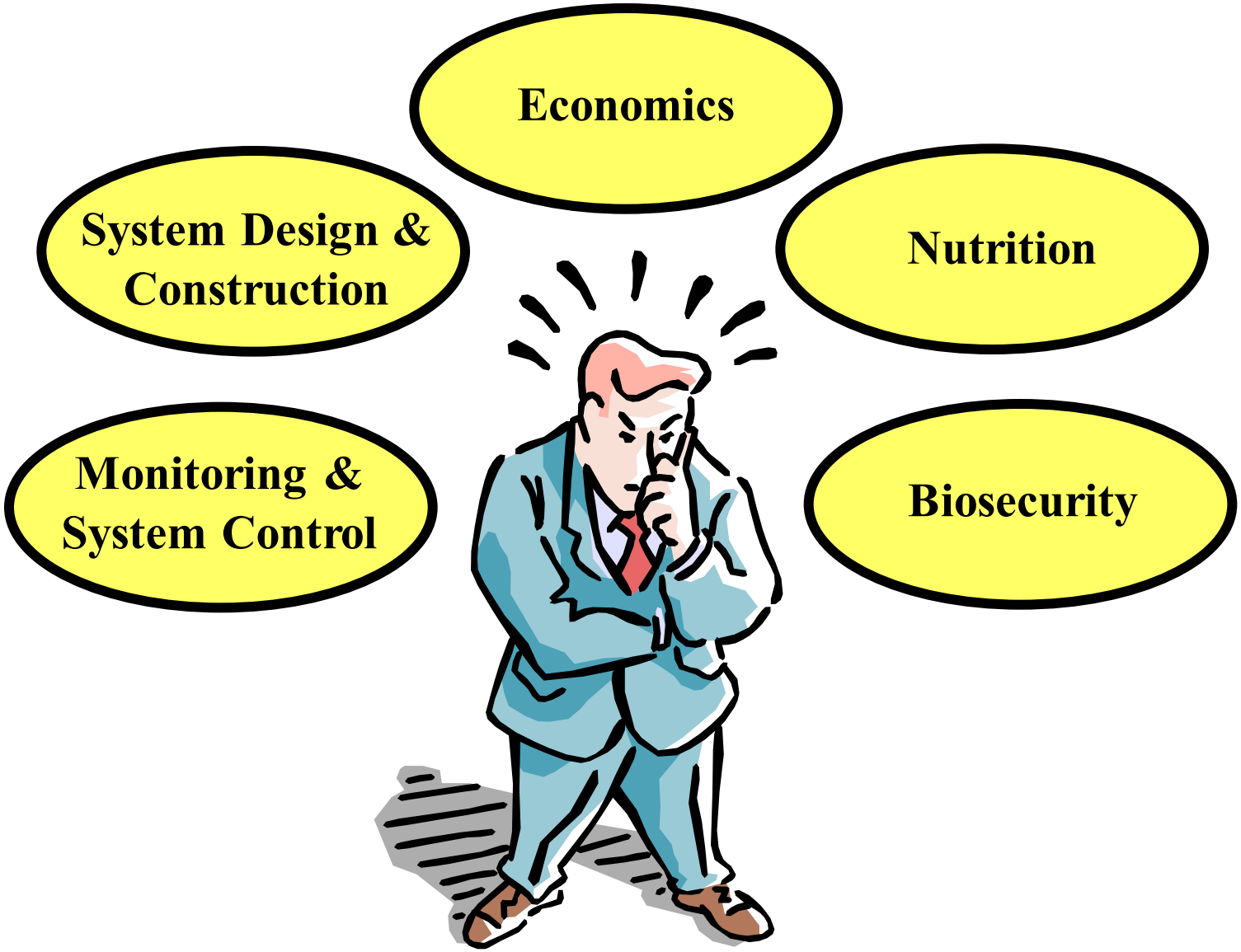
Exist for the grow out of the most tolerant species that show vigorous growth under moderately deteriorated water quality conditions

- **Dissolved oxygen levels- economic optimum level**
- **Ammonia & Nitrite – less than 2mg-N/L**
- **Water quality – marginal**
- **Species evolved under similar natural conditions prosper in these conditions**

Components of Recirculating Fish culture system



Management Decisions



Temperature

Impacts the rates of chemical and biological process at the most fundamental level

Affects: **bacterial growth, respiration, nitrification efficiency**

Cool-water species: below 20º C

Warm-water species: above 20º C





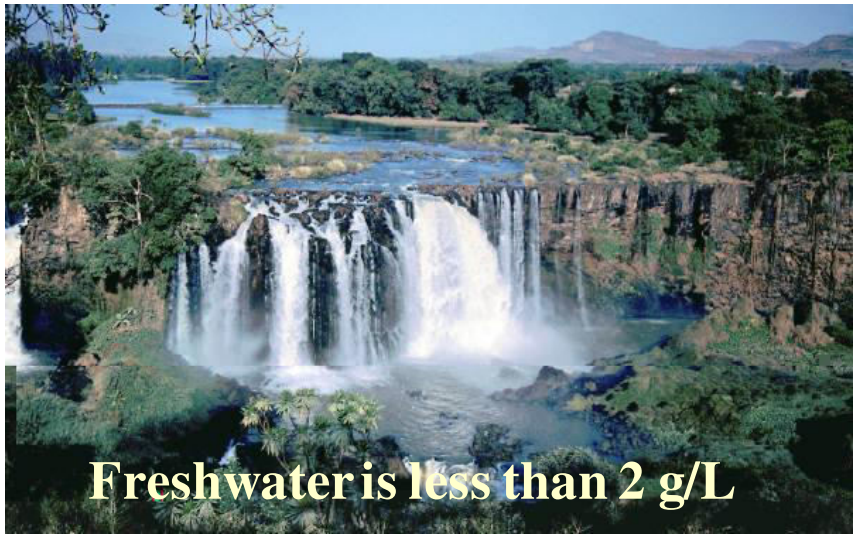
Salinity

Major effect on the oxygen saturation level

<u>Freshwater</u>
Less than 10 ppt

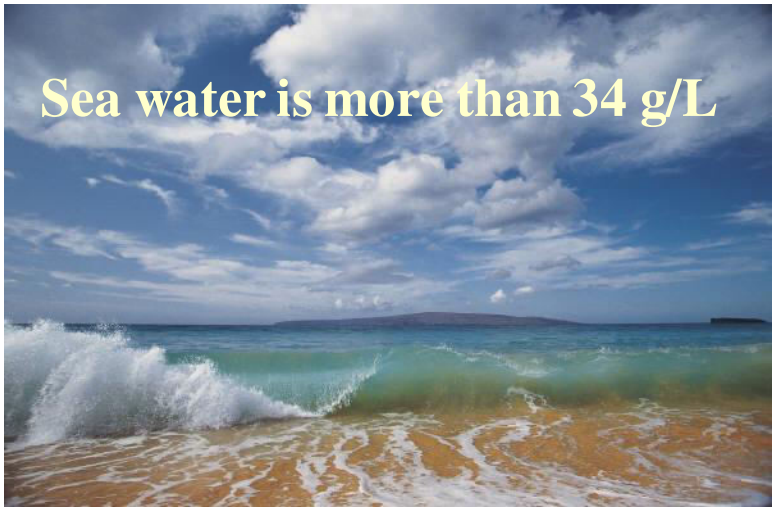
<u>Marine</u>
Greater than 10 ppt

Salinity


Freshwater is less than 2 g/L


Brackish water is 2 g/L to 34 g/L


SALT
NaCl

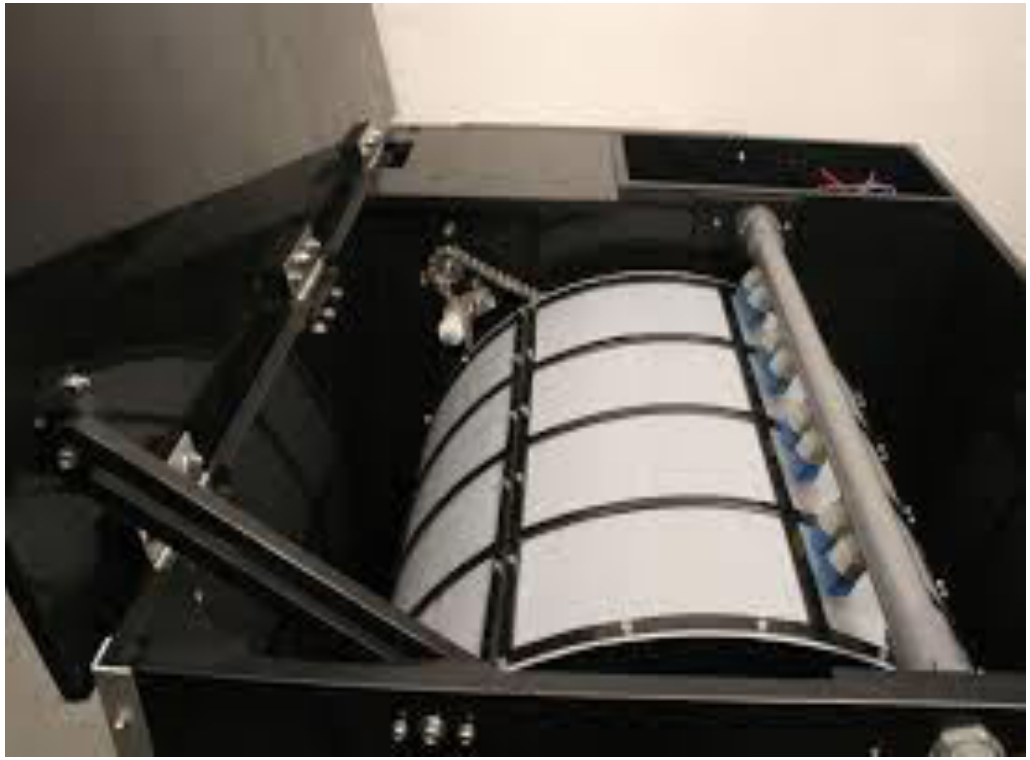

Sea water is more than 34 g/L

Types of Filtration

- Mechanical filtration
  1: Screen filters
  2: Granular Media filters

- Biological Filtration

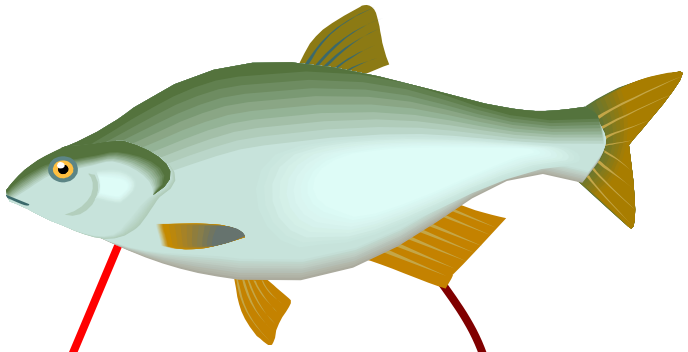
- Screen Filters
- Used with various mesh screens



Biological filtration

- Nitrogen is an essential nutrient for all living organisms and is found in proteins, nucleic acids, adenosine phosphates, pyridine nucleotides and pigments
- In aquaculture there are 4 primary sources of nitrogenous wastes:
- 1: Urea, uric acid and amino acids excreted by fish
- 2: Organic debris from dead and dying organisms
- 3: uneaten feed and feces and
- 4: Nitrogenous gas from atmosphere

Nitrite/Nitrate



feces

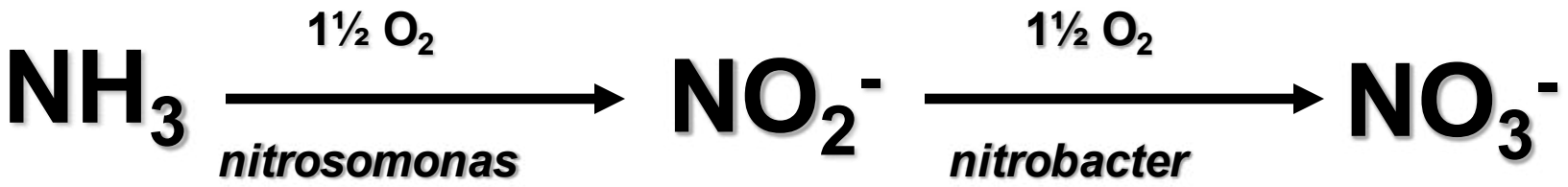
$NH_4^+ + 1.5\ O_2$ + Nitrosomonas → $NO_2^- + 0.5\ O_2$ + Nitrobacter → NO_3^-

Bacterial decomposition

- NH_3, NO_2 and NO_3 are highly soluble in water
- NH_3 exist in water in two forms dependent on temperature and pH of water
- NH_3 is converted to NO_3 by oxidation process and overall process is called Nitrification while reversal is called denitrification

- $NH_4^+ + 1.5O_2 \geq NO_2^- + 2H^+ + H_2O$
- $NO_2^- + 0.5O_2 \geq NO_3^-$
- These reactions need 4.57 g of O_2 and approximately 7.14 g of alkalinity as $CaCO_3$ for complete oxidation of 1 g of NH3

Nitrogenous compounds from Nitrification



- Requires 3 moles oxygen to convert one mole of ammonia to nitrate
-

- Maximum safe unionized ammonia level is 0.025 ppm
- or 1 mg/L or ppm for total NH_3-N for cool water fish and 2-3 mg/L for warm water fish
- production rate of NH_3 (TAN) is

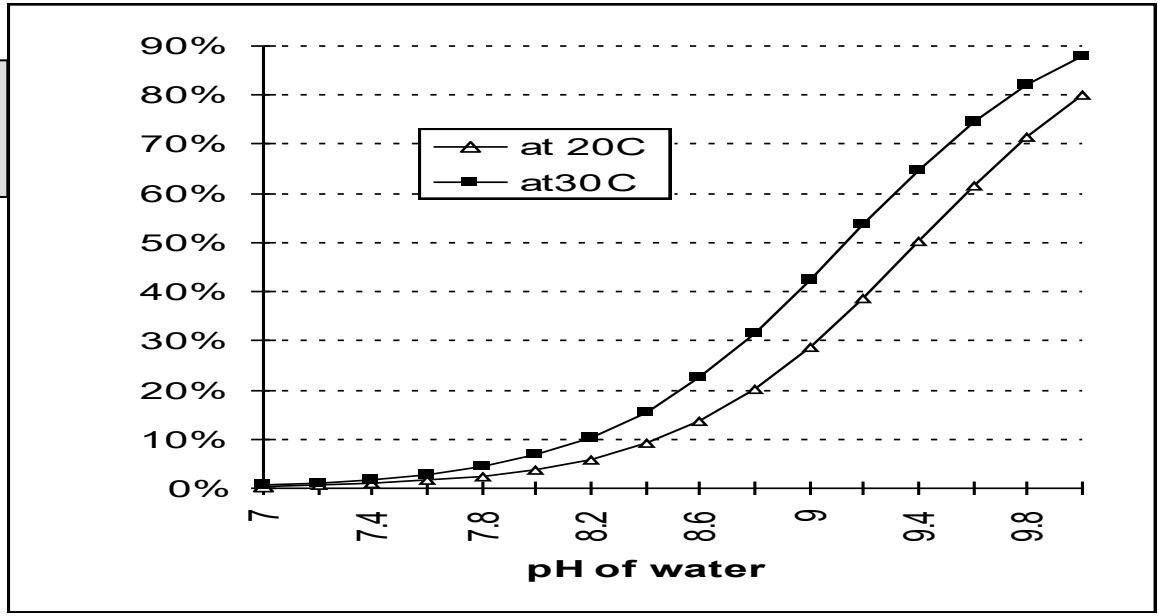
  $P(TAN) = F \times PC \times 0.092$
- Where F is daily feeding level and PC is protein content in feed for one day period

Total Ammonia Nitrogen concentration varies with Temperature and pH

Total ammonia nitrogen (TAN) is a measure of the NH_3 and ammonium levels (NH_4^+) in the water

The ratio of ammonia and ammonium varies in an equilibrium determined by pH and water temperature.

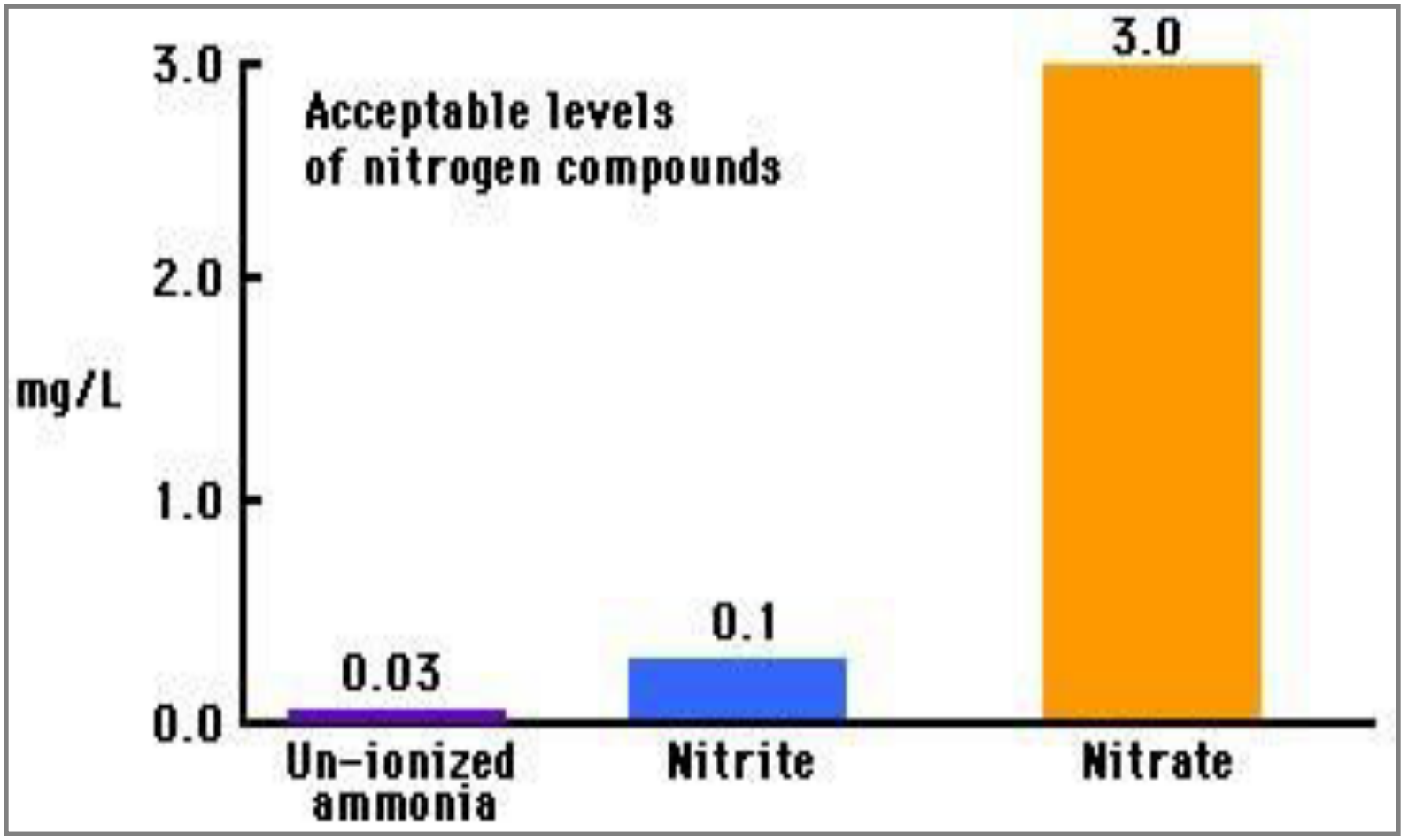
Ammonia as a % of total ammonia nitrogen



Percent of Total Ammonia in the Un-Ionized Form at Various Temperatures and pH

		Percent Ammonia		
Temperature (F)	(pH)	7.0	8.0	9.0
50		0.19	1.83	15.7
68		0.40	3.82	28.4
86		0.80	7.46	44.6

Nitrogenous compounds
Acceptable ranges



- Types of Bio filters Used
-
- commonly used in commercial intensive recirculating aquaculture systems are:
- Trickling filters
- Floating bead filters
- Fluidized bed filters
- Down flow micro-bead filters

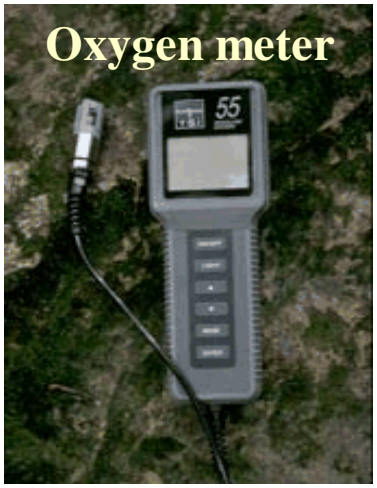
Aeration and Oxygenation

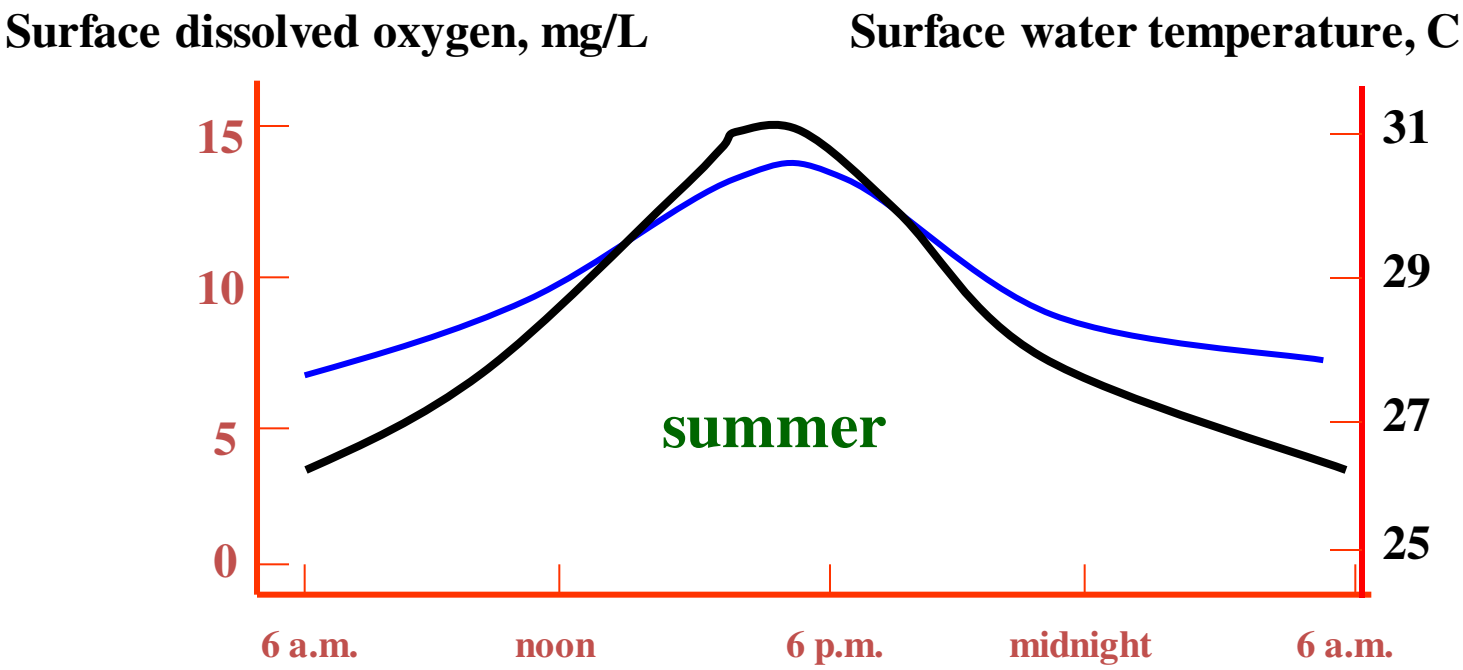
- Oxygen first limiting factor in recirculation fish culture system
- O_2 is consumed by fish and bacteria inhabiting bio filter
- In fish tank O_2 should not be less than 5 ppm and in outgoing bio filter water not less than 2 ppm
- When these levels drop below these concentrations aeration becomes necessary
- Induction air into water is called aeration while induction of pure oxygen is oxygenation
- For stocking density up to 45kg/m^3 aeration is enough but for higher stocking densities oxygenation is must

- Following production term is used for oxygen
- $P_{(Oxygen)}$ = -0.25 kg per kg feed consumed by fish – 0.12 kg per kg feed consumed by nitrifying bacteria -0.13 kg per kg feed consumed by heterotrophic bacteria(can be as high as 0.5) = -0.5 kg(sum of the above) per kg feed for system
- Poor mechanical filtration will let the solid accumulate in the system which will increase O2 consumption
- Safe level is 1 kg O_2 consumption for each 1 kg of feed used

Dissolved oxygen and water temperature

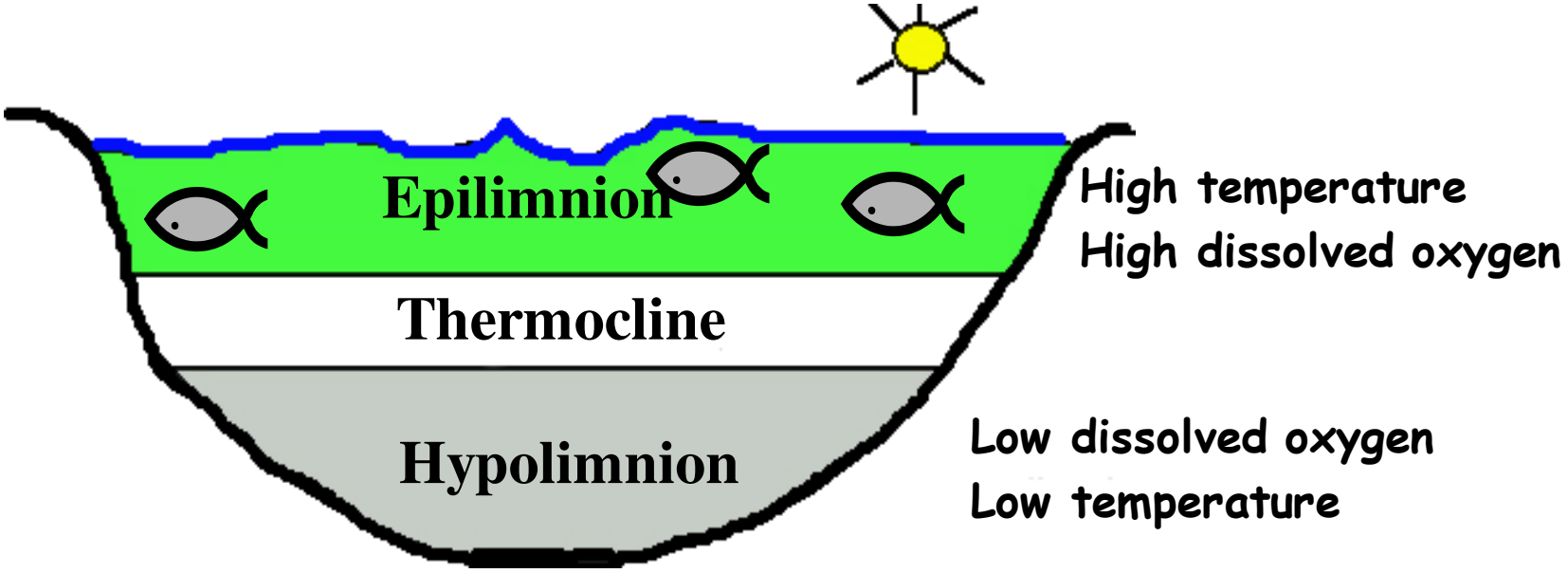
dissolved oxygen and water temperature usually vary over a 24 hour cycle.

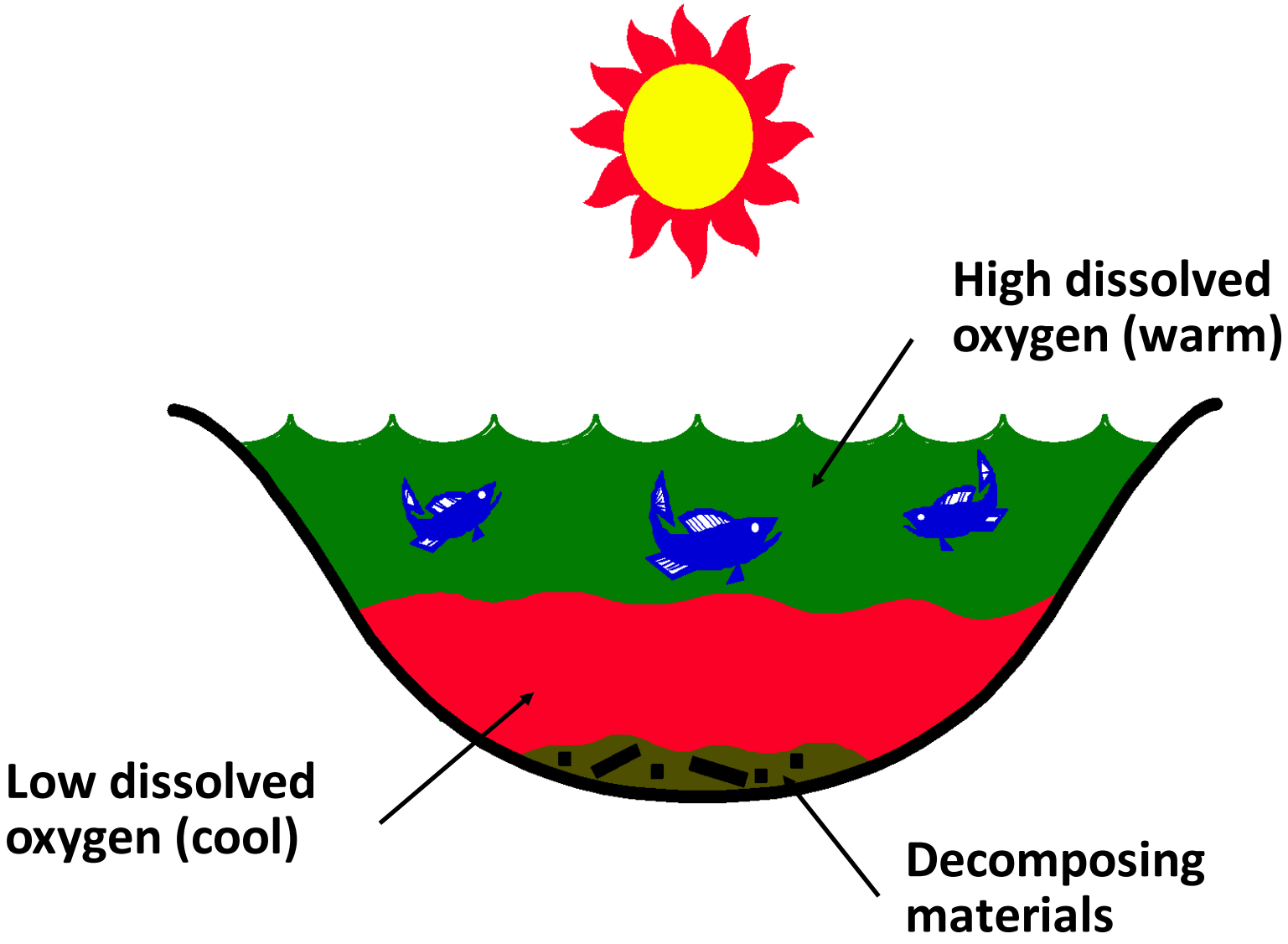




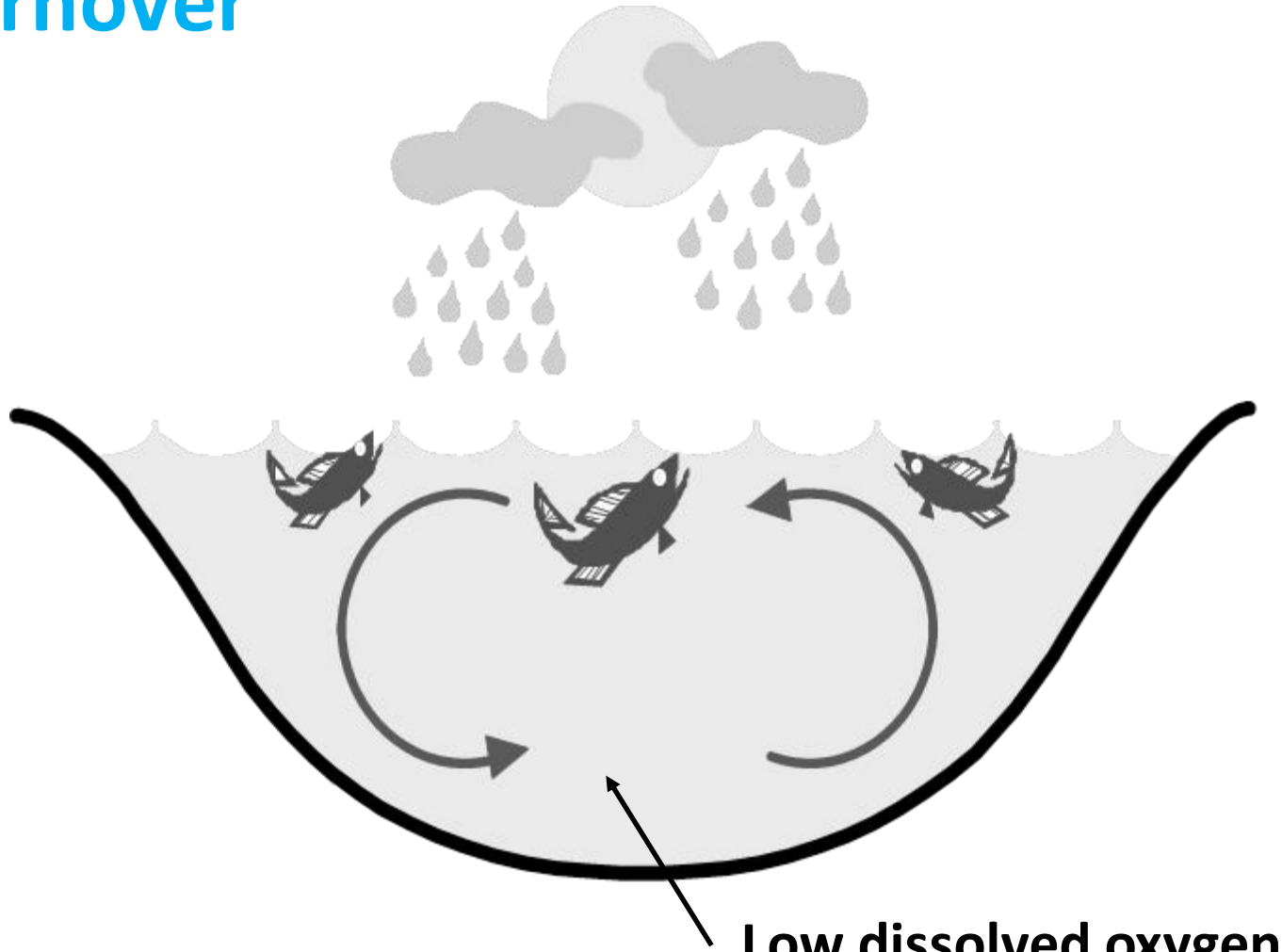
Dissolved oxygen and water temperature

Stratification can cause dissolved oxygen and temperature to vary at different depths in the same system.




High dissolved oxygen (warm)
Low dissolved oxygen (cool)
Decomposing materials

Turnover



Carbon Dioxide Removal

- CO_2 is product of fish and other organisms in the system
- P_{CO2} = 1.375gram produced for each gram of O_2 consumed
- CO_2 obstructs fish respiration by reducing capacity of blood to transport O_2 by Bohr effect
- It decreases pH of the system and decreases bacterial efficiency
- It can be removed by trickling filter by keeping gas/liquid ratio of 5:1.

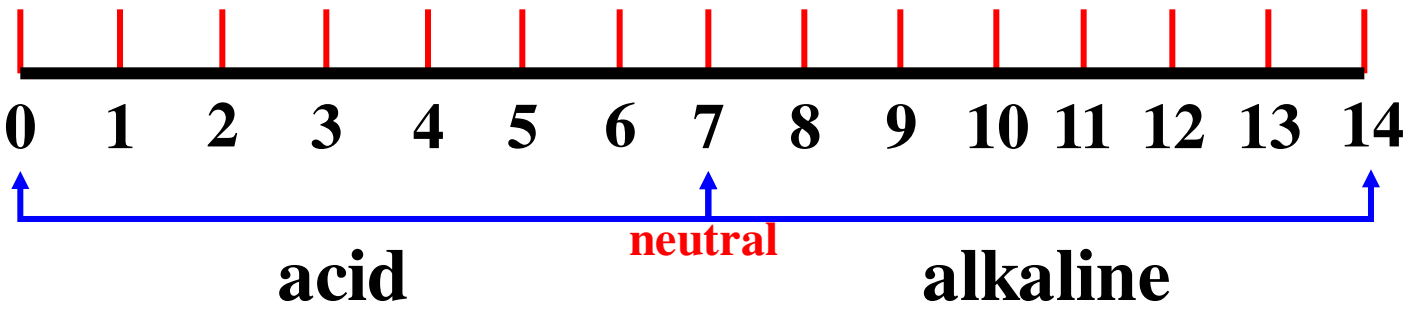
Suspended solids

- Effective control of suspended solids probably most important task that must be accomplished to ensure long term successful operation of RAS
- P_{solids} = TSS = 0.25 x kg feed fed (dry matter basis)

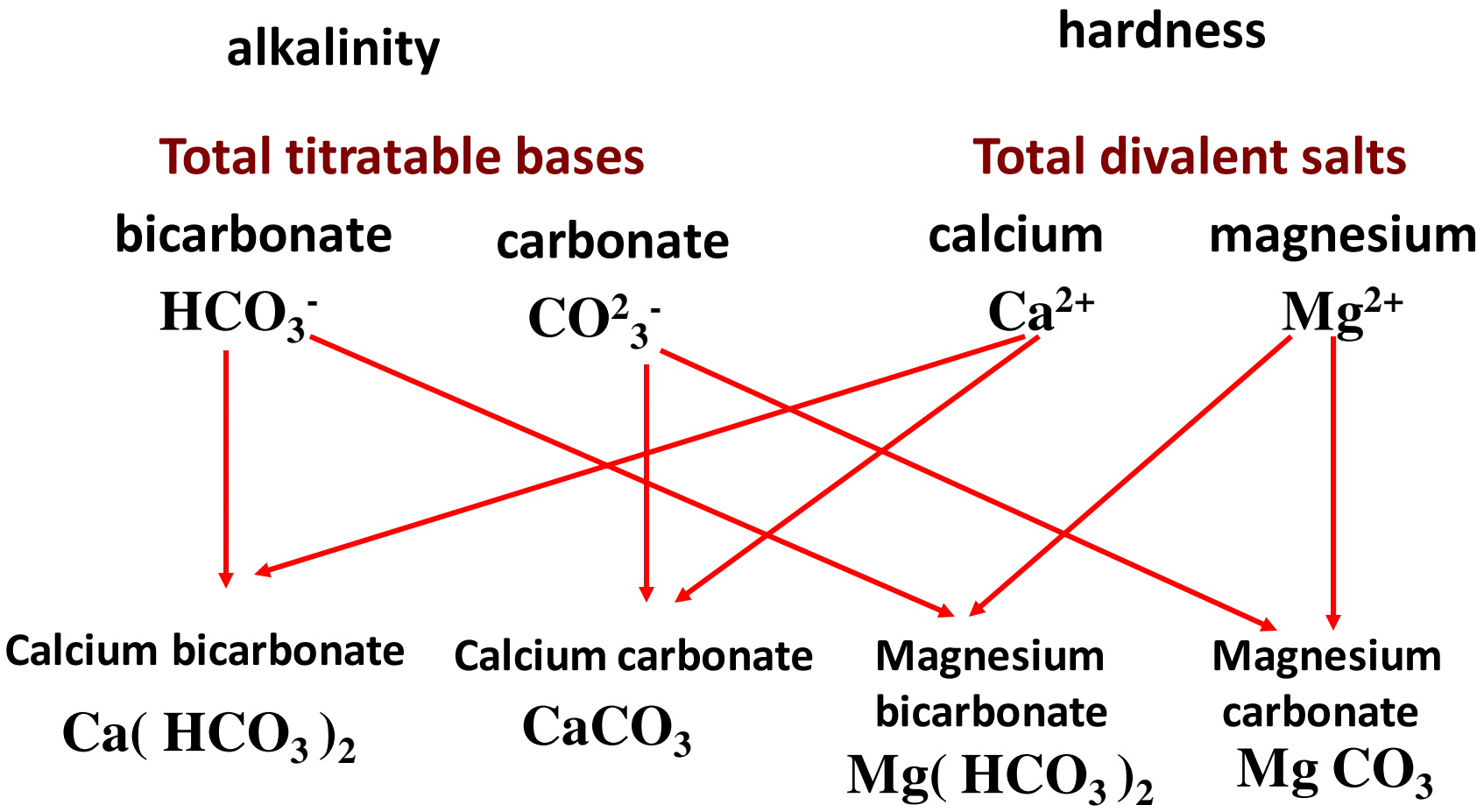
pH

pH is a measure of acidity (hydrogen ion concentration) in water or soil.

$$pH = - \log [H^+]$$



Alkalinity and Hardness



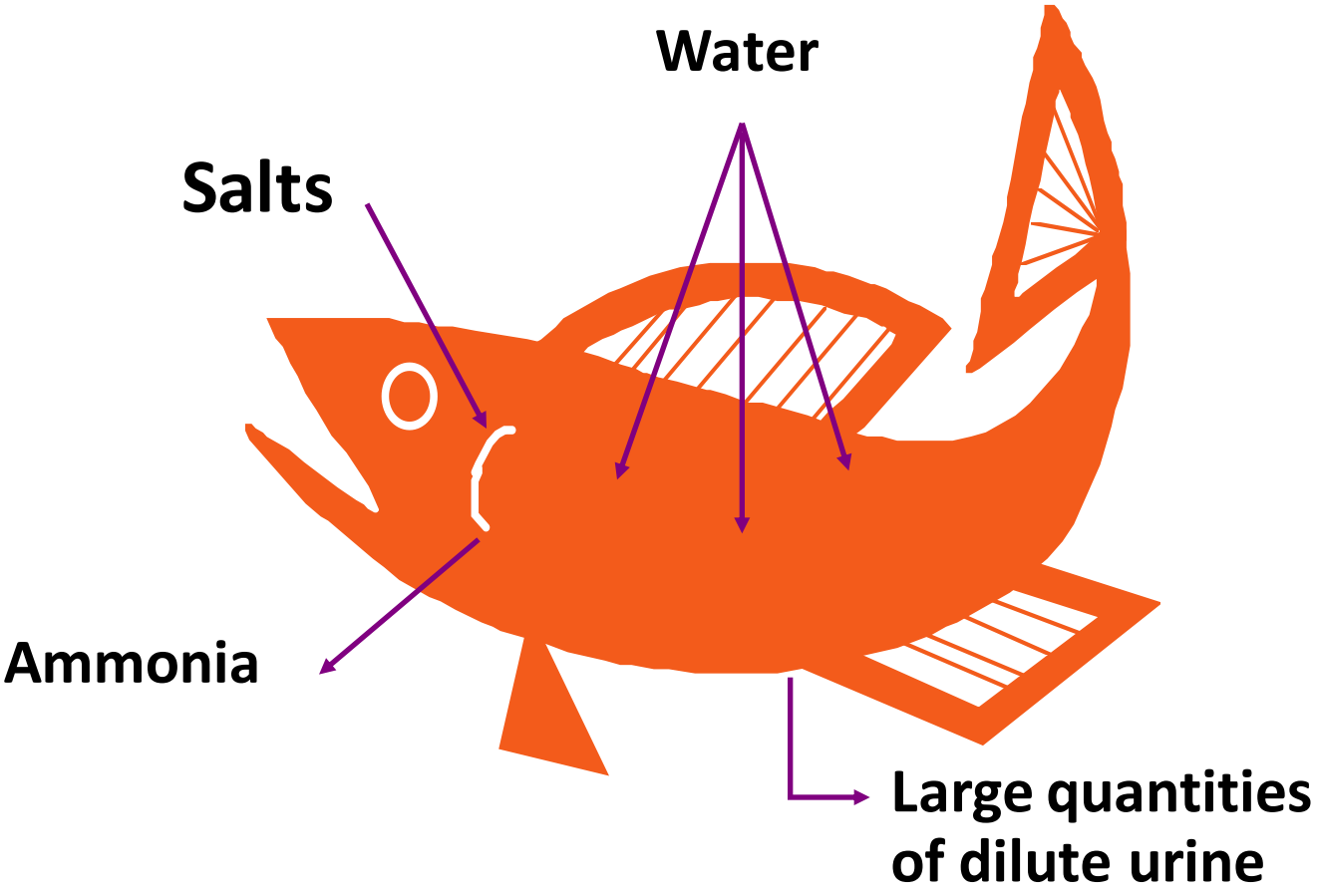
Why water quality for fish

Because fish need good water quality to;

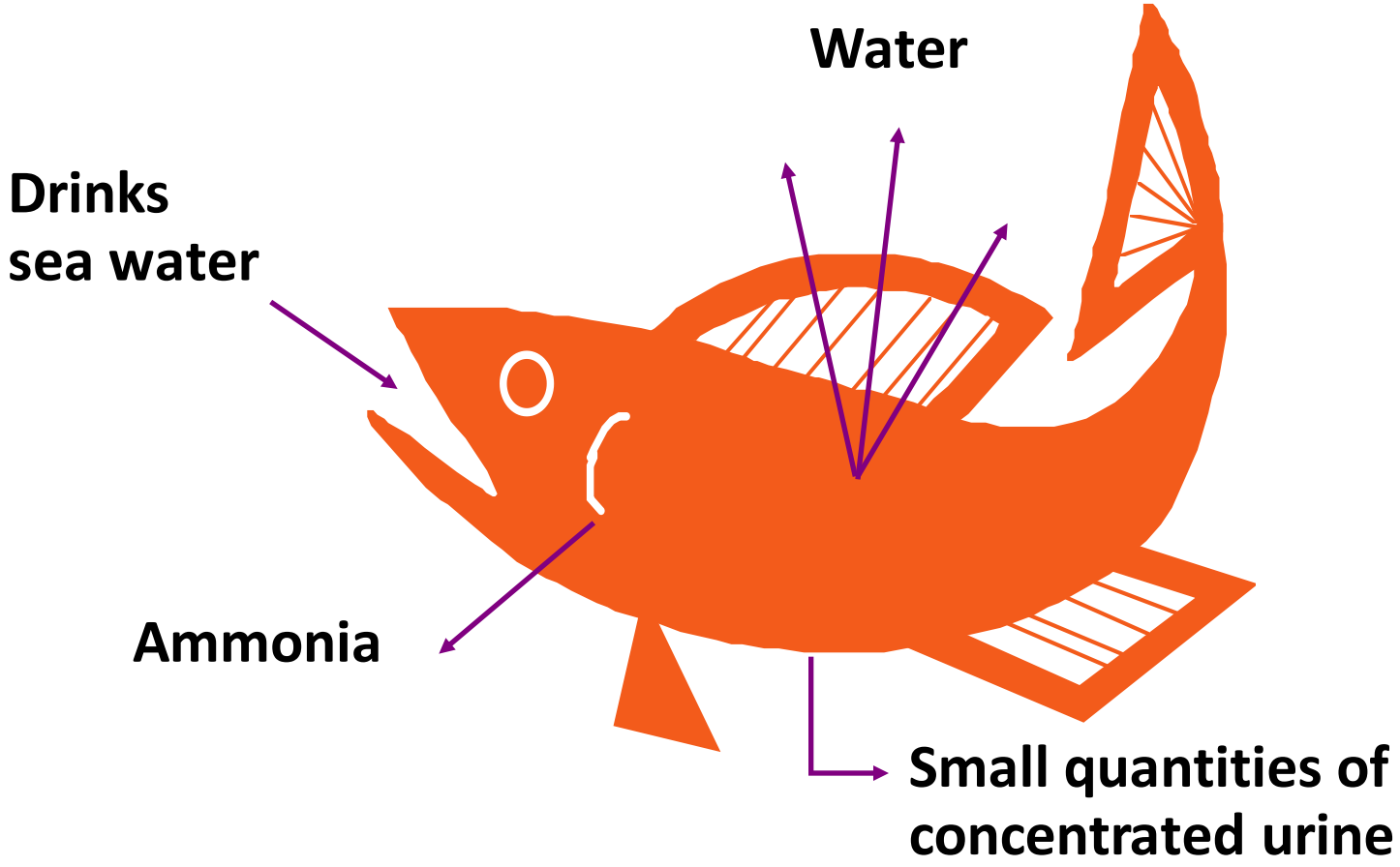
perform all bodily functions in water like

- Eating
- Breathing
- Taking in and losing salts

Water Balance in Freshwater Fish



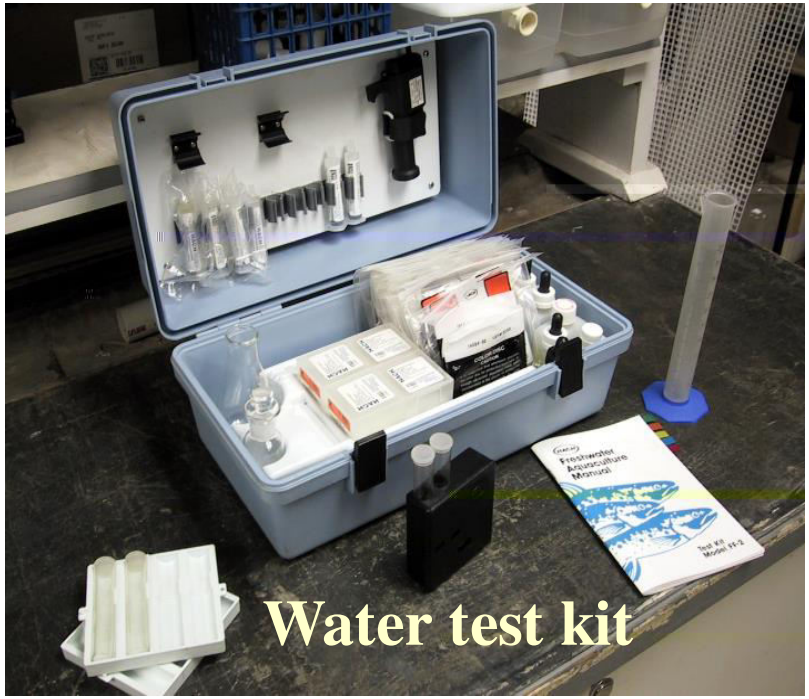
Water Balance in Saltwater Fish



Testing Water Quality

Water quality parameters often tested are:

- **Dissolved oxygen**
- **Water temperature**
- **pH**
- **Total Ammonia Nitrogen**
- **Nitrite/Nitrate**
- **Alkalinity/Hardness**
- **Salinity**



Water test kit

How water quality values are expressed

Parameter	Value
Dissolved oxygen	ppm or mg/L O_2
Water temperature	C (Celsius)
pH	
Total ammonia nitrogen	mg/L N
Nitrite	mg/L NO_2^-
Nitrate	mg/L NO_3-
Alkalinity/Hardness	mg/L $CaCO_3$
Salinity	g/L salt /ppt(‰)

FEEDING OF FISH

FEEDING OF FISH

B-According to environment:

1-Warm water fish: Tilapia & Carp

2-Cold water fish: Rainbow trout

C-According to water:

1-Fresh-water fish

2-Marine fish

Introduction to Aquaculture

- The art of cultivating the natural produce of water.
- Farming implies some form of intervention in the rearing process to enhance production, such as regular stocking, feeding, protection from predators, etc.
- Aquaculture therefore is Farming of aquatic organisms in natural or controlled marine or freshwater environments
- Both under controlled or semi-controlled conditions.
- Mariculture – (old name) marine or brackish water“
- Aquaculture encompass farming of aquatic organisms, including fish, mollusks, crustaceans and aquatic plants.”

FEEDING OF FISH

Nutrient requirements of fish:

- The Standard Environmental Temp. (SET) for cold-water fish about 10C (salmon) & warm-water fish 30C (channel catfish).
- If water temp. deviates upward than SET , nutrient requirements increased & vice versa
- Most requirements for nutrients that have been published focus on juvenile fish/shrimp
- many represent single lab experiments, unchallenged, unsupported by others
- optimum performance can be affected by management, environmental factors and fish/shrimp size
- in formulating diets for a species for which nutrient requirements are unknown, those for a related species are used

Nutrient Requirements

- Most variation among aquatic species can be associated with whether the animals are:
- 1) cold water vs. warm water; 2) freshwater or marine; 3) finfish vs. crustaceans as mentioned earlier
- values in nutrient requirement tables only represent minima, don't allow for processing or storage losses
- AA's, minerals stable with reference to heat, moisture, oxidation
- vitamins and lipids are not stable (affected by heat, oxidation, light, moisture, etc.; store in cool area)
- 50% of ascorbic acid is lost in processing, half-life of 2-3 months in storage
-

FEEDING OF FISH

- Most important nutrient
- Main source of essential amino acids
- The most expensive source
- Carnivorous fish consume foods with ~50% of protein
- Herbivorous & omnivorous = less

1-Protein requirements:

- Function of protein for fish:

A-Provide energy

B-To supply amino acids

C-Functional proteins, hormones & enzymes

-

- Functions:
- 1) Component for muscle tissue formation
- 2) Component for enzyme, antibody, hormone & blood protein serum
- 3) Tissue recovery
- 4) Growth
- 5) Reproduction

Sources of protein:

A-Animal products (fish & meat meal)

B-Agriculture by-products (algae, cereals, legumes)

C-Industrial waste products

- Protein from animal

Fish meal, meat & bone meal, feather meal, by-product slaughtery (bones, organs)

- Protein from plants

Soybean meal, maize meal etc.

Types of Aquaculture

- Extensive Aquaculture: Minimal control, lower density
- E.g., ponds, prevalent in third world countries





- Intensive Aquaculture: Highly controlled, high density, RAS, raceways, confined
- E.g. industrialized

Protein Requirements of Fish

- Fish contain 60-93% crude protein, thus, fish diets are higher in protein than birds & animals.
- Fish appear to be relatively efficient utilizer of protein to energy with 84%.
- Fish can not synthesize their amino acids & obtain chiefly from the diet & absence affects growth of fish

FEEDING OF FISH

Item	Starter	Grower	Finisher
Cold water fish	40-50	35-40	30-40
Warm water fish	35-50	25-50	28-32

Protein requirements

- Species
  - Tilapia - 28-40% diet
  - Baung - 42-55%
  - Keli - 30-40%
  - Lampam - 30-50%
- Size (e,g,, Tiger shrimp)
  - Larva-PL_9 - 50-55%
  - PL_{10-24} - 46%
  - PL_{25}-1g - 40%
  - 1-10g - 38%
  - >10g - 35%
  - Minimum dietary requirement = balanced mixture of amino acids

- Fish synthesize body proteins from amino acids

Table 1. Two major classes of amino acids.

Indispensable (essential)	Dispensable (nonessential)
Arginine	Alanine
Histidine	Asparagine
Isoleucine	Aspartic acid
Leucine	Cystine
Lysine	Glutamic acid
Methionine	Glutamine
Phenylalanine	Glycine
Threonine	Proline
Tryptophan	Serine
Valine	Tyrosine

Deficiency in essential acids = limit protein synthesis

=> Consequences: Reduced weight gain etc

More Waste Limited Digestion

- About 36% of feed is excreted as organic waste
- Live fish biomass = 5x more waste than human
- Why?

- Limited digestion
- Large fraction of feed remain undigested & is excreted

FEEDING OF FISH

Factors affecting protein requirement:

1-Species

2-Age

3-Stocking density

4-Water temperature

5-Production stage

FEEDING OF FISH

Deficiency symptoms of amino acids:

- Def. of lysine $\Rightarrow$ dorsal & caudal fin erosion & increased mortality
- Def. of methionine $\Rightarrow$ cataract
- Def. of tryptophan $\Rightarrow$ scoliosis, renal calcinosis, cataract, caudal fin erosion & decreased carcass lipid content

FEEDING OF FISH

Energy requirements:

A-Carbohydrates:

- Fish digest simple sugar efficiently & decrease digestibility of large molecule.
- Efficiency of carbohydrate in fish 39% compared to 96% in mammals.
- Carbohydrates can be used to spare protein since less protein will be used for energy.
- Fibers as cellulose & hemicellulose controlling passage of feed through gut & not exceed 8%, while 10% $\Rightarrow$ reduction in nutrient intake & digestibility.

Carbohydrates

- Fish do not have a specific dietary requirement for carbohydrates
- Most abundant & least expensive
- Carbohydrates => simple sugars
- Range from easily digested sugars to complex cellulose
- Complex cellulose = Hard to digest
- Intake of carbohydrates depend on enzyme production

-Carbohydrates also gives pellets integrity & stability (less dense)

Prepared feed

- Carnivorous fish = <25%
- Omnivorous = 25- 45%

- Amylase
- In herbivores => amylase occurs through entire digestive tract

Requirements in Fish feed

- 20% carbohydrates = cold-water fishes
- 30% carbohydrates = warm water fishes

- Carbohdyrates from plants

- Starch in seeds, tubers, wheat, paddy

FEEDING OF FISH

Energy metabolism in fish:

- Fish do not spend much energy towards maintaining body temperature (cold blooded).

Factors affecting energy requirement:

1-Species

2-Size

3-Light exposure

4-Composition of diet

5-Physiological activity

6-Age

7-Temperature of water

8-Other environmental factors (water flow, water composition, pollution)

FEEDING OF FISH

Fat:

- Fat-soluble compounds occurring in plant & animal tissues
- Consist of fats, phospholipid, sterols, fatty acids etc.
- Excessive dietary lipid = nutritional disease e.g., fatty liver or visceral fat

Fish can utilize energy of fat by 84% efficiency

Functions of fat:

1-Providing energy

2-Cushions for vital organs

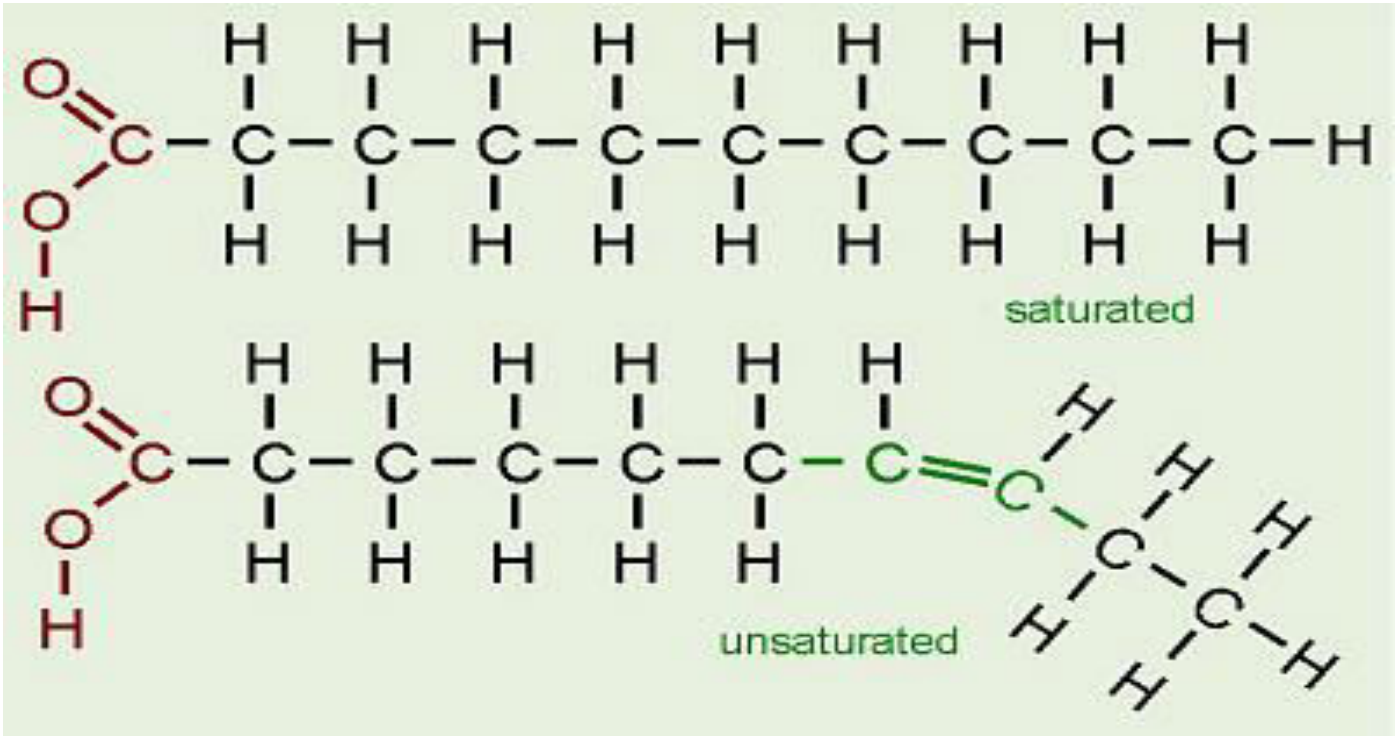
3-Energy reserves

4-Insulators & lubricants

5-Transports of fat-soluble vitamins

Fatty acids

- Saturated FA = no double bonds
- Unsaturated FA = double bonds

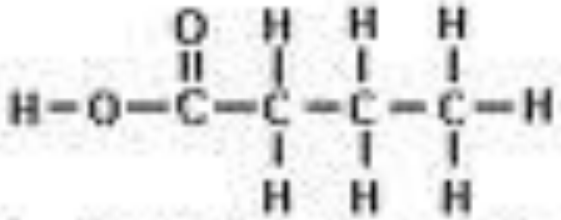


- Most important = EFA
- EFA = Organisms need it but can’t synthesize it

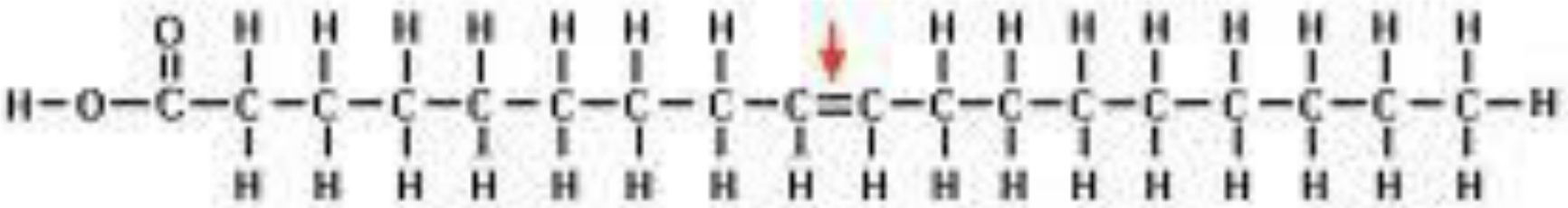
<u>Example</u>

- PUFA =

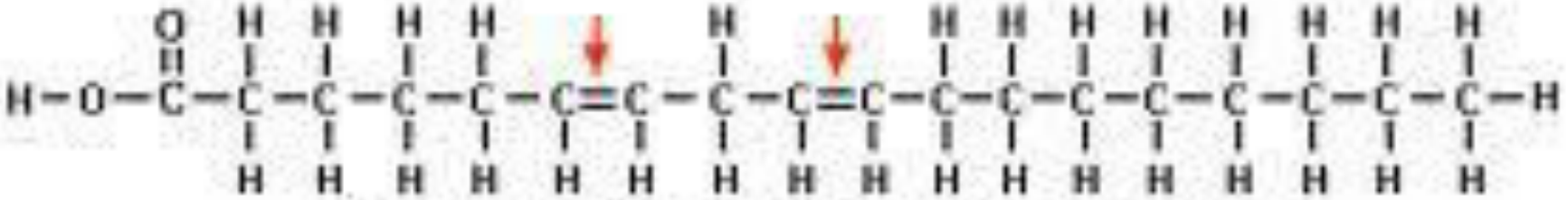
Saturated and Unsaturated Fatty acids



Butyric Acid-Saturated Fatty Acid



Oleic Acid- Monounsaturated Fatty Acid



Linoleic Acid- Polyunsaturated Fatty Acid

FEEDING OF FISH

- The best supplement of fat are fish oils & SBOM.

Fish oil (5%) is the best source of EFAs

Def. of EFAs:

1-Reduced growth & poor feed conversion

2-Increased mortality

3-Elevated muscle water content

4-Increased susceptibility to caudal fin erosion

5-Fatty infiltration in liver

Lipid for tiger shrimp

Size	lipid (% diet)
Larva-PL9	10-15
PL10-24	7.5
PL 25- 1 g	> 3.3
1-10 g	>3.0
> 10 g	> 2.8

Vitamins

- Only needed in small amount
- Critical for the maintenance of normal metabolic & physiological functions
- Water soluble & fat soluble
- Problem:

- Leaching of water soluble vitamins
- Hypervitaminosis, rare but causes enlargement of liver & spleen, abnormal growth, bone formation etc.

Vitamin requirements:

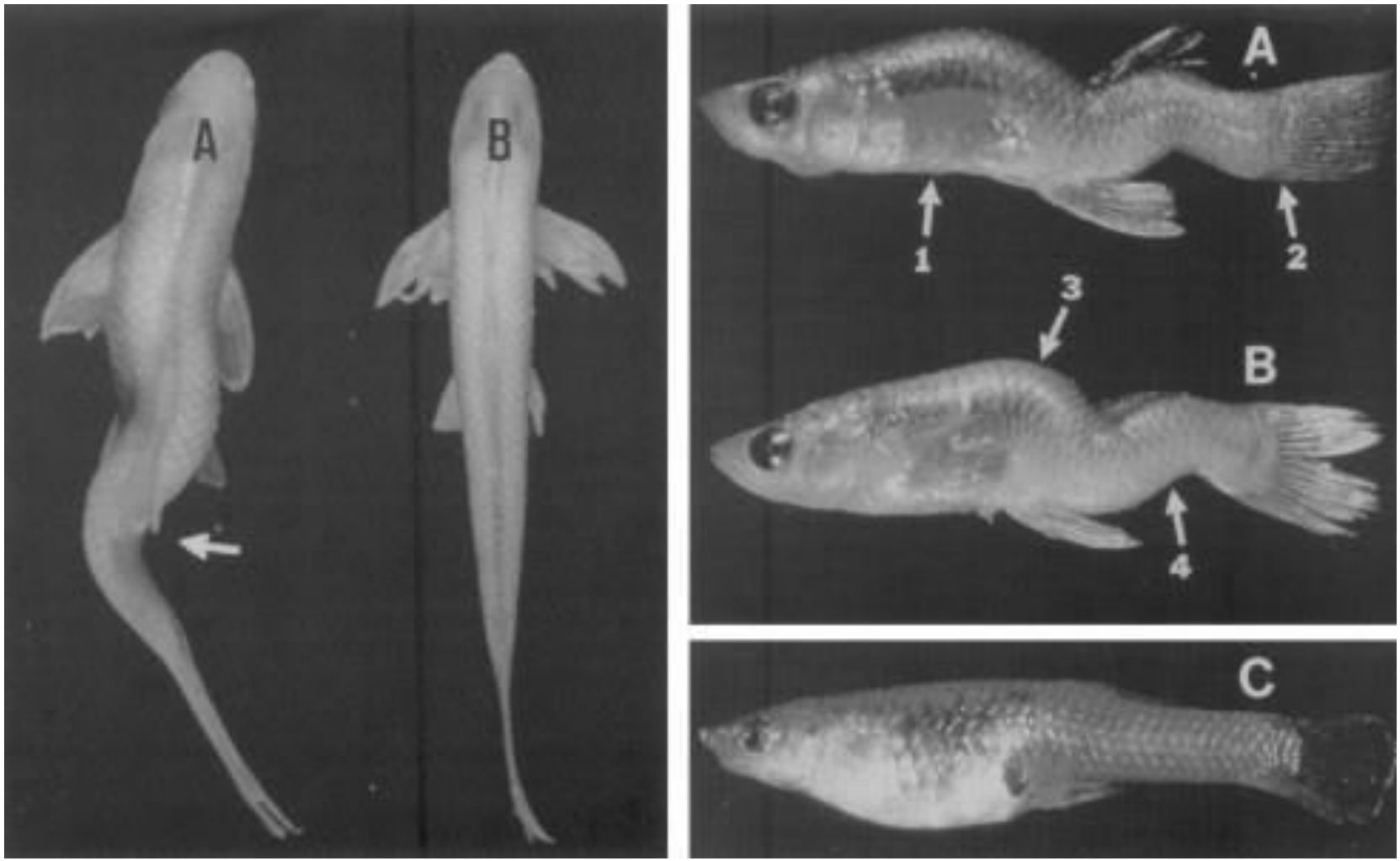
- Vitamin req. of fish resemble those of poultry.
- There is not enough bacterial activity in gut to satisfy B-complex & vit.K requirements
- Fish feed contain high levels of oils $\Rightarrow$ oxidation may result inactivation $\Rightarrow$amounts must be in excess to ensure adequate levels of requirements.

- Vitamins can be of two types
- Fat-soluble vitamins : Vit. A, D, E, K
- Water- soluble vitamins : Vit. C, Vit B12, thiamin, riboflavin etc.
- Vitamin in feed = Sold as vitamin premixes

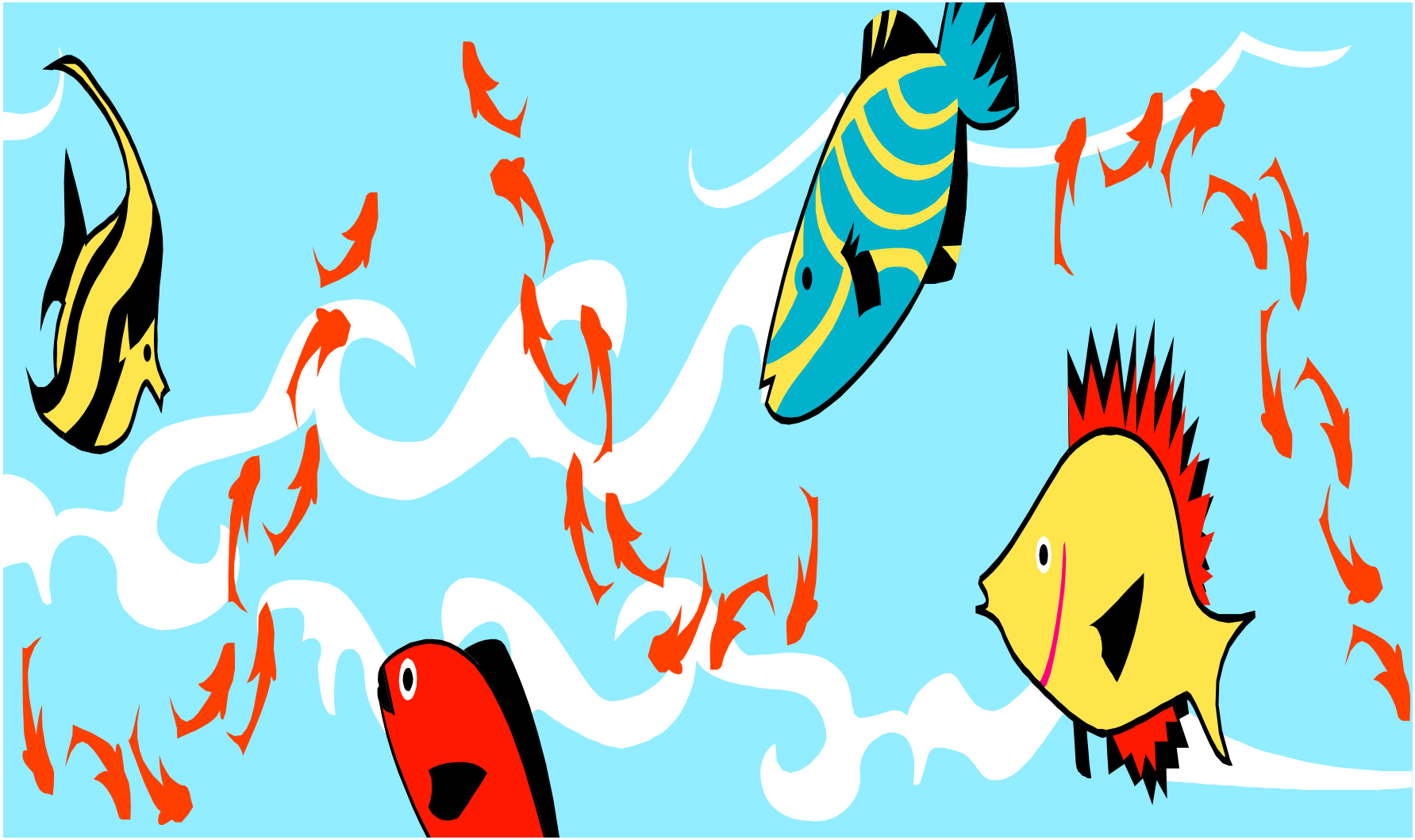
Table 3. Vitamins and some of their major functions as established in fish.

Fat-soluble vitamins	**Function**
vitamin A, retinol	epithelial tissue maintenance, vision
vitamin D, cholecalciferol	bone calcification, parathyroid hormone
vitamin E, tocopherol	biological antioxidant
vitamin K	blood clotting
Water-soluble vitamins	
thiamin, B_1	carbohydrate metabolism
riboflavin, B_2	hydrogen transfer
pyridoxine, B_6	protein metabolism
pantothenic acid	lipid & carbohydrate metabolism
niacin	hydrogen transfer
biotin	carboxylation & decarboxylation
choline	lipotrophic factor, component of cell membranes
folic acid	single-carbon metabolism
cyanocobalamin, B_{12}	red blood cell formation
ascorbic acid, vitamin C	blood clotting, collagen synthesis
inositol	component of cell membranes

Scoliosis (Vit. C deficiency)



FEEDING OF FISH
Mineral Requirements



Minerals

- Inorganic elements for Calcium, phosphorus
- Function:

- Formation of skeletal tissue, respiration, digestion & osmoregulation
- Macromineral & micromineral (based on quantities required in the body)

- Calcium, phosphorus, magnesium etc. are Macrominerals

- While cobalt, chromium, copper, Zn, Selenium are Microminerals (trace minerals)

-Minerals requirements:

- Fish have the ability to absorb a number of minerals directly from water $\Rightarrow$ reducing mineral requirements in the diet.
- Fish require all macro-and micro-elements required by other animals for enzymes & cofactors.
- Fish in soft water (low mineral content) require additional supplements in diet.

Table 2. Trace minerals and some of their prominent functions.

Trace mineral	Function
Copper	metalloenzymes
Cobalt	vitamin B_{12}
Chromium	carbohydrate metabolism
Iodine	thyroid hormones
Iron	hemoglobin
Manganese	organic matrix of bone
Molybdenum	xanthine oxidase
Selenium	glutathione peroxidase
Zinc	metalloenzymes

Type of fish feeds

- Finely ground meals
- Crumbles
- Flakes
- Pellets of various size & density

⇒ Sinking

⇒ Floating

⇒ Slow sinking

- Microdiet









www.semachinery.net

Diet preparation

- Grinding, mixing & pelleting

- Grinding

reduce ingredient particle size to

1) Easy handling of ingredient

2) Improve feed digestibility, acceptability, mixing properties & pelletability

- Mixing

Achieve uniformity of ingredients

- Pelleting

Transformation from soft, dusty mixture to compacted form

- Compression pelleting = sinking pellet





Cooking extrusion = floating pellet

Extrusion process

⇒ Feed mixture subjected to heat & moisture

⇒ Pellets expand, reducing density

⇒ Advantages = Expanded pellet able to absorb more lipid

Proper Feed Storage

- Because feeds contain ingredients that are susceptible to degradation, you are concerned with storage conditions and shelf life
- **What breaks down?** Vitamins, lipids, proteins
- fats and oils break down via rancidification
- proteins can become deaminated: do not use any feed over 3 months old
- big problem for those who import feed

Proper Feed Storage

- Proper storage = protection against high temperature, humidity, moisture, insect & rodent infestations

- Minimum length of time

- Feed sacks should not touch the floor or side walls

- Store = 100% waterproof
- Cool, dry environment
- Formulations affect storage – fish meal and oil
- Nutrient losses – fatty acids, vitamins
- Contamination – rodents, fungal, bacterial

Proper Feed Storage

- Watch out if your are importing!! = delays
- Delays can turn feed into high-priced fertilizer or make it downright toxic!
- Feed typically shipped in 100 lb bags
- sea freight or over-land trucking
- **normal shipment**: 450 x 100 lb bags in one 40 ft container
- **if mill is nearby**: shipment is *a granel* or loose-pelleted
- *a granel* would imply that the farm has a silo and bagging system

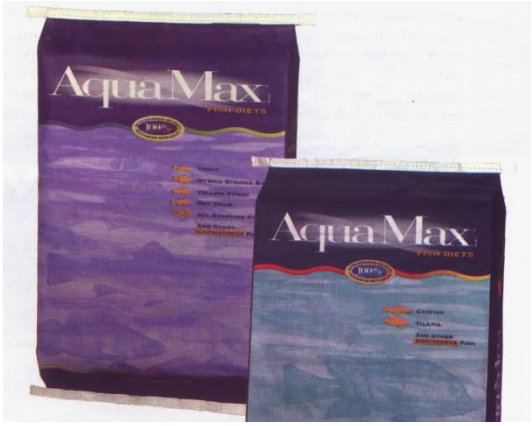




Proper Feed Storage

- Feed bags are made of many materials:
- paper on outside, plastic liner
- continuous plastic (no weave, no air holes)
- woven polymer
- typically contain labels stating feed type, pellet size, proximate analysis, ingredients, date of manufacture, etc.
- must be unloaded immediately and placed in proper storage





Proper Feed Storage

- Direct sunlight will adversely affect the vitamin and lipid quality of the feed
- do not store feed more than 3 months *post manufacture*
- feeds should be purchased, delivered, and utilized on a monthly basis (2-3 containers per month for large farms)
- spoiled, wet or old feeds cannot be used
- economic loss of feeding deficient feed may be greater than cost associated with discarding it







Proper Feed Storage

- Feeds should be stored in a dry, cool and well-ventilated area
- spoilage will occur immediately if feeds become wet: temperature needs to be consistent
- bags stored on wooden pallets, not on floor
- no more than 5 bags high between pallets
- allows for adequate air circulation between bags, constant or similar moisture, temperature



Proper Feed Storage

- Do not store bags directly on concrete floors or touching walls of building
- surfaces are often cooler than the bag: moisture migration
- feed moisture (around 8-12%) will migrate to the cool area, accumulate
- this encourages growth of molds (REM: *Aspergillus flavius*, aflatoxin??)
- also avoid direct sunlight: diurnal temperature flux



Storage problem

- Oxidative damage

- Oil rancidity, oxidation PUFA

- Microbial damage

- Fungi & bacterial growth

- Insect/rodent damage

- Ingestion & contamination (feces)

- Chemical damage

- Breakdown of fatty acids, reduced amino acids & vitamins (esp. vit C)

What are some common methods of feeding fish?

- Feeding fish is very important.
  - Provide the appropriate nutrition.
  - Opportunity to observe fish for any abnormal signs.
- Feeding systems differ based on size of the operation.
  - Small operations
  - Large operations

What are some common methods of feeding fish?

- Small Operations
  - ***Hand feeding***
    - Feed is distributed by hand, shovels, or other non-automated means.
    - Most common method.
    - Used for feeds not suited for automated systems.
    - Meat scraps are hand fed.

Advantages:

1-Operator can note feeding behavior $\Rightarrow$gauge the feed required

2-Operator can ensure that feed is dispersed over wide area.

Disadvantages:

1-high labor cost

2-increased handling of the feed.

What are some common methods of feeding fish?

- Large Operations
  - ***Automated feeding systems***
    - Utilize machinery to distribute the feed.
    - From relatively simple to very complicated, computer-controlled.
    - Blowers
    - Demand feeders
    - Auger feeding systems

Advantages:

1-Reducing labor cost

2-Known quantity of feed dispersed to fish

Disadvantages:

1-Less observation of the fish

What are some common methods of feeding fish?

- ***Blowers***
  - Drive along the side of the pond and blow feed out into the pond.
  - Attached to trucks or tractors.
  - Common in levee pond and watershed pond operations.



www.sweeneyfeeders.com/aquaculture.htm

What are some common methods of feeding fish?



www.missourifishfarms.com/.../automatic_feeders.htm

- ***Demand feeders***
  - Allow fish to bump a rod and release feed when they are hungry.
  - Used in raceways.
  - Fish must be trained how to use them.

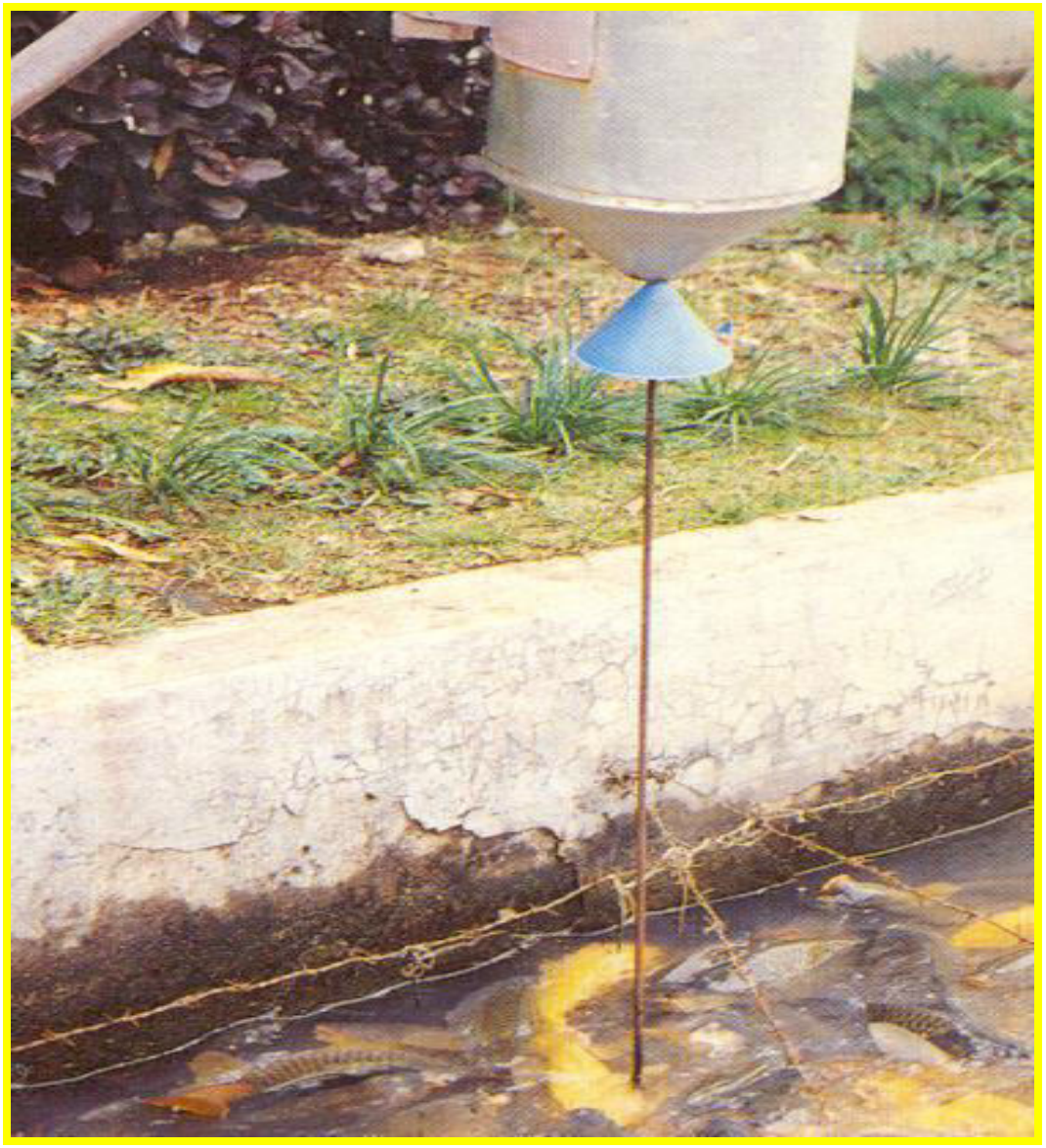
Fish can obtain food on demand by depressing a trigger.

Advantages:

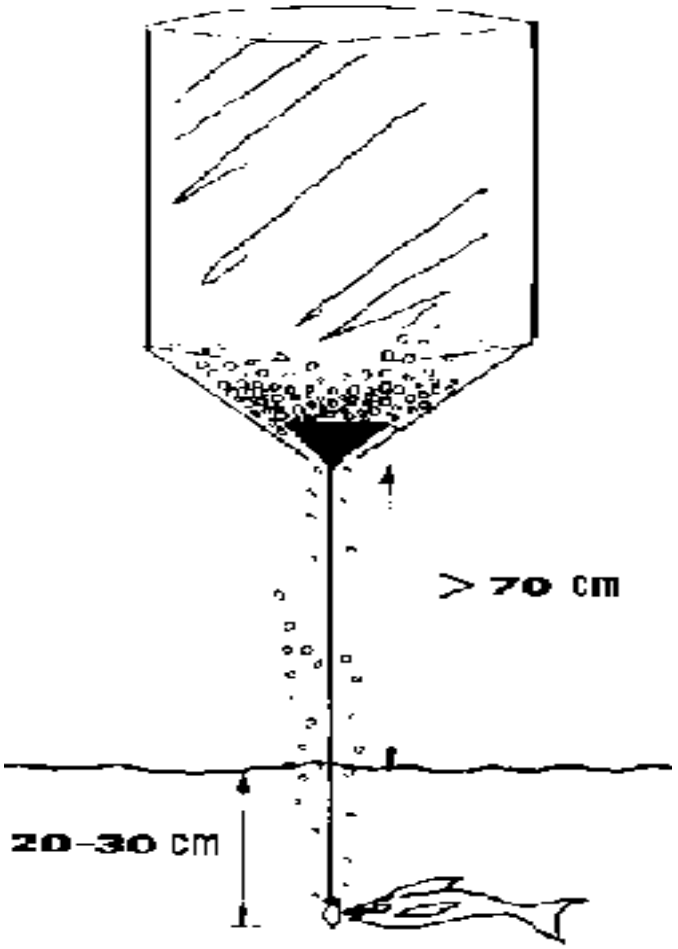
Fish can obtain much food as required

Disadvantages:

Trigger happy $\Rightarrow$ feed wastage & water pollution



- Demand feeders



What are some common methods of feeding fish?

- ***Auger feeding***
  - Similar to systems used by swine and poultry farmers.
  - Feed stored in large tanks
  - Moved by augers that drop feed in pond
  - Common in tanks and raceways



www.ars-grin.gov/ars/PacWest/Aberdeen/overturf.htm

Nitrogen excretion

- Animal = Urea
- Fowl = Uric acid
- Fish = Ammonia

Feeding Frequency

- Small = 4-8 times/day
- Big = 2-3 times/day

Fish = One specific place at a time

Shrimp = whole or half of the pond at a time

Fish Health and Disease

- Health or well being important for
- feeding
- Growth
- Reproduction
- Diseased fish results in poor growth and ends up in losses
  - Further when diseased fish is sent to consumer, it changes
  - Texture
  - Appearance
  - Taste

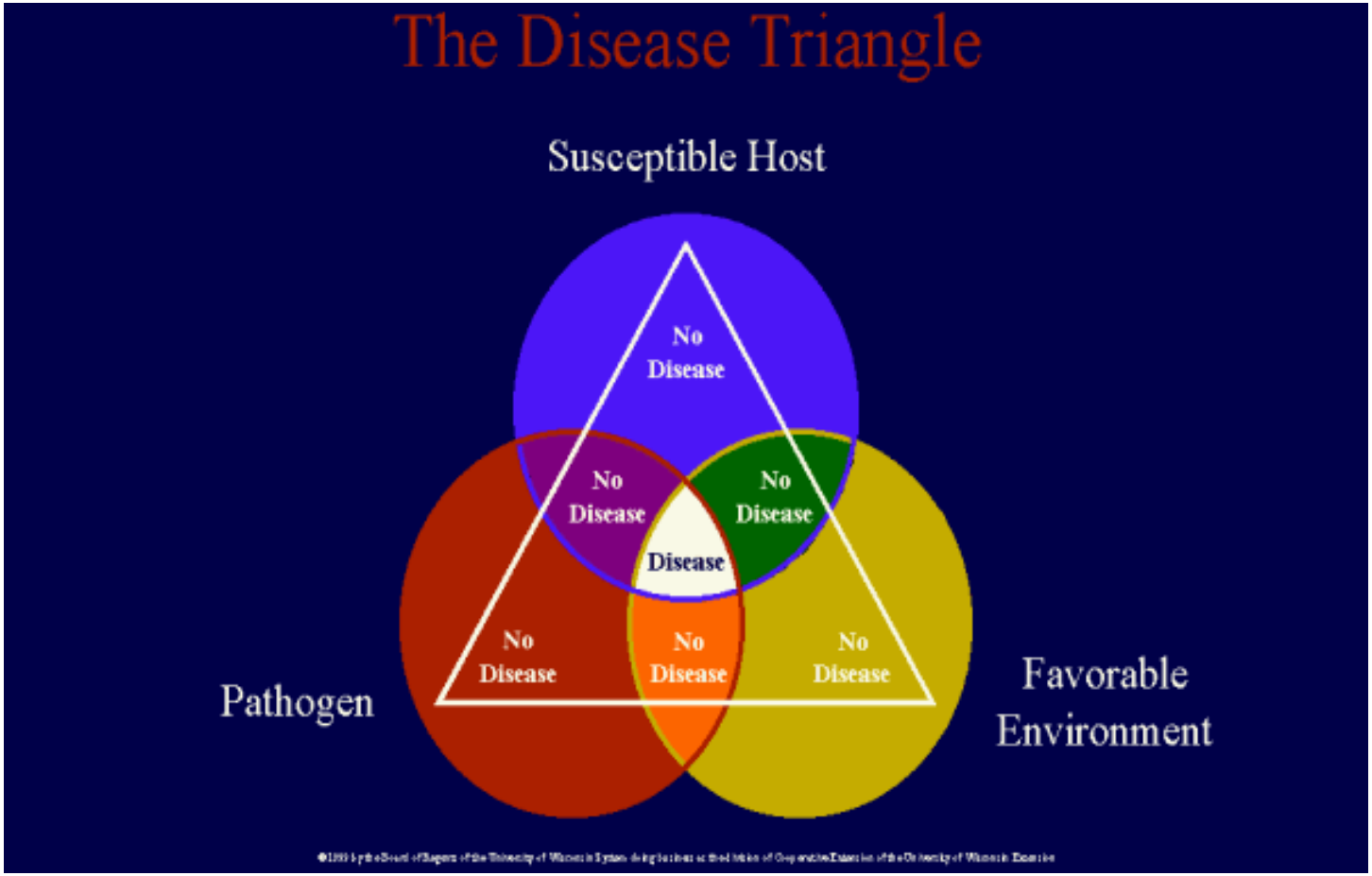
- Now what causes disease
- Pathogens

- Disease causing organisms
- If present in fish environment, fish is always at risk
  - pathogens attack fish,
  - It shows abnormal behavior,
  - Stop feeding and
  - growth
  - if conditions prevail fish succumb(surrender) to death

- Which system has maximum possibilities to accept disease

- extensive , semi-intensive and intensive
- Intensive system
- Reasons
  - Crowding
  - Competition
    - Food
    - Space

Requirements for Disease



- All 3 must be present for an outbreak

Introduction

- **We will focus on**
- **How** do you recognise that a fish might be ill?
- **What** are the causes of fish disease?
- **How** do you know that a fish has a parasite?
- **What** can you do to prevent a disease / parasite?
- **How** do you treat diseased fish?

Fish Diseases

How do you recognise that a fish might be ill?

- Colour may fade out / change
- Body shape, condition and / or behaviour will be abnormal
- The fish may refuse to feed or overfeed and trailing faeces appear at vent.
- Condition of the fins and gills will deteriorate.
- Fins may be clamped close to body.
- The fish may not keep its swimming position.
- There may be signs of injuries, outgrowths or abnormalities.

Fish Diseases

What are the causes of fish disease?

- Bad water quality
- Inappropriate diet
- Temperature (too high or too low)
- Stress
- Bullying(mistreatment, oppression)
- Viral diseases
- Fungal infections
- Bacterial infections
- Parasites

Parasitic conditions: What is a parasite?

- A parasite is an organism that lives off another often to the detriment of the hosts health.
- Parasites can be internal (endo-parasites) or External (ecto-parasites).

- They can be:
- Protozoan (single celled)
- Nematodes/ cestodes / trematodes (worms)
- Crustacean (e.g. louse)

Parasitic Fish Diseases

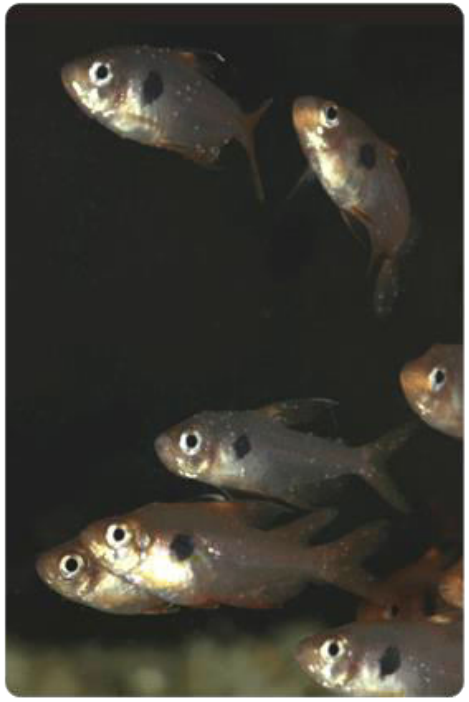
- Itch or White Spot Disease (*Ichthyophthirius)*

  Causes

  Protozoan parasite either free swimming in the water or carried in with new fish or plants.
  Fish under stress from bad water conditions are more susceptible.

Symptoms

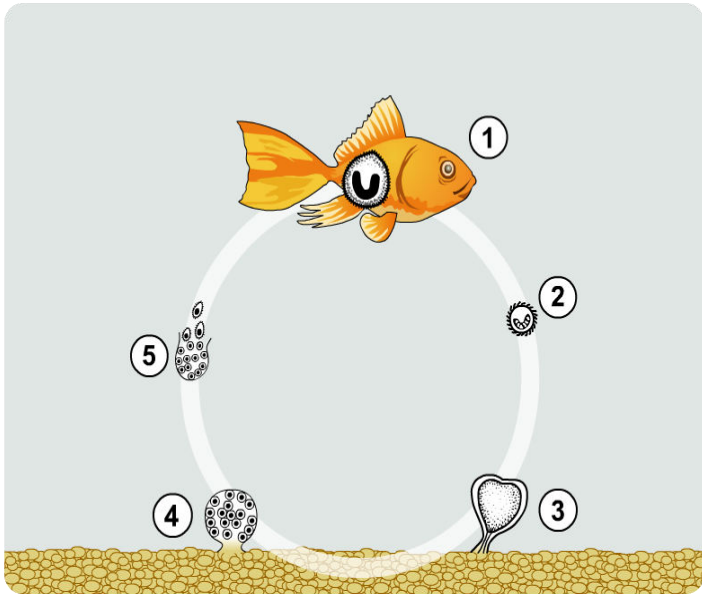
- The fish's skin and fins are covered in tiny white spots
- A badly affected fish may make rapid gill movements



Treatment

- Remove plants and the activated carbon from filters as they can affect / be affected by the medicine.
- Treat with a methylene blue based medicine which kills the free swimming larval stage (theronts)

White Spot Lifecycle



1. After finding their new host, they will eat into the fish's skin.
2. Adults fall from the fish and become free swimming till it settles on the gravel.
3. Once settled it forms a cyst which begins to reproduce by dividing itself up to as many as 2,000 times.
4. The result of this division produces what is called Tomites.
5. These emerge from the cyst when it bursts as free swimming Theronts ready to re-infect the fish.

N.B. If a Theronts is unable to find a new host within 24 – 48 hours they will usually die.

- Whirling disease *Myxobolus cerebralis*
  - Caused by protozoan



Skull deformation from *Myxobolus cerebralis*

TREMATODES
Gill flukes and skin flukes (*Dactylogyrus* & *Gyrodactylus*), monogeneans

Cause

- Trematode (flatworm / flukes) attack by direct contact with contaminated fish, eat gill/skin tissue and blood, free swimming larval stages can attach to the bottom and side of housing.
- **Symptoms**
- The gills may move rapidly and fish may gasp at the water surface
- The fish may scrape itself against objects
- Colours fade as damaged areas are covered in mucus.
- The skin may redden in places
- The fins may become ragged

Treatment

- These parasites can be treated with a formalin based medicine.

- Skin & gill fluke *Gyrodactylus salaris*
  - flatworm



Darkening of skin from anus to tail

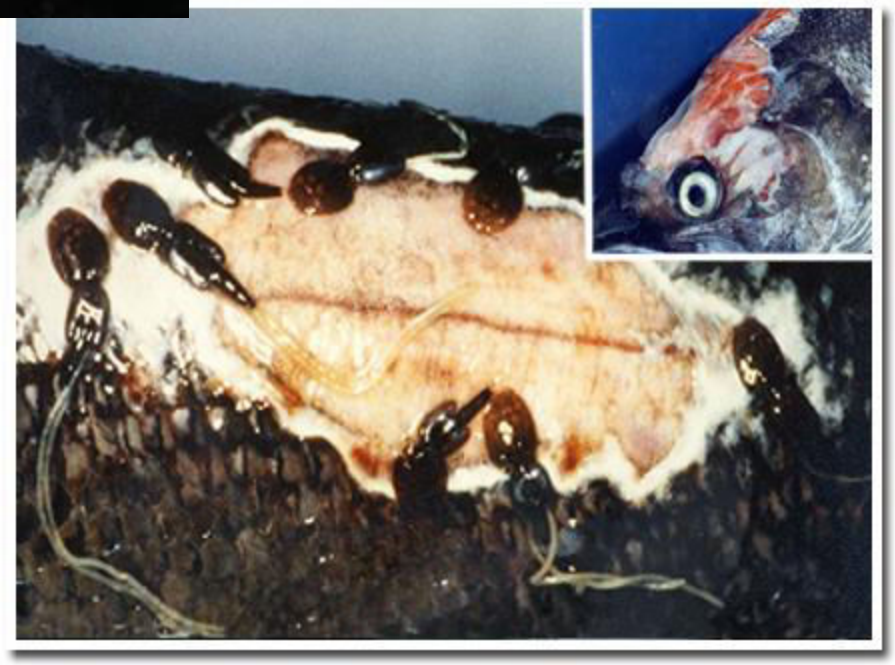


Atlantic salmon with Gyrodactylosis

Ecto-parasites: Crustaceans

- Sea lice







CRUSTACEAN
Anchor Worm *(Lernaea)*



Cause

- The crustacean parasite *Lernaea* it can grow up to 12mm. Usually brought in by non quarantined fish.

Symptoms

- Whitish-green threads hang out of the fish's skin, with an inflamed area or ulcer at the point of their attachment.

Treatment

- The water can be treated with insecticide. The adult parasite can be removed manually and the wound treated with antiseptic to prevent bacterial infection.

Fungal infections: What is a fungus?

- Fungus are multi-cellular, spore producing organisms that live off other organisms, and dead matter, some are parasitic.
- Fungal spores are commonly found in aquarium water.
- Healthy fish have a protective mucus covering which can prevent infection by fungal spores.

Fish fungus

Cause

- Aquatic fungi e.g. *Saprolegnia*.
- Fish that are in poor health and have damaged mucus membranes through bad water quality, rough handling, fighting or physical injury are more prone to infection. Fungus can be a secondary infection to other conditions.

Symptoms

- Grey white or brown cotton wool like growths on the skin or fins.

Treatment

- Medicines containing malachite green can be used and salt baths help recovery.



Bacterial Infections : What are Bacteria?

- Bacteria are microscopic single celled organisms that can reproduce rapidly.
- They are naturally present in aquarium water.
- Fish in good health kept in good water conditions can fight bacterial infections.
- Fish are most prone to such infections if in poor condition as a result of bad or sudden changes in water quality, over crowding or bullying, bad handling or transportation.
- A poor diet lacking in sufficient protein, fatty acids and vitamins can reduce fish resistance to such disease.

Pseudomonas fluorescens

- Causes
- pseudomonas Septicemia (blood poisoning due to bacteria) mainly in general pond fish, seldom in salmonids
- normally a secondary invader, hard to distinguish from Aeromonas Septicemia, not a huge concern in fish
- Agent: ubiquitous bacterium of soil, water,
- culture: standard media, round glistening colonies w/undulating edge, radial striations, easily seen green pigment under UV light

Pseudomonas fluorescens

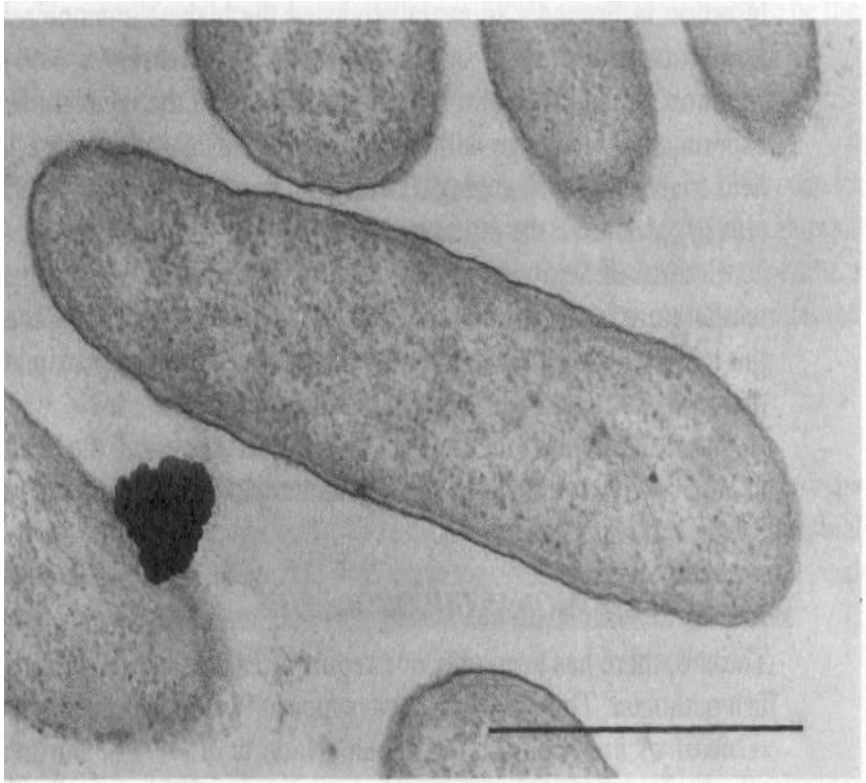


Fig. 12.5 — Transmission electron micrograph of a thin section of *Pseudomonas anguilliseptica*, lacking any signs of an extracellular layer. Bar equals 1 μm. (Photograph courtesy of Dr G. Dear.)

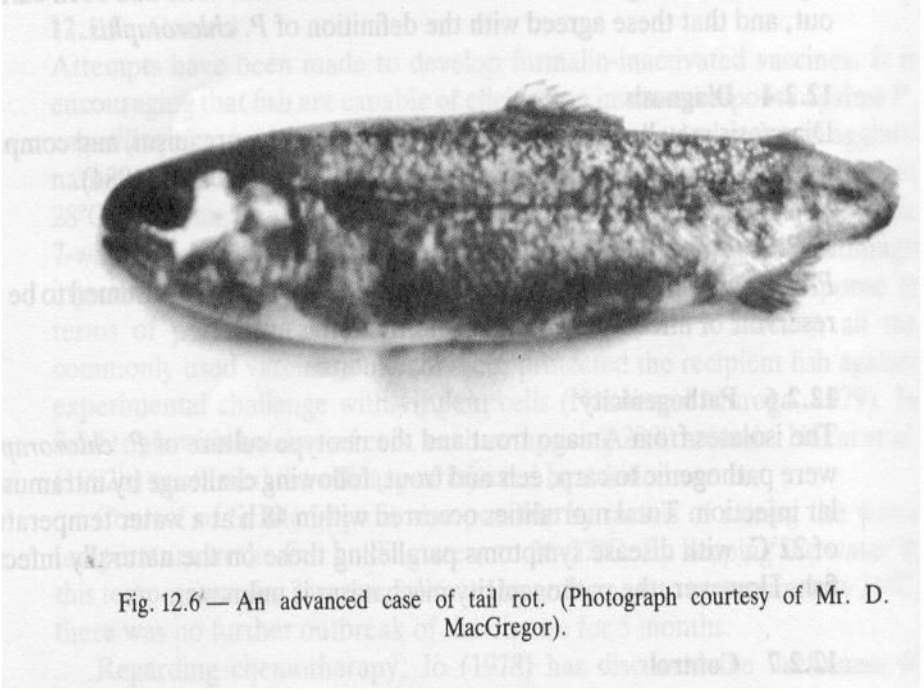


Fig. 12.6 — An advanced case of tail rot. (Photograph courtesy of Mr. D. MacGregor).



Pseudomonas fluorescens

- Epizootiology: worldwide in fw/sw, all fish susceptible but mainly ww, problem for aquarium fish
- reservoirs: mud and water; infected or carrier fish and others (frogs)
- transmission: horizontal, no vertical
- environment: stress, mainly elevated temps
- Pathology: in catfish, largely hemorrhaging and necrosis of internal organs, external lesions, loss of pigmentation
- individuals can withstand large losses, bone exposure

Pseudomonas fluorescens

- Diagnosis: isolation from kidney on TSA Tryptic soy agar
- confirmation via serology
- Control: remove stressor, drug therapy as with other G- (oxytet @50-75 mg/kg/f/day for 10 days)
- no vaccine yet

Fin Rot / Mouth Fungus



Cause

- Bacteria such as *Aeromonas, Pseudomonas (fin rot) and Flavobacterium (mouth fungus)*

Symptoms

- Damaged, split or ragged looking fins (fin rot)
- Cotton wool like tufts around the mouth (mouth fungus)
- May cause loss of appetite and listlessness (lethargy)
- When chronic may develop ulcers on the body.

Treatment

- Aquarium anti-bacterial medicines are available and in serious cases veterinary treatment is needed.
- It is easier to prevent bacterial infections than cure them.

Aeromonas hydrophila (MAS)

- Causes
- Motile Aeromonas Septicemia(blood poisoning), often referred to as a complex of species, mainly affecting warm water fish, opportunistic pathogen, can cause red-leg in frogs
- culture: TSA; can grow at 4°C, but best at 18-25°C;
- white, circular, convex colonies, often confused w/*Citrobacter*

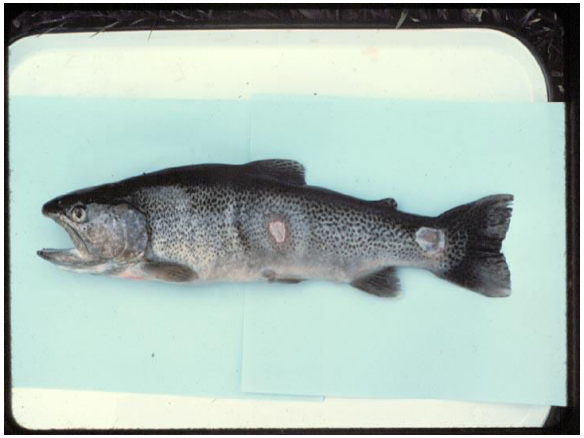
Aeromonas hydrophila (MAS)

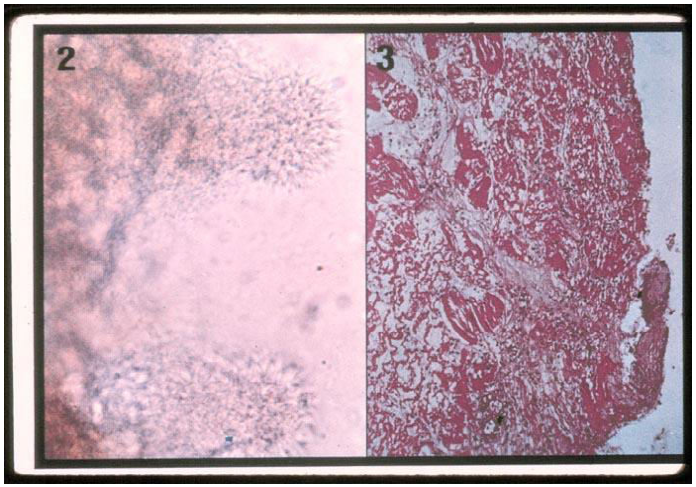
- Epizootiology: worldwide in fw, all fw species susceptible (both ww and cw); others such as frogs, alligators, snails, shrimp and humans
- reservoir: freshwaters w/high organic loads, usually in sewage, normal gut flora of healthy fish; diseased fish/frogs; survivors are carriers
- transmission: horizontal only from intestinal tract, external lesions, through water, via external parasites
- environment: stress from crowding, variable temperatures, changes in weather; rough handling, low DO, high organics

Aeromonas hydrophila (MAS)

- External pathology: usually hemorraghia + necrosis or internal organs + necrotic lesions on skin/muscles
- superficial circular or greyish-red ulcerations
- lesions around mouth similar to ERD(esophageal reflux disease)
- hemorrhaging of fins, exopthalmia
- Internal pathology: swollen, soft kidney;
- petechiae (little red spots) of musculature, intestines free of food

Aeromonas hydrophila







Aeromonas hydrophila (MAS)

- Diagnosis: isolation from kidney into TSA Tryptone or tryptic soy agar
- presumptive: G- motile rod
- ferments in glucose, no fluorescent pigment
- Control: prevention via good management, injection w/chloramphenicol, no vaccines;
- therapy via oxytet (50-75mg/kg fish/day for 10 days), chloramphenicol in Europe

Aeromonas salmonicida (furunculosis)

- First isolated from farmed trout in 1894
- name of disease derived from boil-like lesions known as “furuncles”
- at one time very common, resulted in Furunculosis
- diminished as of late due to better management
- Agent: comes in three subspecies, the most common is salmonicida (produced pigment),
- G- non-motile rod, bipolar staining

Aeromonas salmonicida (furunculosis)

- Culture: TSA, brown pigment in presence of TYR/PHE, grows well at 18-25°C, small white round raised convex colonies
- Pathogenicity: both virulent and a-virulent strains, produces endotoxin
- Epi-zootiology: wherever salmonids are cultured in fw (besides Tasmania/NZ), entered Australia via goldfish, brook trout most susceptible
- reservoirs: obligate fish pathogen, found in waters w/infected or carrier fish

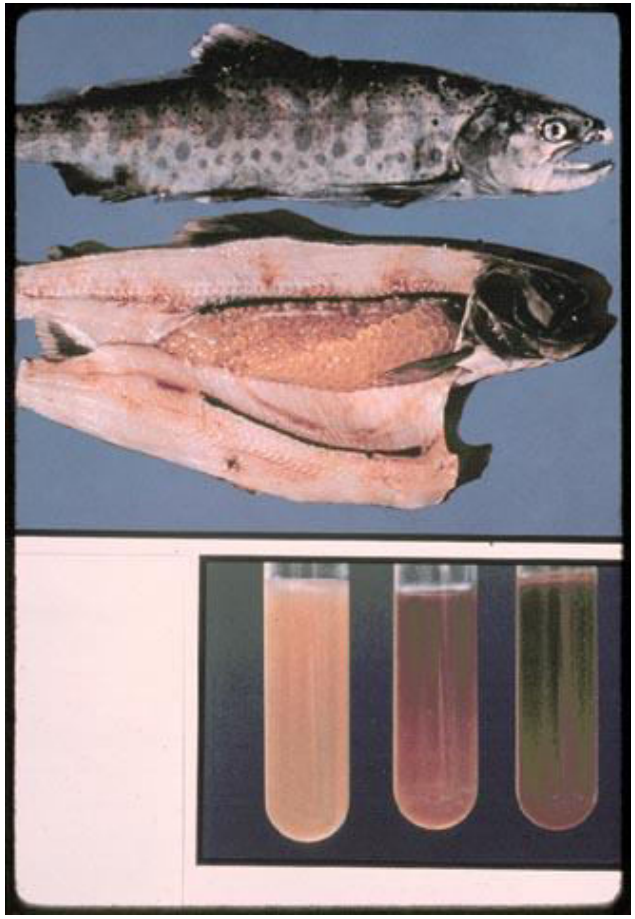
Aeromonas salmonicida (furunculosis)

- transmission: primarily horizontal, contaminated water, eggs, carriers, equipment, clothing, surface of aquatic birds; no vertical demonstrated
- pathogenesis: acute, sub-acute, chronic forms (dose, temp, host resistance, virulence of strain)
- environment: severity increases w/temp, nutrition, handling stress

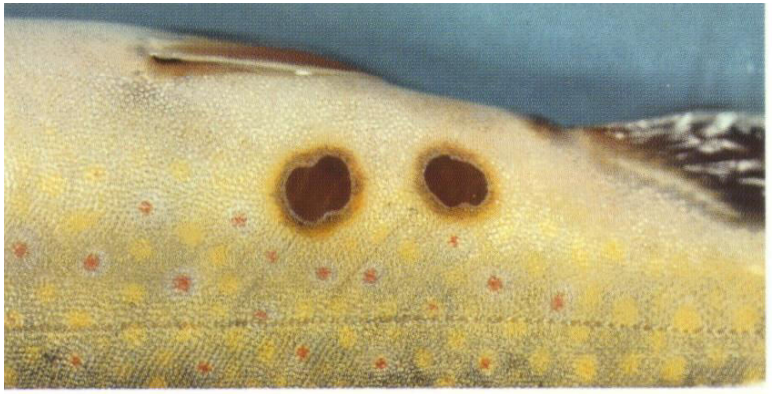
Aeromonas salmonicida (furunculosis)

- Pathology: similar to other G- septicemias w/hemorrhaging, necrosis of internal organs, external lesions
- external: focal necrosis in muscle develops to abscess, hemorhaging and lesions at base of fins, fraying of fins, bloody discharge from vent; blue irridescent sheen on body near eyes, bleeding from gills
- internal: petechiae(purple or red spot due to bleeding) in body musculature, congestion (cramming)of posterior intestine, no inflammatory response

Furunculosis







Aeromonas salmonicida (furunculosis)

- Diagnosis: look at hatchery history with disease; isolation easy from kidney into TSA, BHI(Brain heart infusion for fastidious bacteria);
- presumptive: G-, non-motile rod, brown diffuseable pigment, oxidase +
- definitive: serological (rapid slide agglutination)
- Control: avoidance via clean water/fish; several vaccines on the market; selective breeding??
- therapy: oxytet @ 50-75mg/kg fish/day for 10 days, sulfamerazine, sulfonamide (Romet)

Ulcer Disease & Haemorrhagic Septicaemia

Cause

- A number of different bacteria including *Aeromonas* and *Pseudomonas*. These could be transmitted from other infected fish, and /or bad water conditions.



Symptoms

- Open sores and ulcers, reddening of fins and vent, may lose their appetite and colour may change.

Treatment

- Fish can be fed antibiotic medicine in feed. If severe fish should be isolated and antiseptic applied to infected areas. May require veterinary injection of antibiotics.
- Ensure tank conditions are correct.

Viral infections: What is a virus?

- A virus is a microscopic organism that can only reproduce by inhabiting host cells and using the genetic material in the cells of a host.

- Healthy fish that have a balanced diet and good water conditions have strong immune systems to fight off such infections.

Cyprinid Herpes Virus

Causes

- A herpes virus

Symptoms

- Causes growths that are white or grey in colour and look like melted candle wax.

Treatment

- Fish with a strong immune system can fight off the infection but retain the virus within the body.
- When in poor health the virus symptoms can re – appear.

Spring Viremia of Carp

Cause

- A viral infection caused by *Rhabdovirus carpio*.

Symptoms

- Darkening of skin, pale gills, pop eye, protruding vent, bleeding in gills skin and eyes.
- Lethargy, abnormal swimming positions, sitting on bottom of the tank.

Treatment

- No known treatment .

What is a Swim Bladder problem?

Cause

- This can be caused by a number of things, from internal swellings, tumours, viral and bacterial infections, internal deformities, constipation, parasites etc, to overfeeding.

Symptoms

- Bobbing/jogging to the surface, swimming upside down or listing to one side with abnormal swimming patterns.

Treatment

- Unless the cause can be identified this is difficult to treat. Starvation of fish for a few days might correct the problem if it is as a result of overfeeding.

What can cause growths and abnormalities?

Abnormalities can be brought about by :

- Inbreeding and congenital deformities
- Tumours and swellings
- Viral growths
- Malnutrition or inappropriate diet
- Internal disorders / parasites that cause fluid retention & swelling.

Fish Injuries

- Injured fish have often been bullied by tank mates.
- Injuries can be the site of secondary infections, bacterial and fungal.
- Stress caused by bullying, injuries and infections can lower their immune system further which can in some cases be fatal.

Prevention of diseases and parasites

- Provide an appropriate well filtered tank with suitable water conditions for the species, i.e. correct
- Appropriate ranges of temperature, pH, water hardness, low nitrate levels etc.
- Provide appropriate diet to meet species needs.
- House only suitable species together, make sure they are compatible and not likely to bully or eat each other.
- Only select healthy looking fish to add to the tank and do not overstock.
- Quarantine new fish to ensure they are healthy before introducing them to an established tank.
- Sterilize décor and clean new plants to ensure they are not carrying parasite eggs / larvae.

Prevention, Prevention, Prevention!

- The ultimate way to stop an outbreak is to prevent it
- Prevent stressful situations
  - Proper stocking situations
  - Proper management practices
  - Ideal water treatment



Prevention measures

- Vaccinations
- Stress-treatments (chemical)
- Anti-biotics
- Selective breeding (unintentional & intentional)
  - Disease-free brood stocks
  - Batch culture/ single batches reared to size
  - Fungal control of eggs
- Intensive systems
  - Control of multiple environmental factors



Silk and Rearing of Silkworm
Bombyx mori
or
Sericulture

Overview

- History
- Silkworm: *Bombyx mori*
- Mulberry tree
- Cocoon
- Silk
- Effect of nutrition
- Problems
- Conclusion



History of Silk

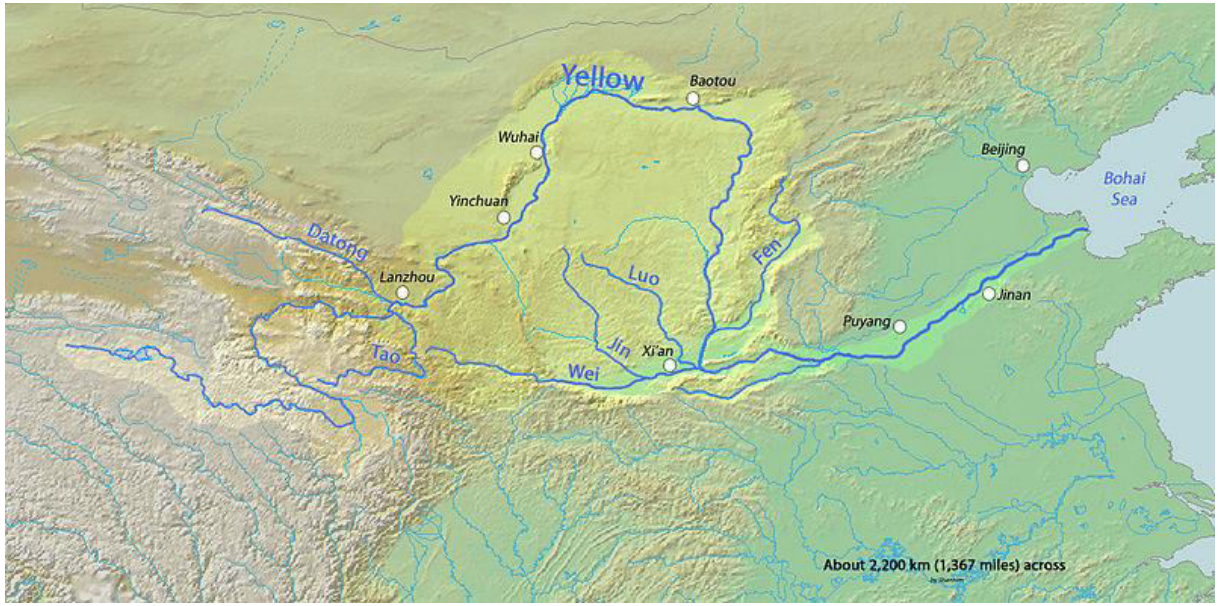
- Begins over 5000 yrs ago
- Debates over the origins of sericulture



http://www.silk-road.com/artl/silkhistory.shtml

History of silk- Origins (China)

- “Chronicles of Chou-King”
- 2200 BC
- Yellow River (Hwang He/Hwang Ho)



http://en.wikipedia.org/wiki/Yellow_River

History of silk- Legends

- Multiple legends
- Empress Ksi-Ling-Shih (2500BC)





History of silk

- China kept the secret of sericulture for 3000 yrs.
- Chinese traded silk for gold
- Smuggling silkworm eggs or mulberry seeds was punished by death

History- From China to Japan

- Chinese emigrants smuggled silk cultivation into Korea
- From Korea, it was taken to Japan



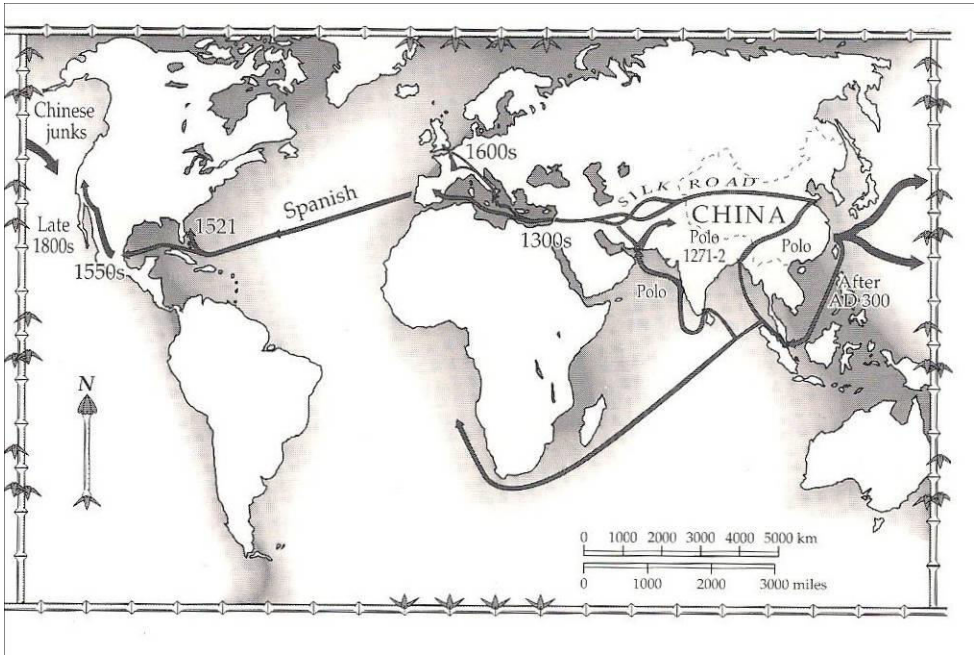
History- India

- Silk culture dates to antiquity
- Mulberry culture spread in India by 140BC from China



History- Silk Road

- Contact between East and West
- Silk, silver, porcelain etc.
- China to Europe
- 6400km.
- Cultural, religious and economic importance



(Feltwell, 1990)

History- Han Dynasty

- 202BC-220AD
- Silk trade prospered
- By 126BC silk was the most important traded material



History- Europe

- In mid 6th cent.
- Emperor Justinian sends 2 Persian monks to steel the secret of sericulture
- The supplies they smuggled supplied the western world for 1200yrs.



History- Europe

- The 1st king of Sicily invaded Greece in 747AD
- Captured silk weavers
- Sericulture began in Italy



History- Europe

- From Sicily to France
- Tours and Lyon
- Henri IV introduced rearing





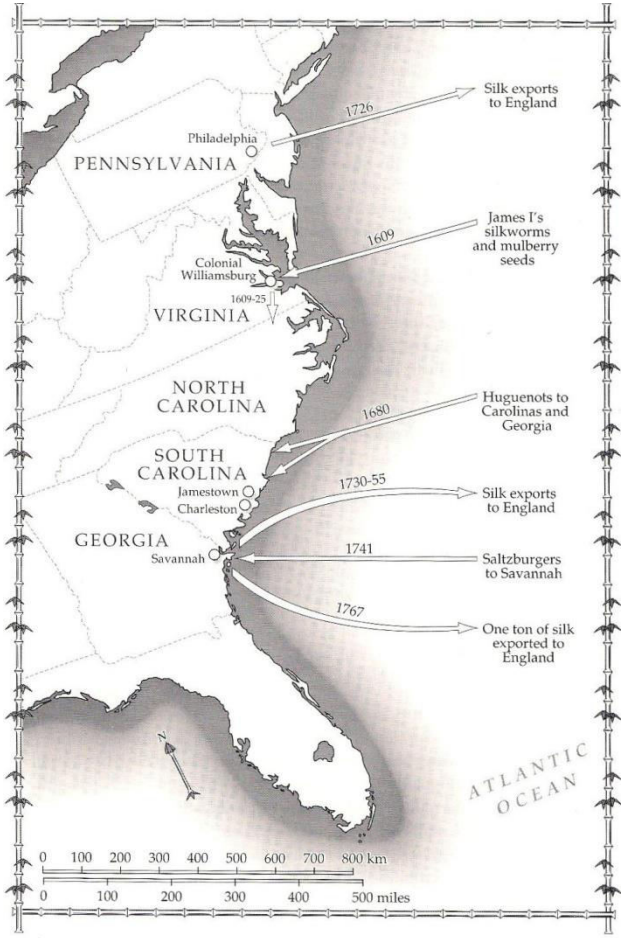
http://www.map-of-europe.us/

History- Europe

- James I introduced sericulture to England
- He also took sericulture to America



(Feltwell, 1990)

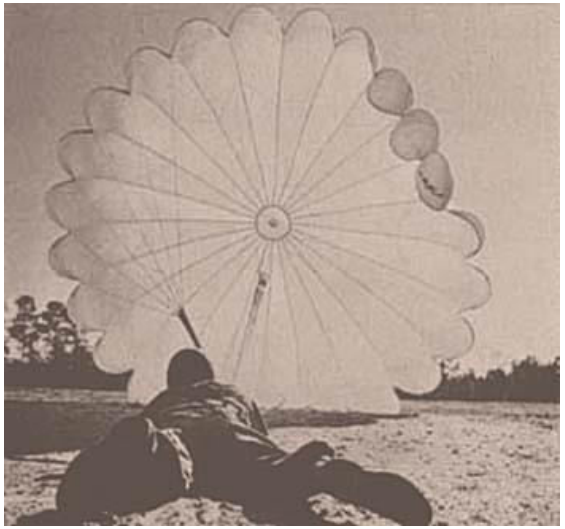


(Feltwell, 1990)

History- 20th century

- Sericulture stopped because of war
- In 1940 it boomed with the use of silk for parachutes
- Silk re-emerged
- China, India and Japan brought it to global market





Silkmoths

Major Silkmoths of the World

Antheraea assamensis	Muga silkmoth
Antheraea mylitta	Tusseh silkmoth
Antheraea peryni	Chinese oak silkmoth
Antheraea polyphemus	Polyphemus silkmoth
Antheraea yamamai	Japanese oak silkmoth
Attacus atlas	Atlas silkmoth
Automeris io	Bullseye silkmoth, Io silkmoth
Bombyx mori	Bombyx silkmoth
Callosamia promethea	Spicebush silkmoth
Hyalophora cecropia	Robin or cecropia silkmoth
Hyalophora gloveri	Glover's silkmoth
Pachypasa otus	Syrian silkmoth
Samia (Eri) cynthia	Eri or Ailanthus silkmoth

Bombyx mori

- Produces over 90% of the world's raw silk
- Common model for research and study
- Genome decoded in 2004
- 432Mb
- 3000 genes with no other homologs
- Important in evolution studies and in combating pests.
- Silkworm classified according to origin and voltinism

Voltinism

- Univoltine
- Bivoltine
- polivoltine

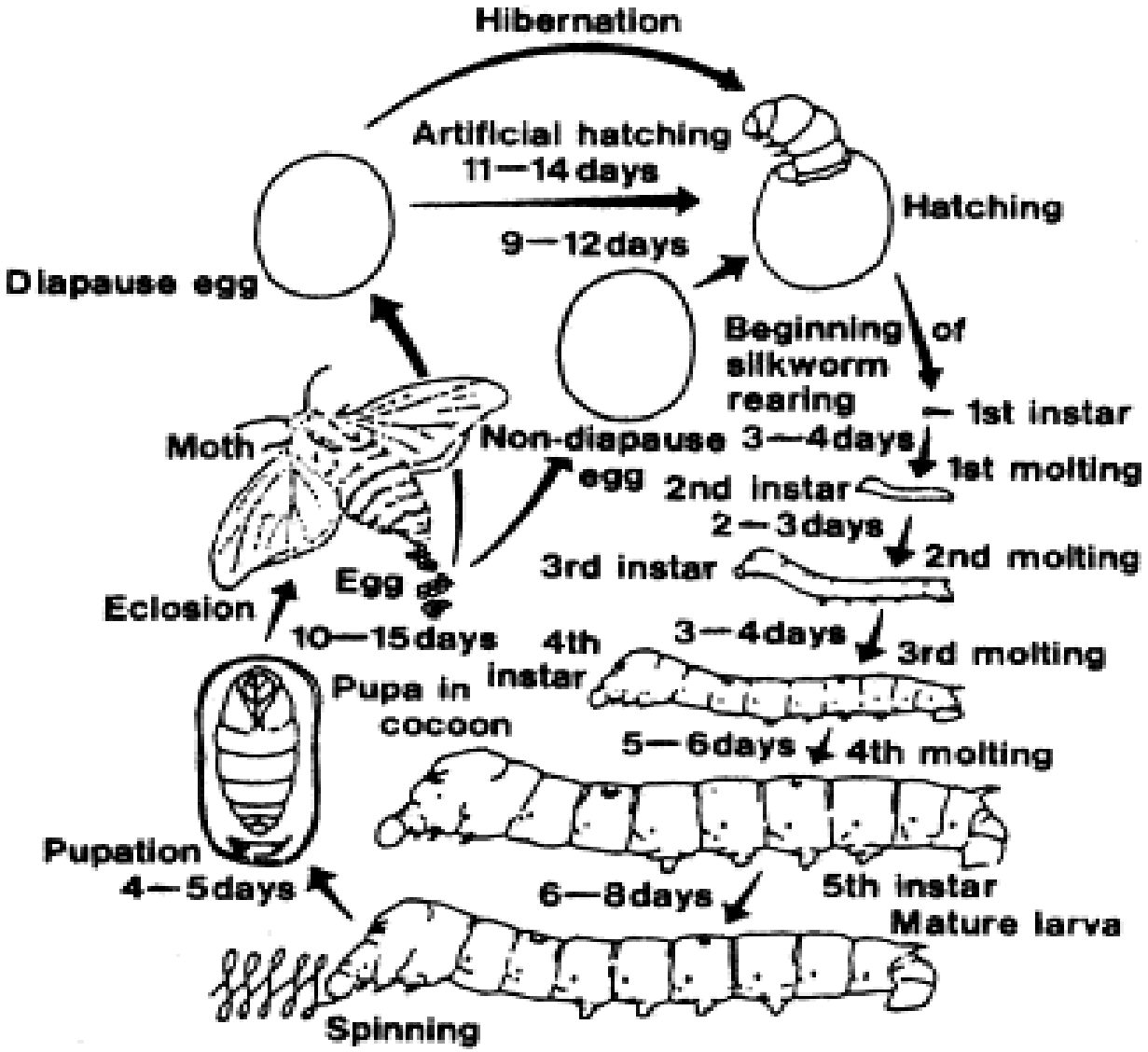
Refer to the number of generations that is produced per year

Characteristics

- It cannot fly
- Caterpillars with no real locomotion
- Female larger than male
- Male has big plumose antennae



Life cycle



(Zhao and Asakura, 2001)

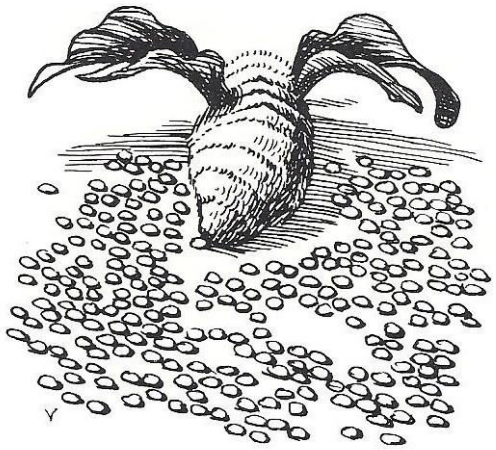
Life cycle

- After mating, the female lays 300-400 eggs
- 1g of eggs = 2000 eggs
- Eggs enclosed in gluten
- Yellowish then darker

- Each egg has a micropyle







(Feltwell, 1990)

Polyvoltine vs. Uni/bivoltine

Polyvoltine

- Hatch after 10 days
- Life cycle repeats itself 6 to 7 times a year

Uni/bivoltine

- Eggs go through diapause
- Hatch in time for spring
- Temperature is the main trigger

Life cycle

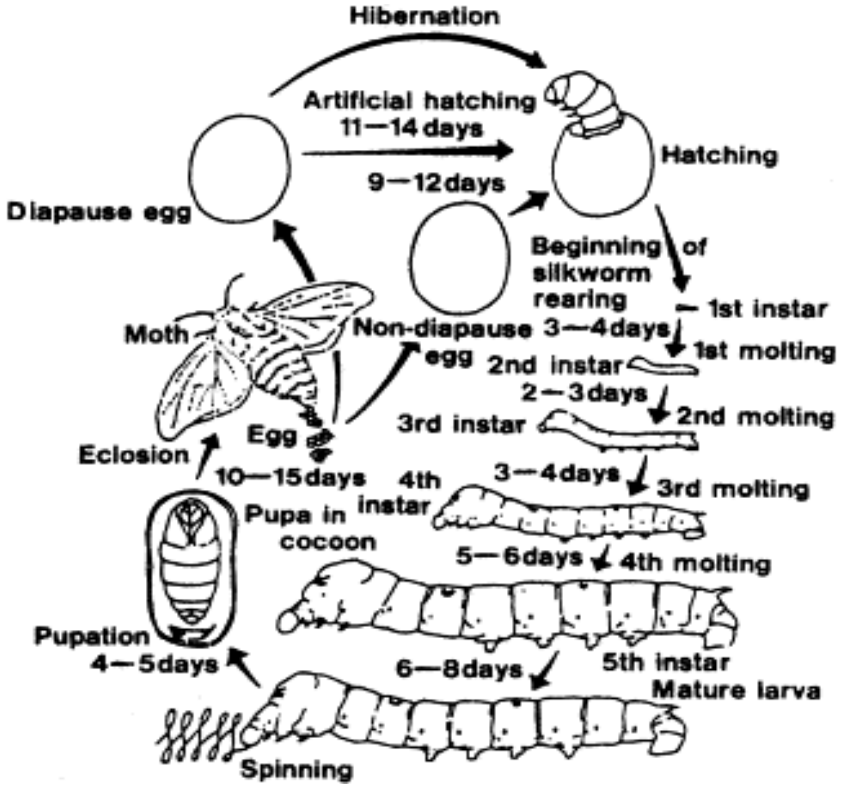
- When first hatched they vary from 1/8 to 3/8 of an inch in length
- Diameter of a human hair
- 1 ounce= 50-60000 worms
- Go from black to brownish yellow to white





Life cycle

- Polyvoltine species go through the 5 instars in 20-24 days
- Uni/bivoltine go through them in 24-28 days
- Total of 4 molts



(Zhao and Asakura, 2001)

Mulberry tree

- Fast growing
- Tropical and temperate habibtat
- *Morus spp.*
- *Moracea* famillly
- 95 species in total
- *M. alba, M. nigra, M. indica, M. rubra etc.*



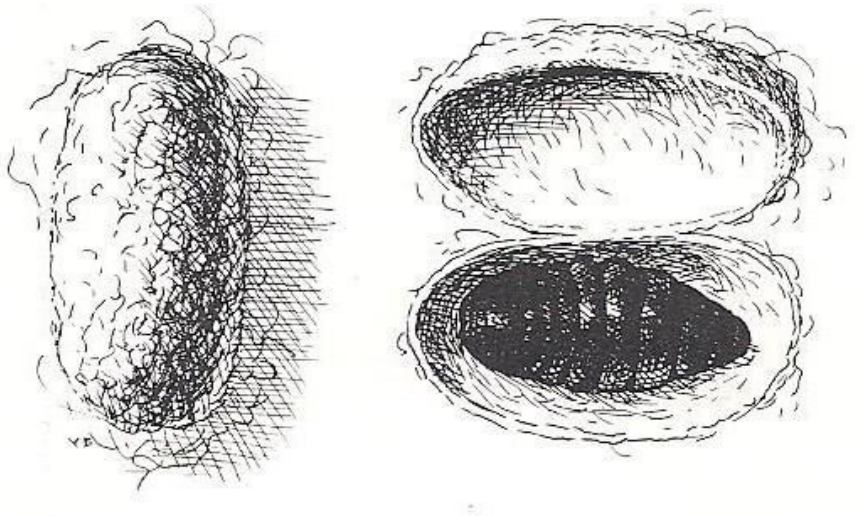


Life cycle

- During the first 3 instars, the larvae feed on chopped leaves.
- When they grow, they become able to chew on whole leaves
- The silkworm goes from
  - 3.0mm to 95mm in length
  - 0.00045gm to 5gm

5th instar

- Silkworm attains max size
- JH decreases
- Ecdysone triggers metamorphosis
- Color change
- Cocoon making
- Pupation occurs



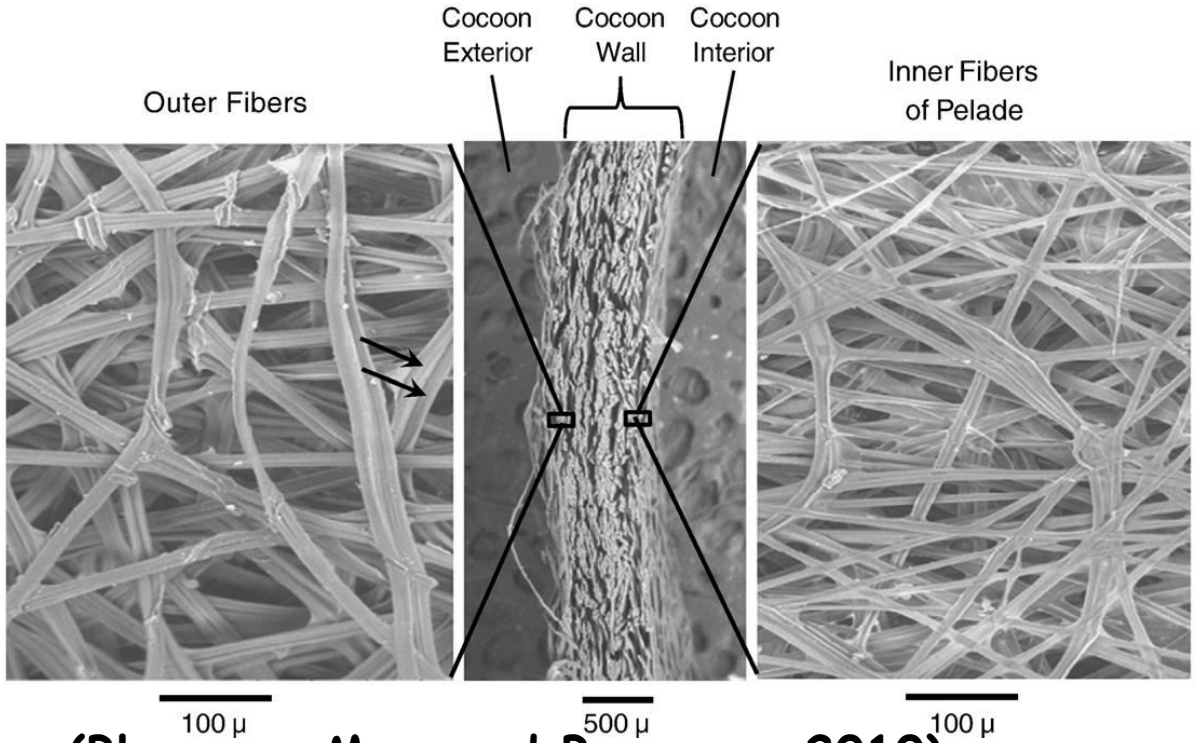
(Feltwell, 1990)

Cocoon

- 5 day process
- Silk laid with figure-8 movement
- Why spin a cocoon?
  - Protection
  - Silk as a way of excretion

Cocoon

- Heterogenous:
  - Inner silk filaments are tightly packed
  - Inner silk filaments have a smaller diameter



(Blossman-Myer and Burggren, 2010)

cocoon

- Shape variable
  - Breed
  - disease



http://www.wildfibres.co.uk/html/silk.html

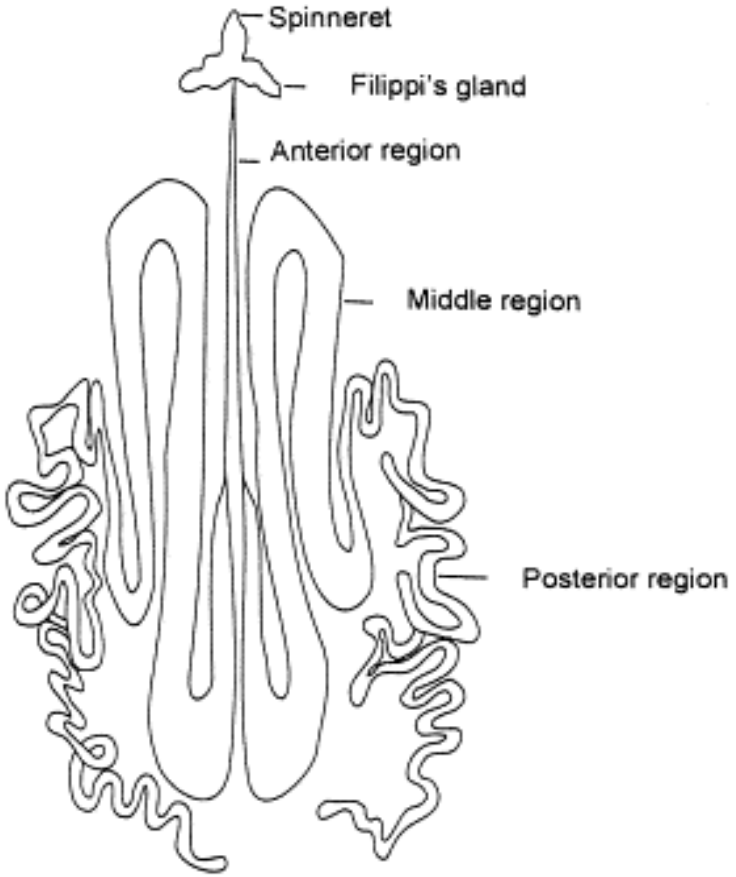
Life cycle

- Metamorphosis occurs
- The moth emerges
- Cocoonase



Silk gland

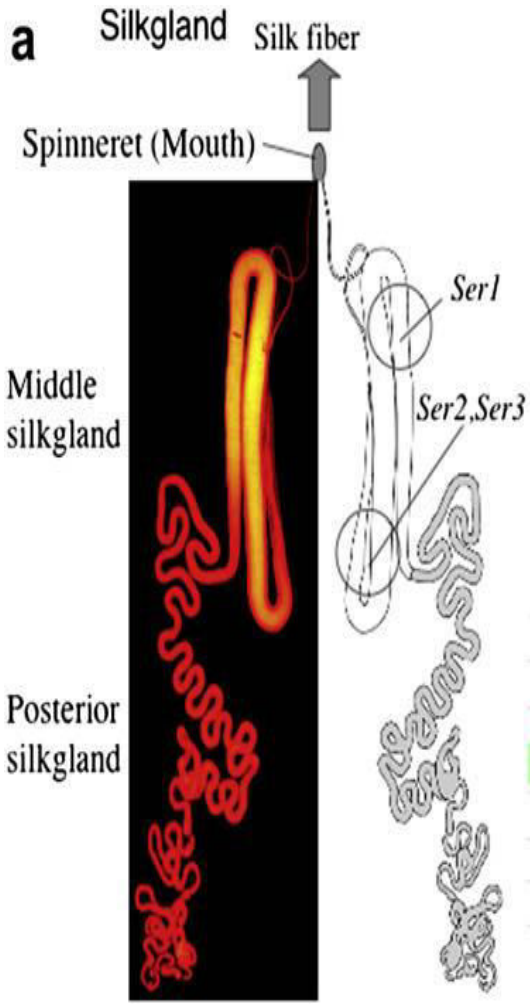
- Runs all along the body
- At 5th instar it becomes the largest tissue
- Ectodermal
- One cell layer
- 2 types of secretory cell
  - Fibroin secretory cells
  - Sericin secretory cells
- 3 parts: ASG, MSG and PSG

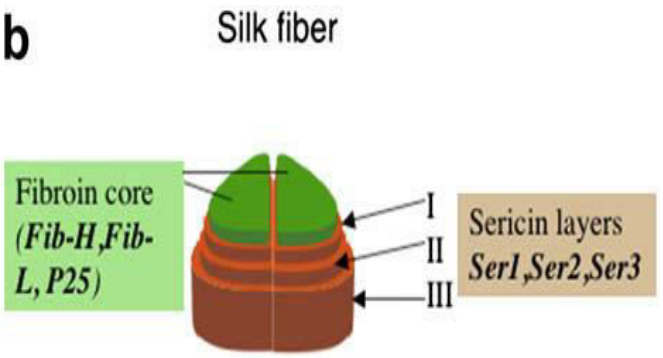


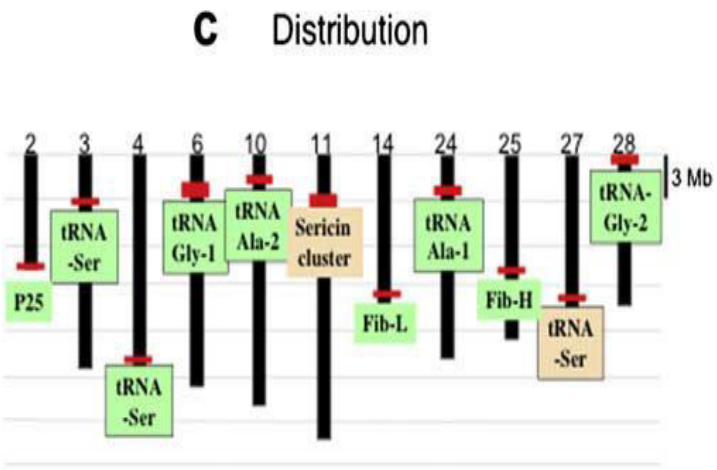
(Zhao and Asakura, 2001)

Silk filament

- Sericin 18-20%
- Fibroin 75-80%
- Carbohydrates 1.2-1.6%
- Lipids and wax 0.5-1%
- Colors and minerals 1.0-1.4$







(The International Silkworm Genome Consortium, 2008)

Effect of nutrition

- Composition
- Quantity
- Mulberry leaves satisfy the min. need of the larvae
- Amount of vitamins found in them varies
- Focus is on giving young and healthy leaves

Artificial diet

- Since processing of leaves is costly
- Introduced in 1960s
- Mulberry leaf powder
- By 1970, they had made a dry powder with no mulberry in it

Problems

- Diseases and pests
- 20 mulberry diseases were isolated
- Fungal, bacterial or viral

Leaf spot



http://msucares.com/newsletters/pests/infobytes/19980324.htm

mildew



http://en.wikipedia.org/wiki/Mildew

Root- Knot



http://aggie-horticulture.tamu.edu/publications/cucurbitproblemsolver/root/root-knot.html

Problems

- Pests

Adult of Bihar hairy caterpillar



Wingless grasshopper



Leaf roller



Problems

- Silkworm diseases: pebrine, flacherie, muscardine

Pebrine, *Nosema bombycis*

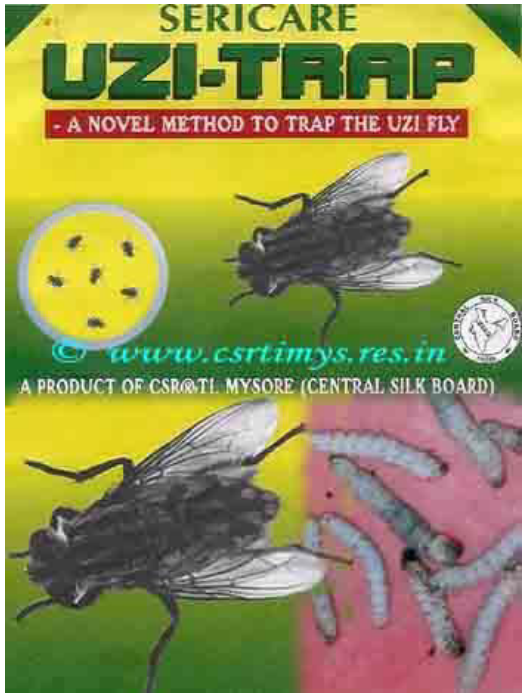


grasserie



Problems

- Pests: ants, wasps, crickets, praying mantis, Uzi fly





Nesolynx thymus



LAC CULTURE



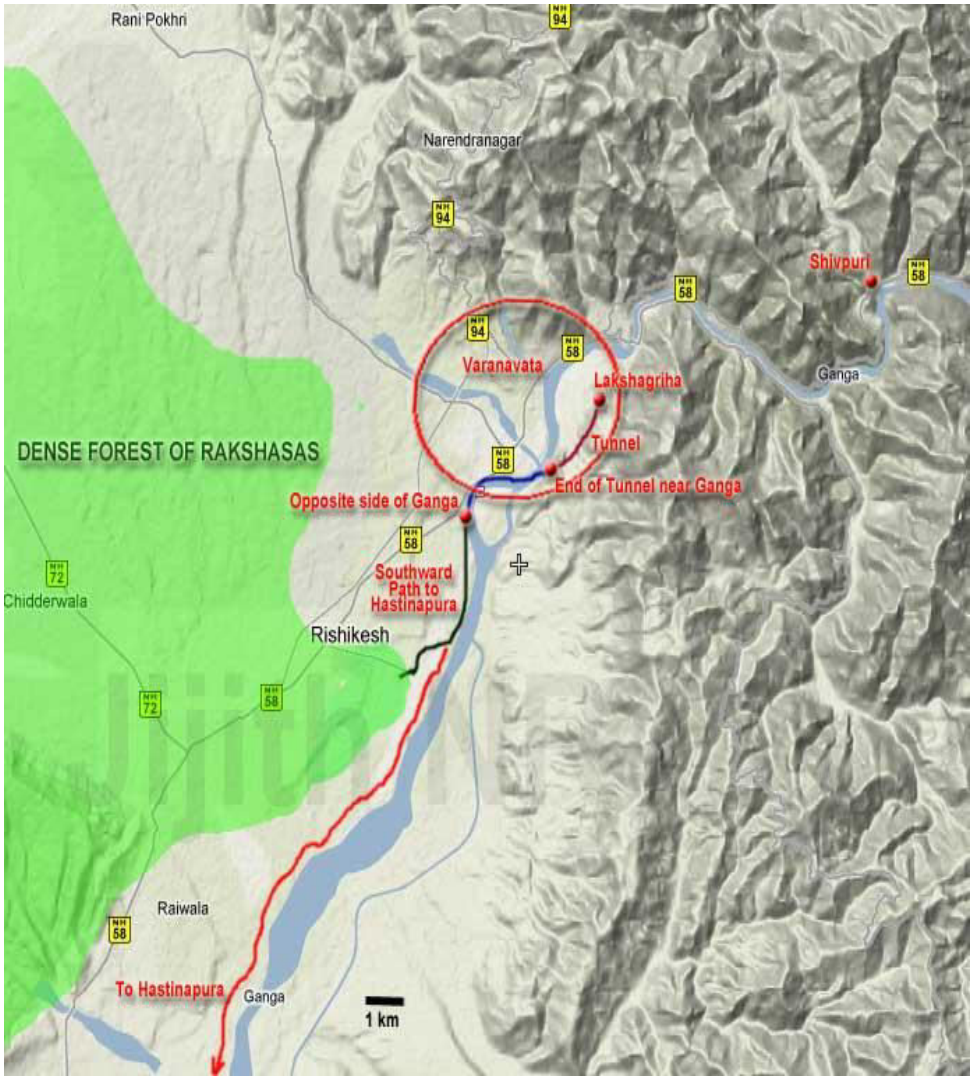
Lac Culture

- **Nature has given much for welfare of human beings through animals and their products.**
- **On the other hand human beings never seem to tire of discovering the mysteries of nature. But the animals seem to be greater experimenters as some of them have astounded most human beings by their complex, strange and at times bizarre performance.**
- **One of such performer known to man from good old days is the tiny insect that has given a very valuable product in the form of lac, to the civilization of man.**
- **Lac is a natural resin of animal origin. It is secreted by an insect, known as lac-insect in order to obtain lac, these insects are cultured and the technique is called lac-culture. It involves proper care of host plants, regular pruning of host plants, propagation, collection and processing of lac.**

Lac Culture

- Lac is a resinous exudation from the body of female scale insect. Since Vedic period, it has been in use in India.
- Its earliest reference is found in Atherva Veda. There, the insect is termed as ‘Laksha’, and its habit and behavior are described.
- The great Indian epic‘Mahabharata’ also mentions a ‘Laksha Griha’, an inflammable house of lac, cunningly constructed by ‘Kauravas’ through their architect ‘Purocha’ for the purpose of burning their great enemy ‘Pandavas’ alive.

‘LAKSHA GRIHA’, AN INFLAMMABLE HOUSE OF LAC,



- **History:**
- Lac has been used in India from time immemorial for several purposes, from the epic of Mahabharat it has been recorded that Kauravas built a palace of lac for the destruction of Pandavas.
- We come across references of lac in the Atharvaveda and Mahabharata, so it can be presumed that ancient Hindus were quite familiar with lac and its uses.
- Scientific study of lac started much later. In 1709 Father Tachard discovered the insect that produced lac. First of all Kerr (1782) gave the name *Coccus lacca* which was also agreed by Ratzeburi (1833) and Carter (1861).
- Later Green (1922) and Chatterjee (1915) called the lac- insect as *Tachardia lacca* (kerr).
- Finally, the name was given as *Laccifer lacca*.

Lac --- Nature's gift to mankind

- Lac is the only known commercial resin of animal origin.
- It is the hardened resin secreted by tiny lac insects belonging to a bug family.
- To produce 1 kg of lac resin, around 300,000 insects lose their life.
- The lac insects yields

1. **Resin**
2. **Lac dye**
3. **Lac wax.**

- Application of these products has been changing with time. Lac resin, dye etc. still find extensive use in Ayurveda and Siddha systems of medicine.







Importance of Lac

- With increasing universal environment awareness, the importance of lac has assumed special relevance in the present age, being

1. An eco-friendly
2. Biodegradable
3. Self-sustaining natural material

- Since lac insects are cultured on host trees which are growing primarily in wasteland areas, promotion of lac and its culture can help in eco-system development as well as reasonably high economic returns.
- It is a source of livelihood of tribal and poor inhabiting forest and sub-forest areas.

LAC INSECTS

- The English word lac synonyms Lakh in Hindi which itself is derivative of Sanskrit word Laksh meaning a lakh or hundred thousand.
- It would appear that Vedic people knew that the lac is obtained from numerous insects and must also know the biological and commercial aspects of lac industry.
- It is also worth to mention that a laksh griha would need a lot of lac which could only come from a flourishing lac industry in that period.



LAC INSECT TAXONOMY

- The first scientific account of the lac insect was given by J. Kerr in 1782 which was published in Philosophical Transaction of Royal Society of London (vol. 71, pp.374-382).
- The first scientific name given to it was *Tachardia lacca* following the name of French Missionary Father 'Tachardia'.
- It was later changed to *Laccifer lacca Kerr.*
- The other name given to it has been *Kerria Lacca Kerr.*

Phylum - Arthropoda

Class - Insecta

Order - Hemiptera

Super family - Coccoidea

Family - Lacciferidae

Genus - *Laccifer*

Metatachardia

Laccifer

Tachordiella

Austrotacharidia

Afrotachardina

Tachardina

- **Systematic Position:** *Laccifer lacca*
- A number of species of lac insects are known, of this *Laccifer lacca* is by far the most important and produces the bulk of the lac for commerce. It belongs to—
- Phylum — Arthropoda
- Class — Insecta
- Order — Hemiptera
- Super-family — Coccidae
- Family — Lacciferidae
- Genus — Laccifer
- Species — Lacca

STRAINS OF LAC INSECT:

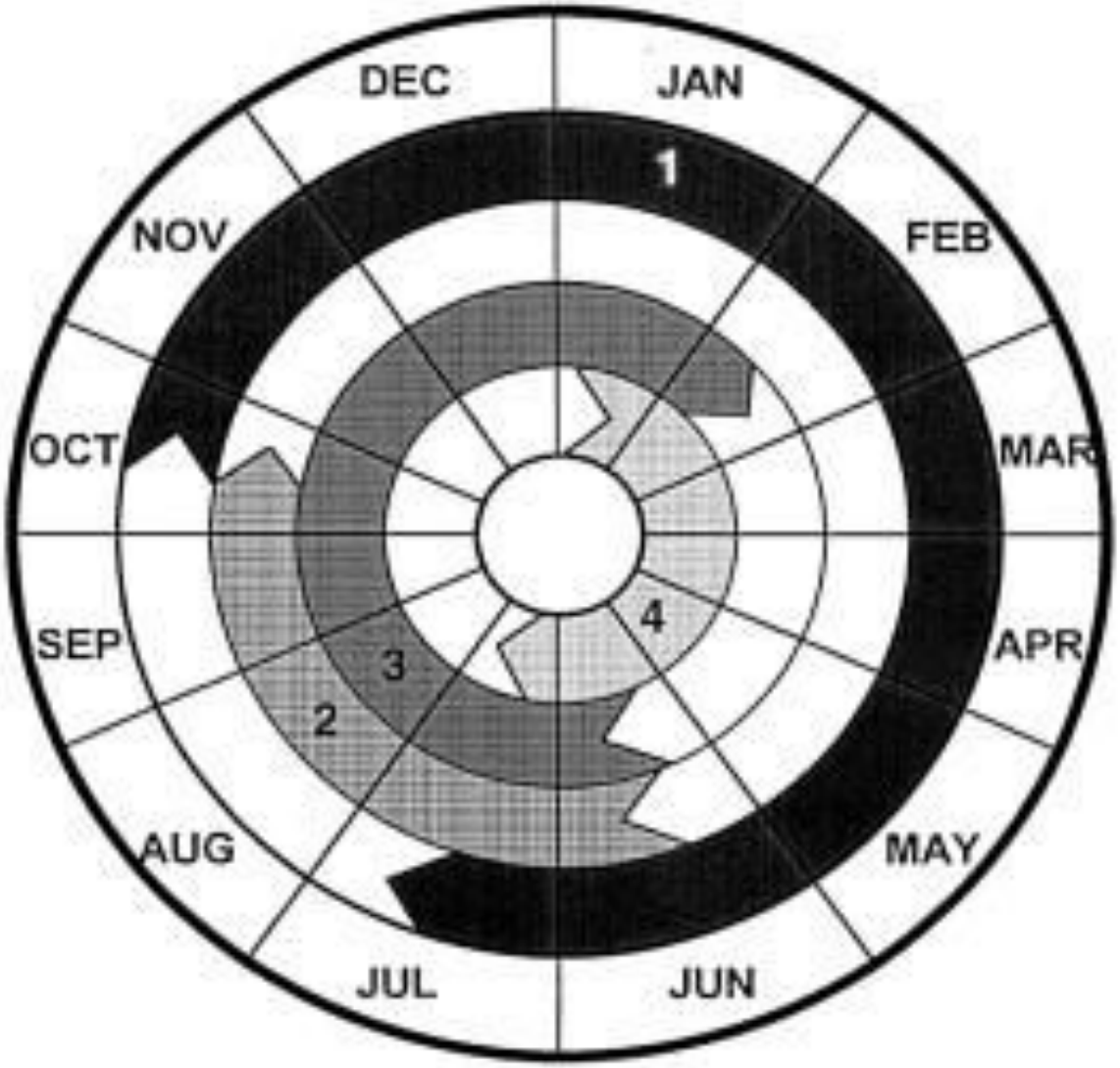
- In India, *lac insect* is known to have two distinct strains: **Kusumi and Rangeeni.** The Kusumi strain is grown on Kusum or on other host plants using Kusumi brood.
- The Rangeeni strain thrives on host plants other than Kusum.
- The life cycle of lac insects take about six months, hence, two crops a year can be obtained.
- **In case of kusumi strain, two crops are:**

I) Jethwi (June / July) and II) Aghani (Jan. / Feb).

II) In case of Rangeeni, two crops are:

I). Karrtiki (Oct. / Nov.) and II) baisakhi (May / June).

- The crops have been named after Hindi months during which these are harvested.
- The lac of Rangeeni crops is harvested while it is still immature.
- Aghani and baisakhi strain are the main corps contributing about 90% of lac production.
- The kusumi crop lac is considered superior resin, because of the lighter color of resin, and it fetches better price.



RANGEENI STRAIN

1 BAISAKHI crop

2 KATKI crop

KUSUMI STRAIN

AGHANI crop 3

JETHWI crop 4

Kusumi lac

- lac is directly related to the host plant and to the strain of lac insects.
- Based on industrial parameters, kusumi lac is better and fetches higher price in market.
- **In this respect, ber tree as** a potential kusumi lac host is already getting momentum.
- This host species is available in plenty and can supplement and fulfill the kusmi brood lac requirement in many areas.
- Similarly, siris (*Albizzia sp.) has also been identified as good host for kusumi brood lac*.
- *The* trees can be raised and utilized within a period of 5-6 years of plantation in comparison to around 15 years for kusum.



ARI LAC AND PHUNKI LAC

If lac crops are harvested by cutting down the lac bearing twigs a little before the larval emergence, that lac is known as ARI LAC (immature lac)

If lac crops are harvested by cutting down the lac bearing twigs a little after the emergence is over, that is called PHUNKI LAC (empty lac).

LAC AND ITS FORMS

STICK LAC : The lac encrustations is separated by knife or broken off with finger from the twig of host plants and is known is STICK LAC or CRUDE LAC or RAW LAC.

SEED LAC : The stick lac, after grinding and washing, is called SEED LAC or CHOWRI.

SHELLAC : The manufactured product prepared from stick lac after washing and melting, which takes the form of yellow colored flakes, is called SHELLAC.

BUTTON LAC : After melting process, lac is dropped on a zinc sheet and allowed to spread out into round discs of about 3" diameter and 1/4" thickness is called BUTTON LAC

GARNET LAC : It is prepared form inferior seed lac or kiri by the solvent extraction process. It is dark in colour and comparatively free from wax.

BLEACHED LAC : It is a refined product obtained by chemical treatment. It is repared by dissolving shellac or seed lac in Sodium carbonate solution, bleaching the solution with Sodium hypochlorite and precipitating the resin with sulphuric acid. Bleached lac deteriorates quickly and should be used within 2-3 months of manufacture.

DISTRIBUTION

- Lac is currently produced in

1. INDIA
2. MYANMAR
3. THAILAND
4. MALAYA
5. YUAN PROVINCE OF CHINA.

- India and Thailand are main areas in the world, while India has prime position in relation to lac production.
- Lac cultivation is introduced into Thailand from India.

HOST PLANTS

- Lac insects thrive on twigs of certain plant species, suck the plant sap, and grow all the while secreting lac resin from their bodies.
- These plants are called host plants. Although lac insect is natural pest on host plant, these insects enjoy the privileged position not being treated as pest.
- This is because:

i) **They yield a useful product,**

Ii) The host plants are economically not so important

Iii) The insects cause only temporary and recoverable damage to the host plants.

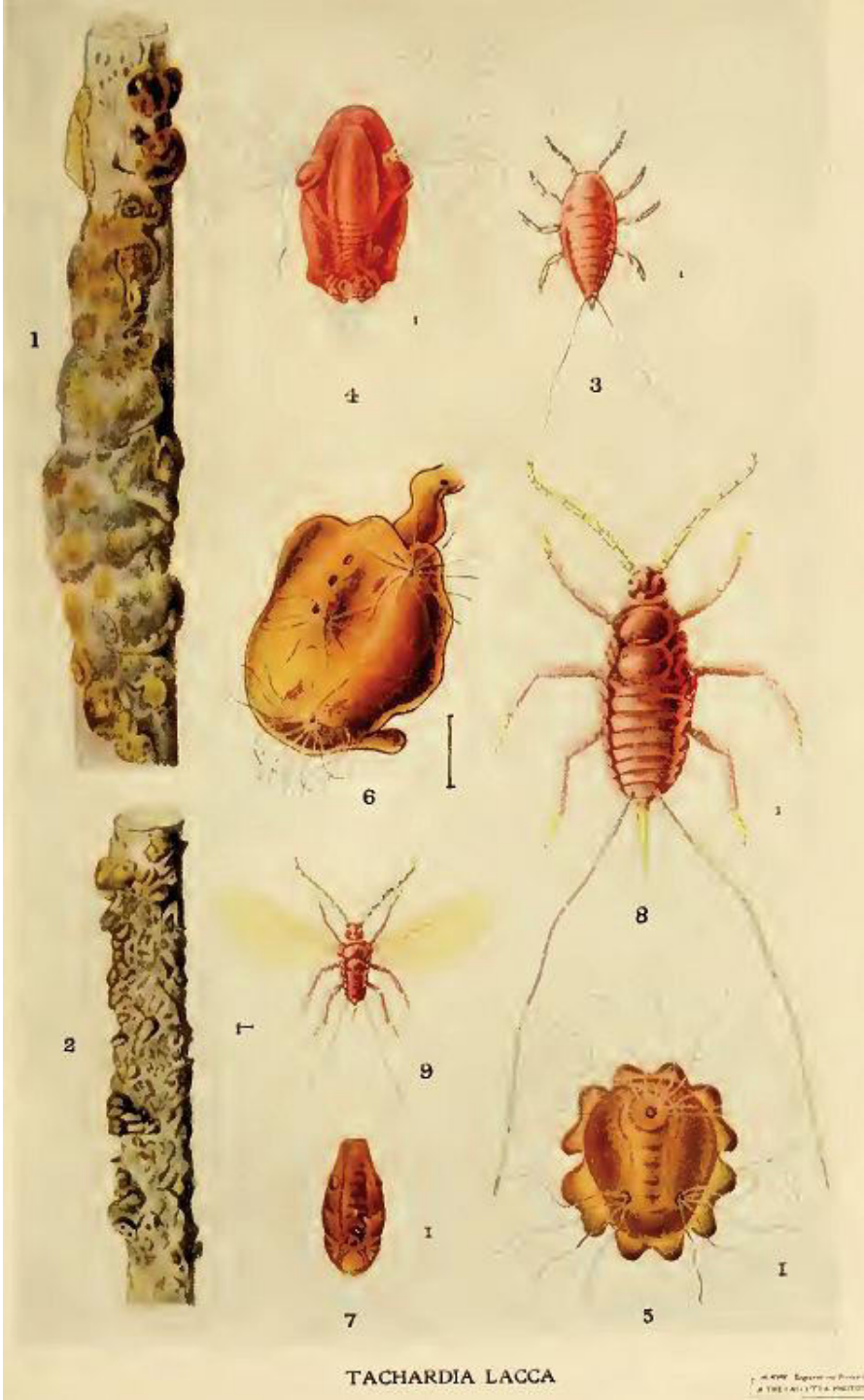
- About 113 varieties of host plants are mentioned as lac host plant. Out of which the followings are very common in India:

1. ***Butea monosperma (Vern. Palas)***
2. ***Zizyphus spp (vern. Ber)***
3. ***Schleichera oleosa (Vern. Kusum)***
4. ***Acacia catechu (Vern. Khair)***
5. ***Acacia arabica (Vern. Babul)***
6. ***Acacia auriculiformis (Vern. Akashmani)***

- Arhar (*Ca]anus indicus*)
- The insects live as a parasite, feeding on the sap of certain trees and shrubs. The lac insects breed and thrive well on them until they start producing lac.
- Before coming to the actual mechanism of lac secretion and its processing, it is advisable for a lac-culturist to have detailed knowledge of lac insect and its life cycle.
- The adult lac insect shows a marked phenomenon of sexual dimorphism.
- The male and female insect varies in shape, size and also in presence or absence of certain body parts.

lac insect

- Lac insect is a minute crawling scale insect which inserts its suctorial proboscis into plant tissue, sucks juices, grows and secretes resinous lac from the body.
- Its own body ultimately gets covered with lac in the so called ‘CELL’.
- Lac is secreted by insects for protection from predators.







Male lac insect

- Male is red in colour and measures 1.2 - 1.5mm in length.
- It has reduced eyes and antennae.
- Thorax bears a pair of hyaline wings.



- **Structure of Male Lac-insect:**

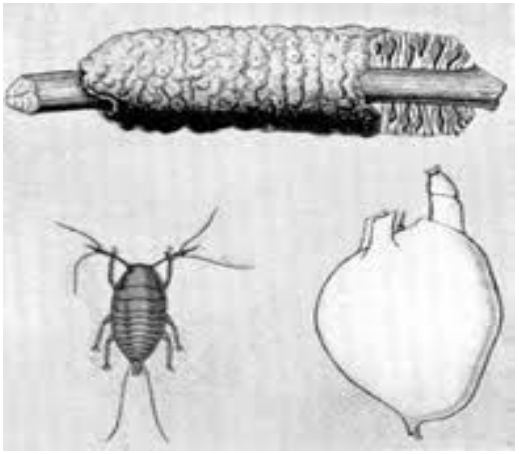
- It is red in color. The body is typically divided into head, thorax and abdomen.
- The head bears a pair of antennae and a pair of eyes. Mouth parts are absent so a male adult insect is unable to feed. Thorax bears three pairs of legs.
- Wings may or may not be found.
- Abdomen is the largest part of the body bearing a pair of caudal setae and sheath containing penis at the posterior end.

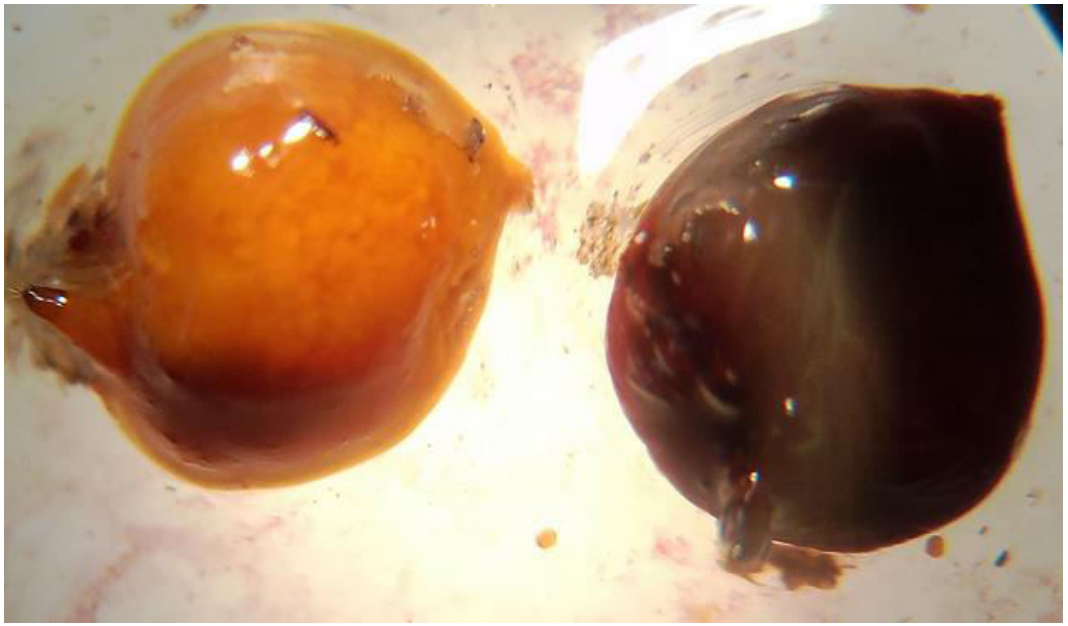
Male Lac Cell

- After the first molt, both male and female nymphs lose their appendages, eye and become degenerate.
- While still inside their cells, the nymphs cast off their second and third molt and mature into adult.
- Both the male and female larvae become sexually mature in about eight weeks.
- Only the male one undergoes a complete metamorphosis or transformation into another form; it loses its proboscis and develops antennae, legs and a single pair of wings.
- It is contained in a brood cell somewhat slipper like with a round trap door (operculum) through which it emerges.
- The adult male is winged and walks over the females to fertilize them.

Female lac insect

- Female is larger than male, measures 4-5 mm in length and has a pyriform body.
- The head, thorax and abdomen are not clearly distinct.
- The antennae and legs are in degenerated form, and wings are absent.

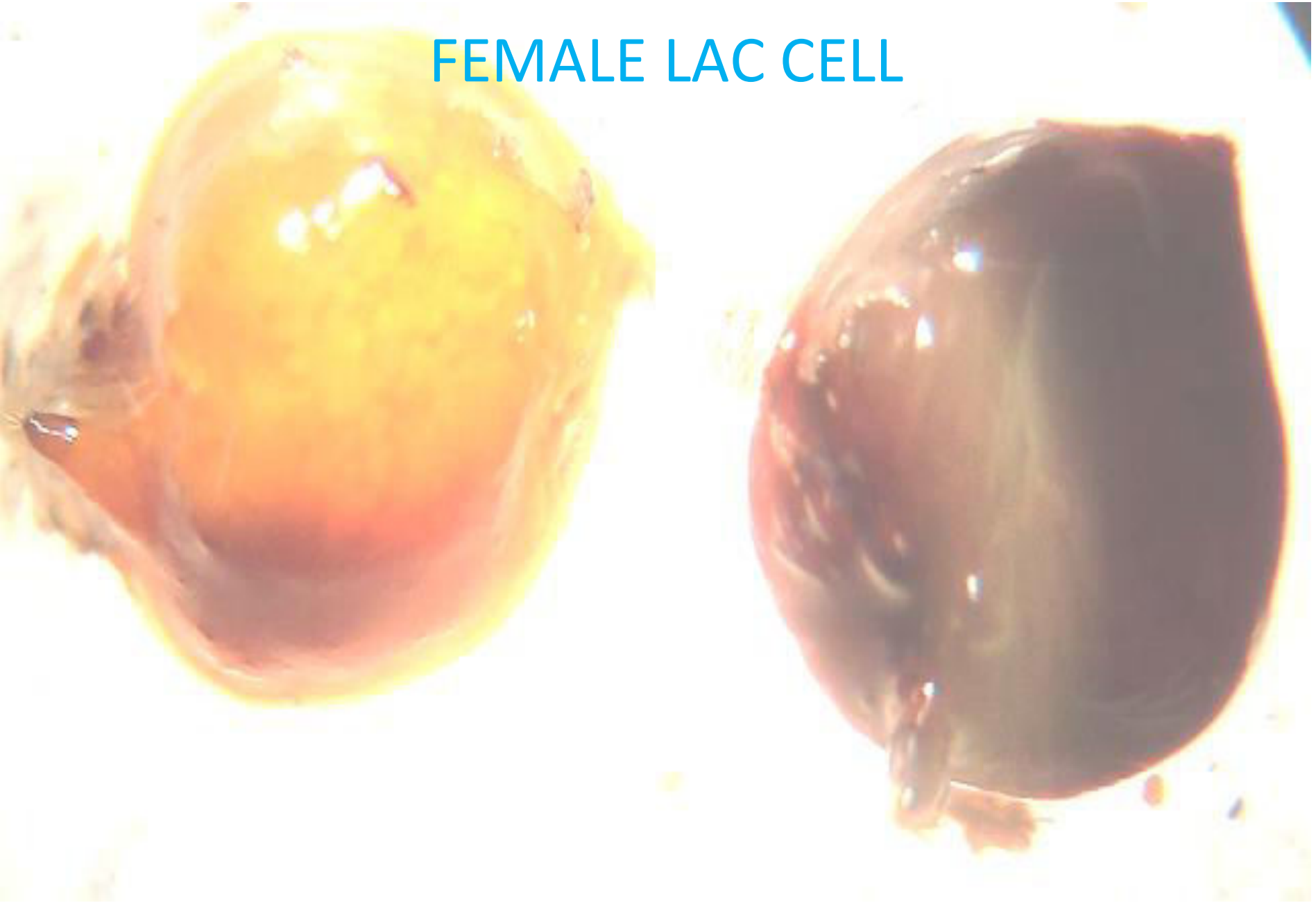




- **Structure of Female lac-insect:**

- Head bears a pair of antennae and a single proboscis. Eyes are absent.
- Thorax is devoid of wings and legs. The loss of eyes, wings, and legs are due to the fact that the female larvae after settling down once never move again and thus these parts become useless and ultimately atrophy.
- Abdomen bears a pair of caudal setae. It is female lac insect which secretes the bulk of lac for commerce.
- The female brood cell is larger and globular in shape and remains fixed to the twig.
- The female retains her mouth parts but fails to develop any wings, eyes or appendages.
- While developing, it really becomes an immobile organism with little resemblance to an insect.
- Females become little more than egg producing organisms.

FEMALE LAC CELL



Lifecycle of lac insect

- The Life cycle of lac insect takes about six months and consists of stages:

1. EGG,
2. NYMPH
3. INSTARS,
4. PUPA AND
5. ADULT.

- The lac insects have an ovoviviparous mode of reproduction.
- Female lays 200-500 ready to hatch eggs, i.e. the embryos are already fully developed in eggs when these are laid.

- **Fertilization:**
- After attaining the maturity, males emerge out from their cells and walk over the lac incrustations(crusts).
- The male enters the female cell through anal tubular opening and inside female cell it fertilizes the female.
- After copulation, the male dies. One male is capable of fertilizing several females.
- Females develop very rapidly after fertilization.
- They take more sap from plants and exude more resin and wax.

Brood lac is used for inoculation

- When the eggs hatch, larvae emerge and the whole process begins all over again
- After the cycle has been completed and around the time when the next generation begin to emerge, the resin encrusted branches are harvested.
- They are scraped off, dried and processed for various lac products.
- A portion of brood lac is retained from the previous crop for the purpose of inoculation to new trees.

- **Life Cycle of Lac Insect:**

- The females after fertilization are capable of producing eggs. But it has been noticed in case of lac insects that the post fertilization developments start when the eggs are still inside the ovary.
- These developing eggs are oviposited into the incubating chambers (formed inside the female cell by the body contraction of females).
- A female is capable of producing about one thousand eggs (average 200-500). Inside incubating chamber, the eggs hatch into larvae.

- Female cell is oval, having a pair of small branchial pores in anterior side and a single round anal tubular opening in posterior side.
- Through the anal tubular opening are protruding waxy white filaments, secreted by the glands in the insects body, which is an indication that the insect inside the cell is alive and is in healthy condition.
- These filaments also prevent the blocking of the pore during excess secretion of lac.

- The cell produced by male and female differ in shape, and can be easily distinguished sometimes later.
- Male cells are elongated and cigar shaped.
- There is a pair of branchial pores in the anterior side and a single large circular opening covered by the flap in the posterior side.
- It is through the posterior circular opening that the matured male lac insect emerges out of its cell.

- **The Life Cycle of Lac Insect**

- The larvae are minute, boat shaped, red colored and measure little over half millimeter in length.
- Larva consists of head, thorax and abdomen. Head bears a pair of antennae, a pair of simple eyes and a single proboscis. All three thoracic segments are provided with a pair of walking legs.
- Thorax also bears two pairs of spiracles for respiration. Abdomen is provided with a pair of caudal setae.
- These larvae begin to wander in search of suitable center to fix them. This mass movement of larvae from female cell to the new off-shoots of host plant, is termed as “swarming”.

Lifecycle of lac insect

- Swarming occurs after the emergence of nymph and it may continue for 5 weeks.
- The nymphs crawl about on branches.
- On reaching soft succulent twigs, the nymphs settle down close together at rate of 200-300 insects per squire inch.
- At this stage, both male and female nymphs live on the sap of the trees.
- They insert their suctorial proboscis into plant tissue and suck the sap.



- The emergence of larvae from female cell occurs through anal tubular opening of the cell and this emergence may continue for three weeks.
- The larvae of lac are very sluggish and feed continuously when once they get fixed with the twig.
- In the meantime the larvae start secreting resinous substance around their body through certain glands present in the body. After some-time the larvae gets fully covered by the lac encasement, also known as lac cell.
- Once they are fully covered, they molt and begin to feed actively.

Lifecycle of lac insect

- After a day or so of settling, the nymphs start secreting resin from the glands distributed under the cuticle throughout the body, except mouth parts, breathing spiracles and anus.
- The resin secreted is semi-solid which hardens on exposure to air into a protective covering.
- The nymphs molt thrice inside the cells before reaching maturity.
- The duration of each instar is dependent on several factors, viz. temperature, humidity and host plant.

Ovoviviparous nature

- The female increases in size to accommodate her growing number of eggs.
- Lac resin is secreted at a faster rate, and a continuous layer coalesces or grows into one body.
- After fourteen weeks, the female shrinks in size allowing light to pass into the cell and the space for the eggs.
- About this time, two yellow spots appear at the rear end of the cell.
- The spots enlarge and become orange colored.
- When this happens, the female has oviposit a large number of eggs in the space called 'Ovisac'.
- The ovisac appears orange due to crimson fluid called lac dye which resembles cochineal.
- It indicates that the eggs will hatch in less time.



- Eggs deposited hatch within a few hours of laying, and a crimson-red first instar nymph called crawlers come out.
- The crawler measures 0.6 x .25 mm in size.

Lac Insect; Nature of Attack

Adult female scales produce a high-domed 'test' or shell with four to six lobe-like projections that anchor the test to the plant surface.
The test is hard and glossy with a reddish-orange tint around the edges, and darker toward the center.




2



- In some specimens, white string-like wax fiber extrusions project from the dorsum of the test, but these may break off.



- In heavy infestations, the tests of multiple females will develop into a single, aggregated mass such that the distinctive appearance of the individual tests is lost.
- The test darkens as the scale matures.
- The female scale, which is a deep red color, lives inside the test and requires a special procedure to remove without damage.




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- The sticklac is a protection for the insects.
- By excreting sugars they also attract the Praetorian Guard (weaver ants, *Oecophylla smaragdina*).

- **Lac Secretion:**
- Lac is a resinous substance secreted by certain glands present in the abdomen of the lac insects. The secretion of lac begins immediately after the larval settlement on the new and tender shoots. This secretion appears first as a shining layer which soon gets hardened after coming in contact with air.
- This makes a coating around the insect and the twig on which it is residing.
- As the secretion continues the coating around one insect meet and fuses completely with the coating of another insect.
- In this way a continuous or semi-continuous incrustation of lac is formed on the tender shoots.

- **Cultivation of Lac:**

- Cultivation of lac involves proper care of host plants, regular pruning of host plant, infection or inoculation, crop-reaping, control of insect pests, and forecast of swarming, collection and processing of lac.

- The first and perhaps the most important prerequisite for cultivation of lac is the proper care of the host plant.
- It is the host plants on which lac insects depend for their food, shelter and for completion of their life cycle.
- There are two ways for the cultivation of host plants. One is that plants should be allowed to grow in their natural way and the function of lac-culturist is only to protect and care for the proper growth of plants.

- Another way is that a particular piece of land is taken for the purpose and systematic plantation of host plant is made there.
- Regular watch is necessary in this case by providing artificial manures, irrigation facilities, ploughing and protecting the plants from cattle and human beings for which the land should be fenced.
- The larvae of lac insects are inoculated on host plants only after the host plants have reached a proper height.
- The lac larvae feed on the cell sap by inserting their proboscis in the tender twigs. The proboscis can only be inserted in the tender young off-shoots. For this before inoculation, pruning of lac host plants is necessary. The branches less than an inch in diameter are selected for pruning.
- Branches half inch of less in diameter should be cut from the very base of their origin. But the branches more than half inch diameter should be cut at a distance of 1 ½ inch from the base.

- **Inoculation:**

- The method by which the lac insects are introduced to the new lac host plant is known as inoculation.
- This may be of two types, namely “Natural infection” and “Artificial infection”. When infection from one plant to other occurs by natural movements of insect, it is called natural infection. This may be due to overcrowding of insect population and non-availability of tender shoots on a particular tree.
- Artificial infection takes places through the agencies other than those of nature. Prior to about two weeks of hatching, lac bearing sticks are cut to the size of six inches. They are called “Brood lac”.
- Brood lacs are then kept for about two weeks in some cool place.

- When the larvae start emerging from this brood lac, they are supposed to be ready for inoculation.
- Strings can be used for tiding the brood lac with the host plant may be of different types in longitude infection the brood lac is tied in close contact with host branches.
- In lateral infection the brood lac is tied across the gaps between two branches.
- In interlaced method, brood lac is tied among the branches of several new shoots.

- **Lac Crops:**
- The lac insects repeat its life cycle twice in a year. There are actually four lac crops since the lac insects behave in two ways either they develop on Kusum plants or develop on plants other than Kusum.
- The lac which grows on Non-Kusum plants is called as “Ranjeem lac,” and which grows on Kusum plant is called as “Kusumi lac.
- Four lac crops have been named after four Hindi months in which they are cut from the tree. They are as follows:

- **Ranjeeni Crop:**
- **(i) Katki:**
- Lac larvae are inoculated in June-July. Male insect emerges m August-September. Female give rise to *swarming larvae* in October-November and the crop is reaped in Kartik (October and November).
- **(ii) Baisakhi**:
- Larvae produced by Katki crop are inoculated in October-November, male insects emerges in February-March, females give rise to swarming larvae in June-July, the crop is reaped in Baisakh (April-May).

- **Kusumi Crop:**

- **(i) Aghani**:
- Lac larvae are inoculated in June-July, male insect emerges m September, female give rise to swarming larvae in January-February and crop is reaped in Aghan (December-January).

- **(ii) Jethoi**:
- The larvae produced by Aghani crop is inoculated in the month of January- February, male emerges in March-April, female give rise to swarming larvae in June- July and the crop is reaped in the month of Jeath (June-July).
- The time of infection with swarming larvae, the time of emergence of male insects, the time of reaping the crop, and the time of producing swarming larvae by female etc., are shown m tabular form below
-

- **Scraping and Processing of lac:**

- Lac cut from the host plant is called as "stick lac". Lac can be scraped from the twigs before or after the emergence of larvae. If it is used for manufacturing before the emergence of larvae, the type of lac produced is called as "Ari lac" and if it is used for manufacturing purpose after swarming of larvae has occurred, the lac is said to be "Phunki lac".

- The scraping of lac from twig is done by knife, after which they should not be exposed to sun. The scraped lac is grinded in hard stone mills. The unnecessary materials are sorted out In order to remove the finer particles of dirt and color, this lac is washed repeatedly with cold water.

- Now at this stage it is called as “Seed lac” and is exposed to sun for drying. Seed lac is now subjected to the melting process. The melted lac is sieved through cloth and is given the final shape by molding. The final form of lac is called “Shellac”.
- Colour or different chemicals may be mixed during melting process for particular need.

- **Lac Enemies and Their Control:**

- A lac enemy imposes a challenge to the lac culturist, as they not only decreases the population of lac insects, but also retard the production and quality of lac. Damage caused to lac insects may be grouped under two heads, (a) damage caused by insects (b) damage caused by animals other than insects. Insect enemies of lac crop may be predators and parasites.

- The common parasites of lac insect are known as "Chalcid." They are small, winged insects which lay their eggs inside the lac coat either on the body of the lac insect or inside the body of the lac insect.
- The larva which hatches from these eggs feed upon the lac insects, thereby causing mortality of their host. Damage done by this parasite constitute about 5-10% of the total destruction of the lac crop.

- Damage done by the predators is of greater intensity (35% of the total destruction).
- The major predators of lac insects are *Eublemma amabilis* (the white moth) and *Holococera pulverea* (the blackish grey moth).
- They not only feed on lac insects but also destroy the lac produced by term. Squirrels, monkey, rat, bat, birds (wood peckers), man etc., are the enemies other than insects which destruct the lac crop in different ways.
- Damage is also done by climatic factors such as excess heat, excess cold, heavy rain, and storm and partly by the faulty cultivation methods.

- **Control:**
- Damage caused by the above mentioned animals can be reduced to certain extent by the use of the following methods.

- **Cultural Method:**
- The amount of damage by infection can be reduced to a greater extent by taking care during the culture of lac insects, especially at the time of inoculation.
- The brood lac showing the minimum enemy attack should be selected for inoculation and should be cut from the host plant very near to the time of emergence of larvae (about one week before the emergence).
- This will reduce the chances of parasite attack on the emerging larvae at new place (host).

- The brood lac used for inoculation should be removed from the new host's branches as soon as the emergence of larvae stops (approx. 3 weeks after inoculation). It reduces the chance of transference of enemies to the new host plant from the brood lac.

- The infected brood lac not fit for inoculation or the used up brood lac should not be retained for long. The lac should be scrapped at once and the rest may be crushed or dropped into fire in order to destroy the predators and parasites.

- The delay in processing also gives chances to the enemy insects to escape into field.
- So the manufacturers should try to convert stick lac into seed lac as soon as possible.
- By these cultural methods the future production can be saved from infection to some extent.

- **Artificial Method:**

- During the crop reaping, it is not always possible for the manufacturers to convert the huge amount of stick lac to seed lac at a time.
- To avoid the spreading of enemies at this time from stocked stick lac simple artificial method can be used. Bundles of stick lac should be tied with stones and immersed in fenced water (river or ponds) for about a week.
- This kills all the parasitic and predator insects as they cannot survive in water.

Biological Method:

- **It is an indirect method for killing the parasitic and predator insects.**
- **For this purpose hyper-parasitic insects are used which attacks the parasitic insects of lac and kill them.**
- **These hyper-parasitic insects are however, not harmful for lac crop.**

- **Use of Lac:**

- Lac has been used for the welfare of human beings from the great olden days
- No doubt the development of many synthetic products have made its importance to a little lesser degree, but still it can be included in the list of necessary articles.
- Lac is used in making toys, bracelets, sealing wax, gramophone records etc.
- It is also used in making grinding stones, for filling ornaments, for manufacturing of varnishes and paints, for silvering the back of mirror, for encasing cable wires etc.,
- Waste materials produced during the process of stick lac is used for dying purpose. Nail polish is a good example of the by-product of lac.

- **Composition of Lac:**

- Lac is a mixture of several substances, of which resin is the main constituent. The approximate percentage of different constituents of lac is given below:
- Resin – 68 to 90%
- Dye – 2 to 10%
- Wax – 5 to 6%
- Mineral matter – 3 to 7%
- Albuminous matter – 5 to 10%
- Water – 2 to 3%

- **Present Position of this Industry**

- Lac is produced in a number of countries including India, Thailand, Mayanmar, China, Indonesia, Vietnam and Laos.
- India and Thailand are the major producers, producing on the average 1700 tons of lac annually, followed by China.
- India alone, accounts for about 70 % of global lac production.
- Former Bihar is the most important lac producing state of India. The Indian council of Agriculture Research has established Indian Lac Research Institute at Namkum in Ranchi district of Jharkhand.

- The average of different states in the total quantity of stic lac produced in this country is given below:
- Bihar – 55.5%
- Madhya Pradesh – 22%
- West Bengal – 10%
- Maharashtra – 7.1%
- Gujrat – 2.7%
- Uttar Pradesh – 1.8%
- Assam – 0.6%
- Orissa – 0.1%

- Total annual global production of pure lac is estimated to be 20,000 tons. The average total production of stick lac in India is about 24,000 tons, while the annual average pure lac produced in the country is 11,890 tons. About 6000 tons of pure lac produced in India is exported to different countries of the world, with an average earning of Rs. 202.38 million in term of foreign exchange. It has been estimated that 3-4 million people mostly tribal are engaged in the cultivation and several thousands in addition are engaged in the trade and manufacture of lac.
- Two main competitors of Indian lac are (i) Thailac, which accounts 50% of the total lac exported, and (ii) Synthetic resin, which have replaced lac in certain field. Shellac being a versatile resin, there is immense scope of increasing its utilization in various fields and there is also scope to modify it to meet particular need.

Beekeeping

- Beekeeping is Applied Bee Biology
- Beekeeping is Anticipation not merely reaction
- Beekeeping is Colony Population Management
- Beekeeping is Science and Art

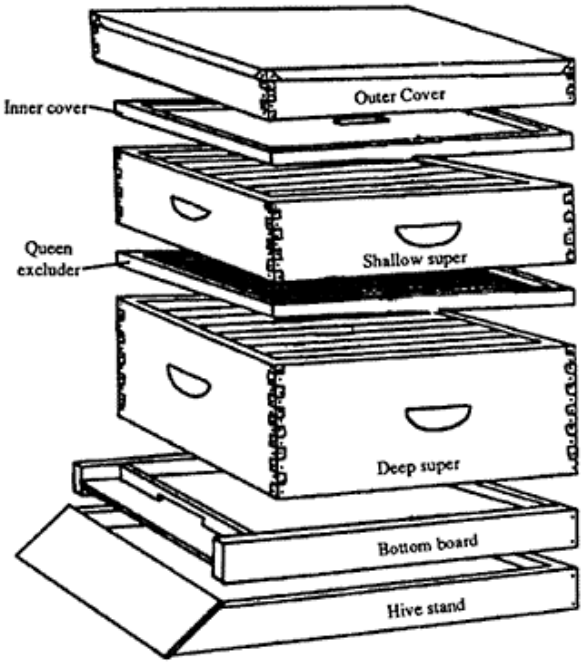




The place where a beekeeper keeps his bees is called an apiary or a bee yard.

- The bee colony is kept inside a hive that is made from a series of wooden boxes and frames that hold wax sheets for the bees to use as a starting point when building honeycomb.
- The top box contains honey, while the bottom box is used to hold the queen bee and most of the worker bees.
- In the United States, the most popular type of hive design used for beekeeping is known as a Langstroth bee hive.





Beekeeping, the practice of artificially maintaining honey bee colonies, one of the oldest forms of food production.

- Honey is a sweet yellow to rich amber colored fluid produced by bees. Other insects can also produce honey, but bee honey is the product which most people are familiar with, since it has been consumed for centuries as a sweetener.





Formally known as *apiculture*, beekeeping is thought to have been practiced as early as 13,000 BC.

- The ancient Egyptians were particularly skilled in the art of beekeeping, since they considered honey to be an important part of their diet.
- Temples kept bees in order to satisfy the desire of the gods for honey and for the production of medicines and ointments.





Tomb of Pabasa (25th dynasty)

Beekeeping History

- Human as Hunter



Beekeeping History

- Human as Primitive Beekeeper





Beekeeping History

Revolutionary war-era beekeeping – 2 gums & 2 skeps



Skep beekeeping

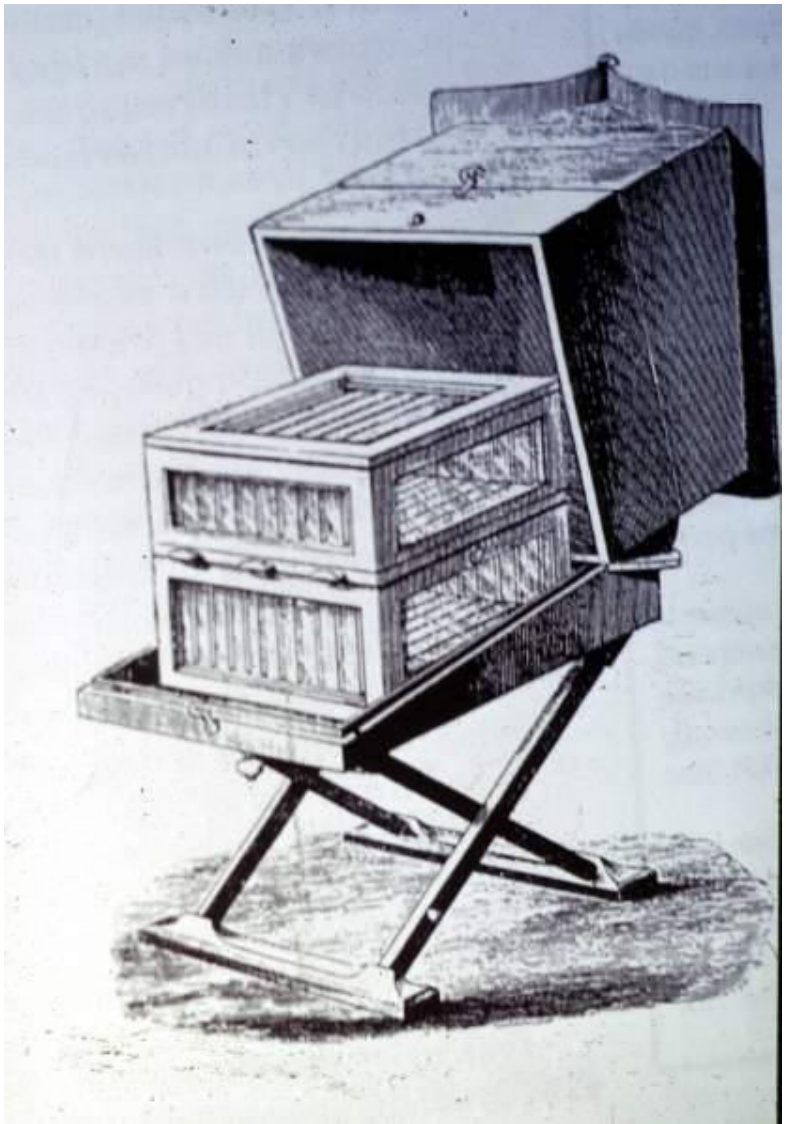


Beekeeping History

- Human as Beekeeper



Rev L.L. Langstroth

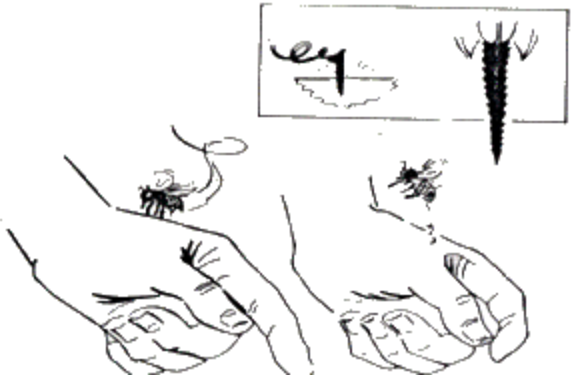


Good beekeeping basics that can help.

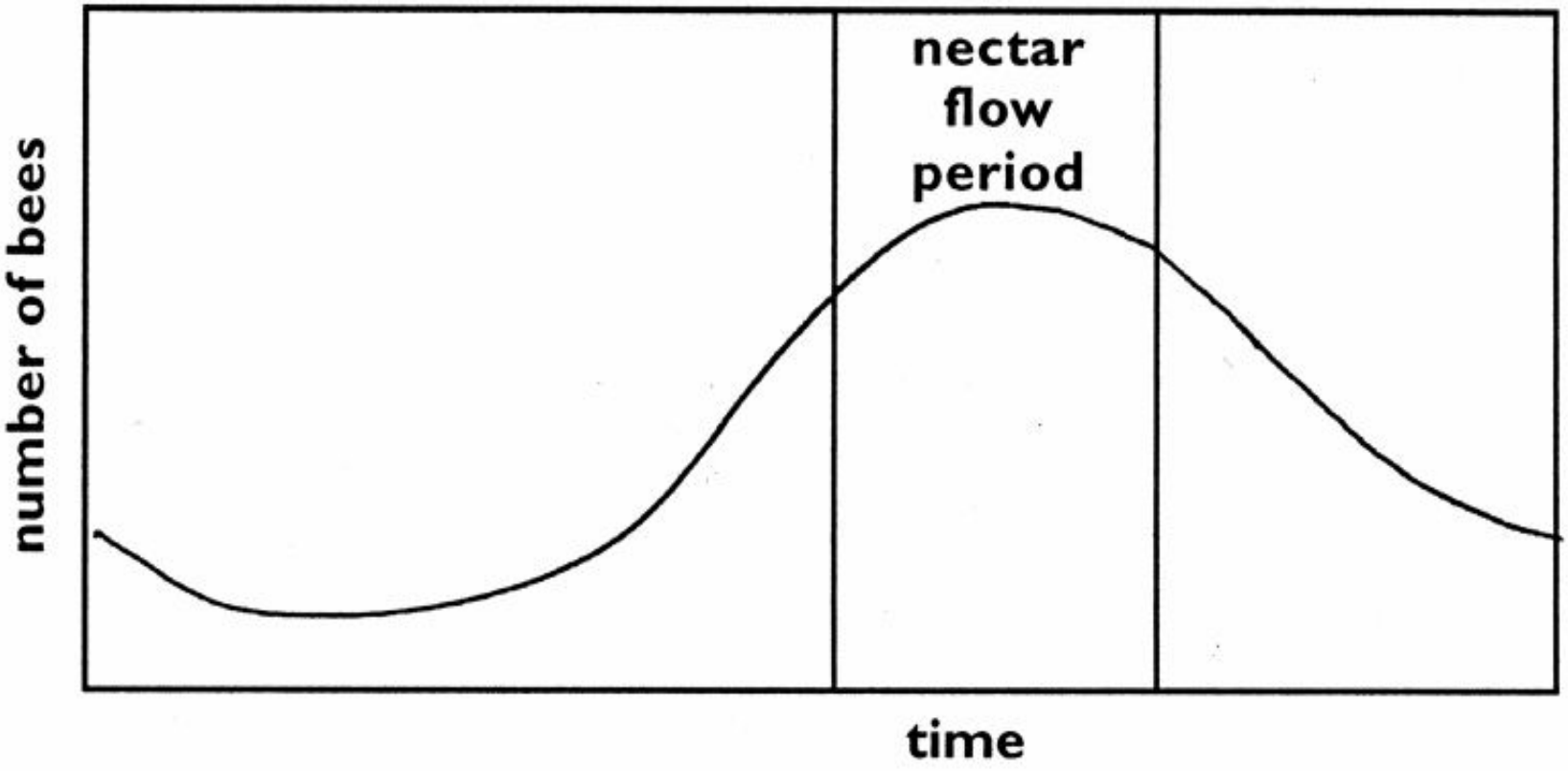
- Knowledge of bee science.
- Knowledge of bee botany.
- Familiarity with modern agriculture.
- Some elementary economics.
- Tools and the ability to work with wood.
- Ability to keep vehicles running
- Fortitude(courage) to take bee stings.

  A bee's front end is sweet and kind,
  But never trust a bee's behind.
  A bee can sting if it can sit,
  So always stay in front of it!

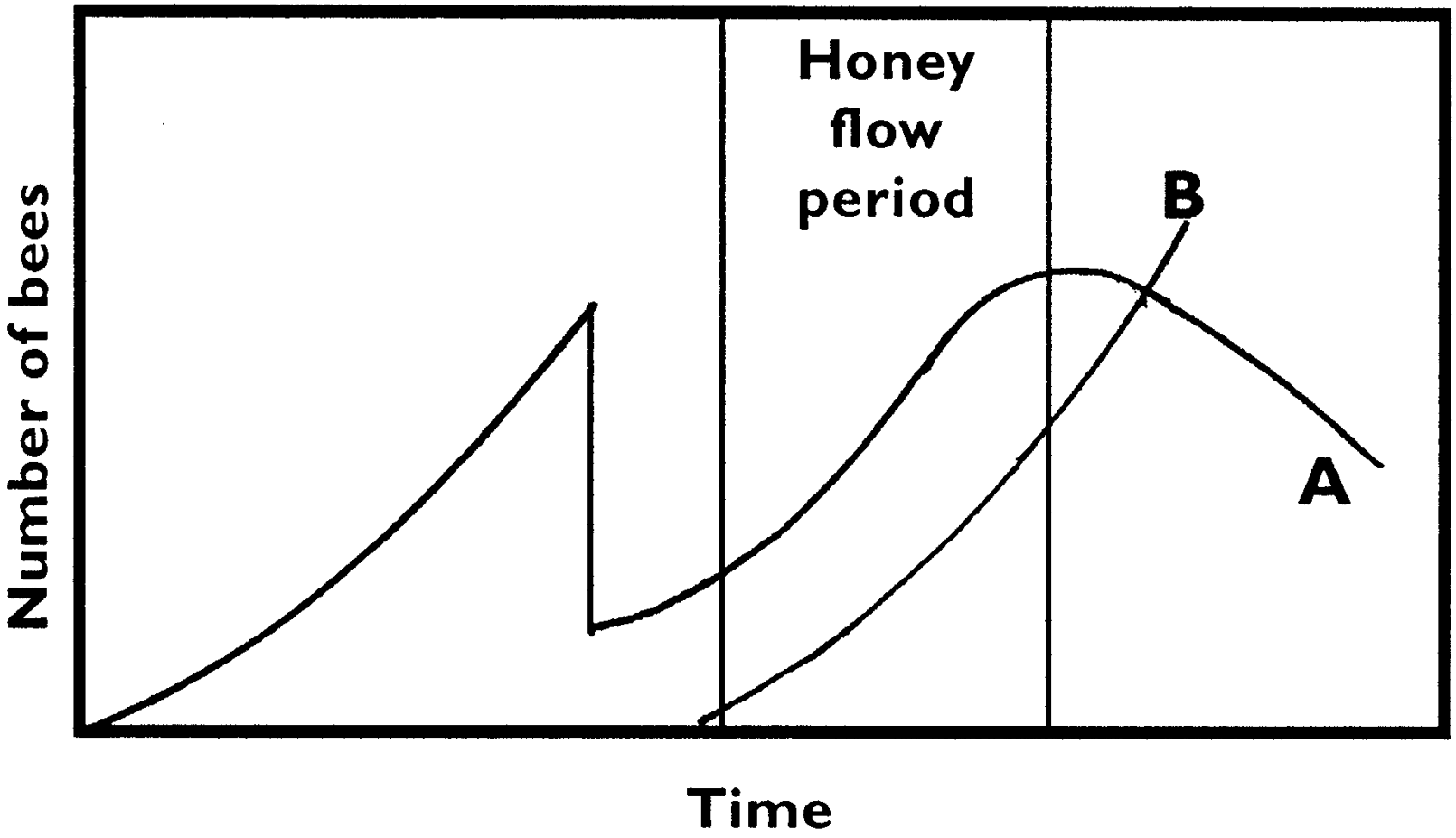




Beekeeping – the real KEY!



Beekeeping – the real KEY!



Beekeeping – the KEY!

Manage colonies intensively



or



extensively

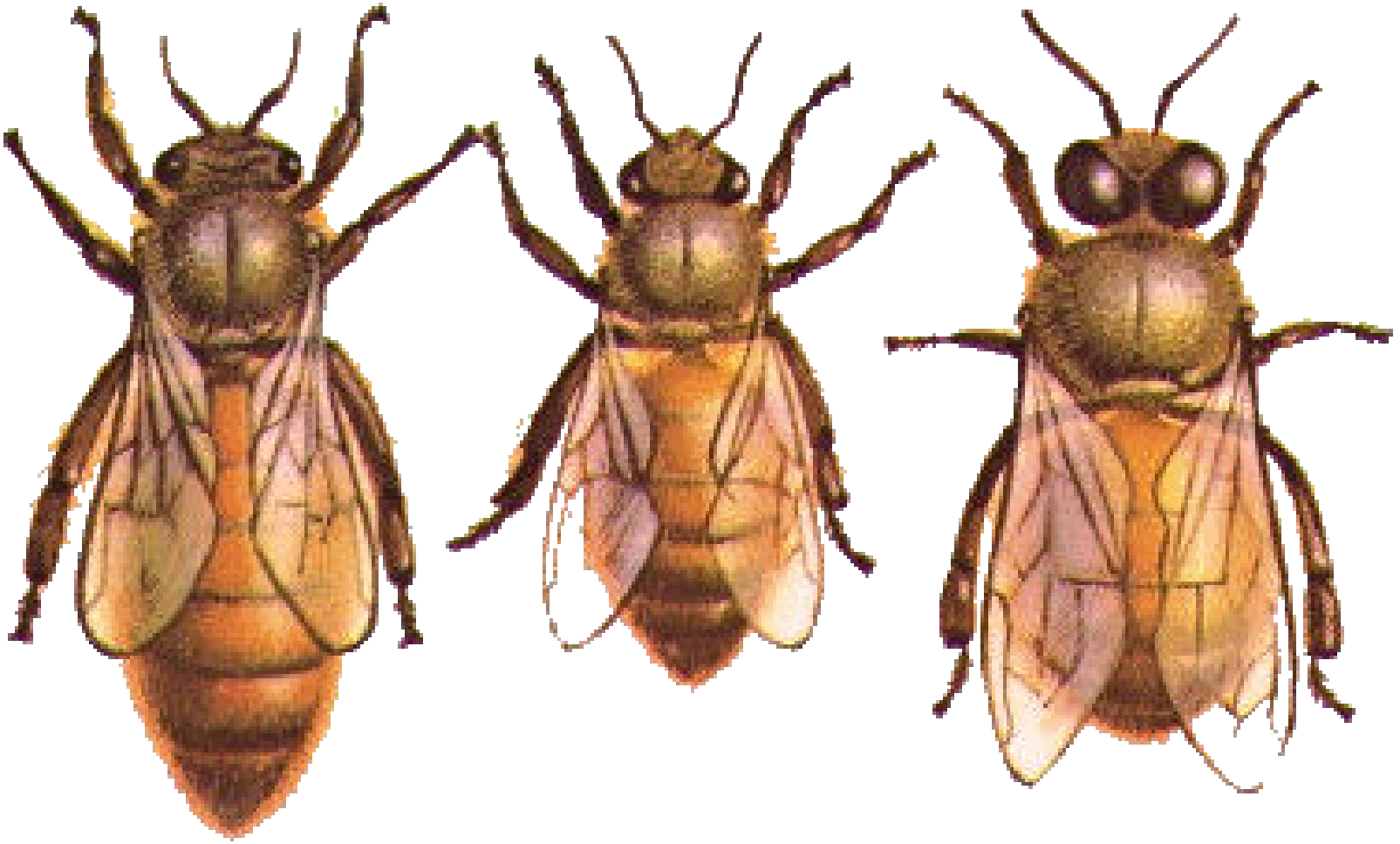
Beekeeping

Picture yourself here

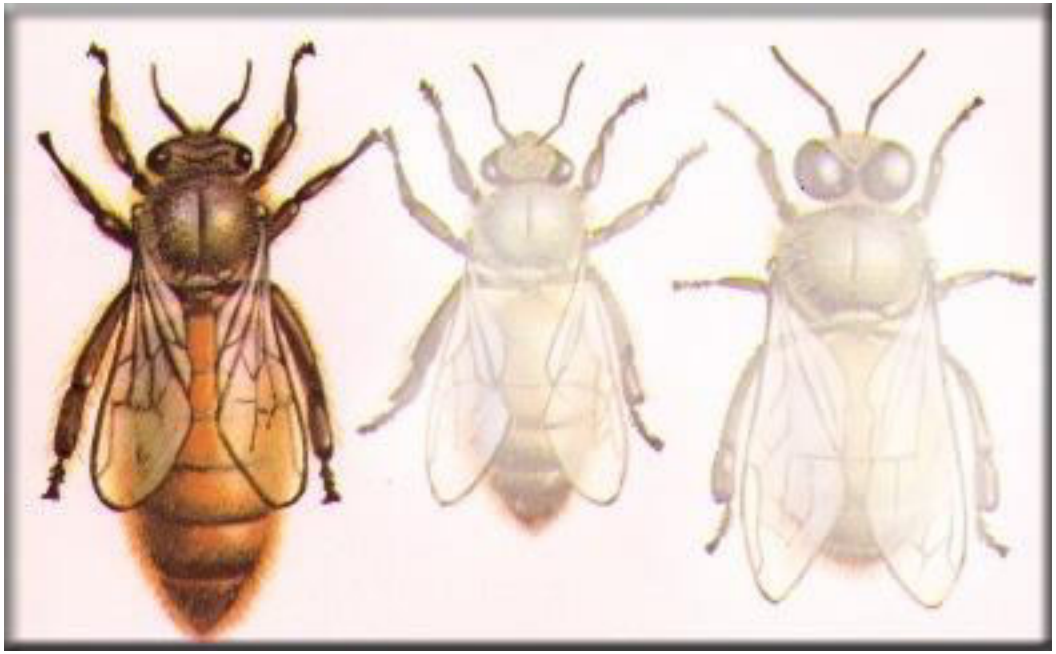


So, let’s meet these wonderful creature’s.

Queen **Worker** **Drone**



The Queen



- The queen is the only sexually developed female in the hive.
- She is the largest bee in the colony.
- A productive queen can lay 2,000 eggs in a single day.

The Worker



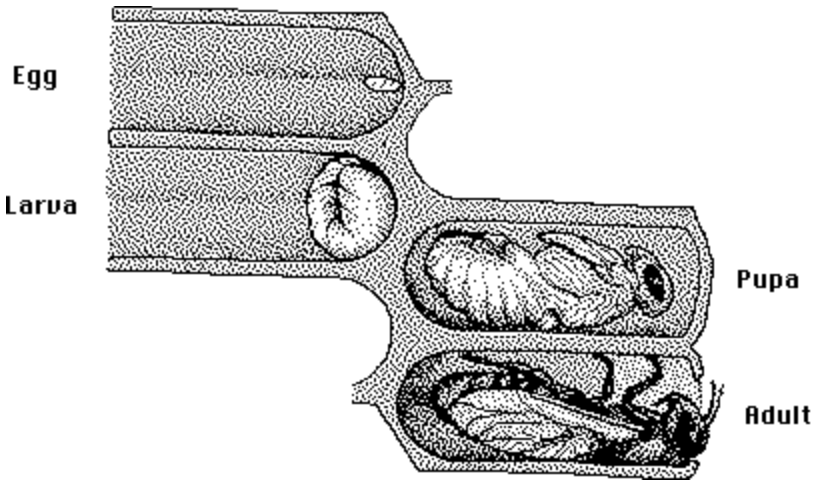
- Workers, the smallest bees in the colony, are sexually undeveloped females.
- A colony can have 50,000 to 60,000 workers. The life span of a worker bee varies according to the time of year. Her life expectancy is approximately 28 to 35 days.
- Workers that are reared in September and October, however, can live through the winter.
- Worker bees also collect nectar to make honey.
- Bees produce honey as food stores for the hive during the long months of winter when flowers aren't blooming.

The Drone



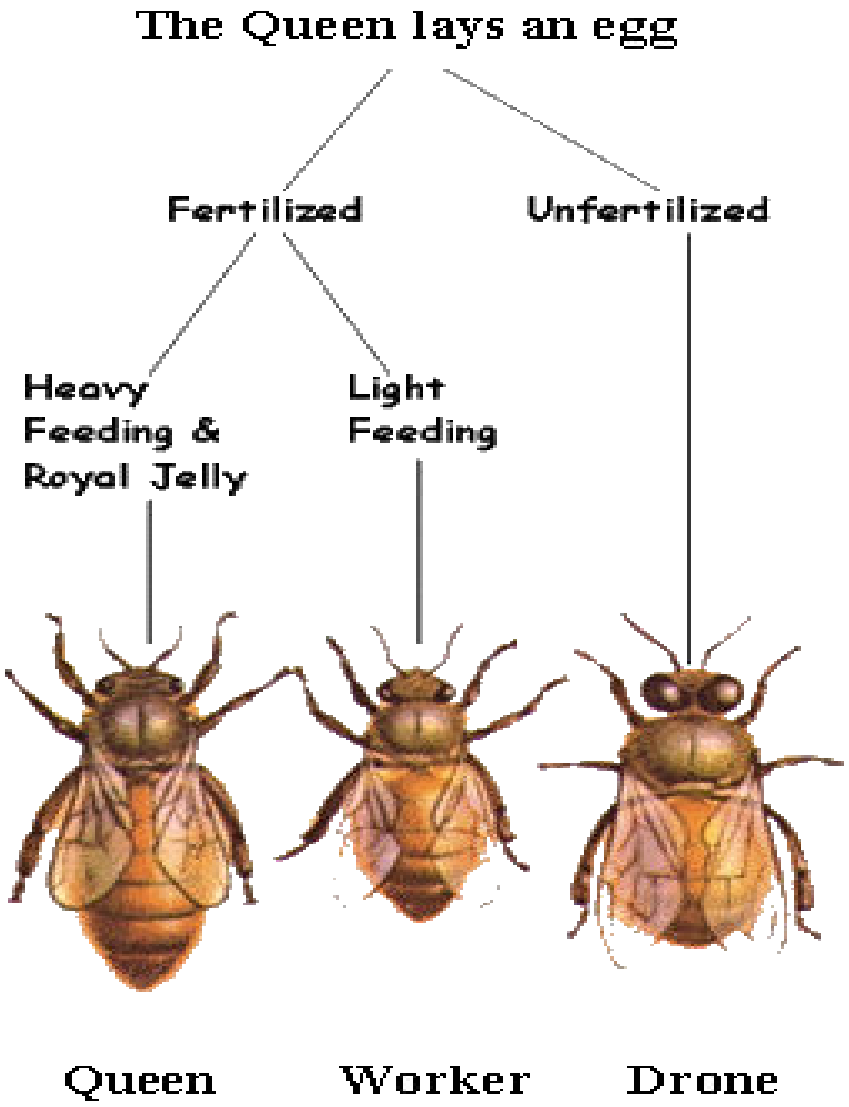
- Drones are stout male bees with large eyes and no stingers.
- Drones do not collect food or pollen from flowers.
- Their sole purpose is to mate with the queen.
- They die upon mating.
- If the colony is short on food, drones are often kicked out of the hive.

Bee Stages



Type	Egg	Larva	Pupa	Total
Queen	3 days	5 days	9 days	17 days
Worker	3 days	6 days	12 days	21 days
Drone	3 days	7 days	14 days	24 days

Bee Development



Visit the life of the Worker Bee



(Day 1-2)

- **Cell cleaning** - Brood cells must be cleaned before the next use - cells will be inspected by the queen and if unsatisfactory will not be used. Worker bees in the cleaning phase will perform this cleaning, if not clean worker bee must do it again.

(Day 3-11)

- **Nurse bee -** Feed the worker larvae, worker jelly, secreted from the same glands that produce royal jelly.

Life of a Worker Bee

(Day 6-11)

- **Advanced Nurse -** Bees Feed royal jelly to the queen larva. Drones receive worker jelly for 1 to 3 days at which time they are moved to honey and pollen.

(Day 12-17)

- **Wax production** - Build cells from wax, repair old cells, and store nectar and pollen brought in by other workers. Early in the worker's career she will exude wax from the space between several of her abdominal segments. Four sets of wax glands, situated inside the last four ventral segments of the abdomen, produce wax for comb construction.
- **Honey sealing -** Mature honey, sufficiently dried, is sealed tightly with wax to prevent absorption of moisture from the air by workers deputized to do same.
- **Drone feeding -** Drones do not feed themselves; they are fed by workers.

Visit the life of the Worker Bee

(Day 12-17) Continued

- **Queen attendants** - Groom and feed the queen. They also collect QMP (Queen Mandibular Pheromone) from the queen and share it with the bees around them who also share it spreading its effects through the hive.
- **Honeycomb building** - Workers will take wax from wax producing workers and build the comb with it.
- **Pollen packing** - Pollen brought into the hive for feeding the brood is also stored. It must be packed firmly into comb cells and mixed with a small amount of honey so that it will not spoil. Unlike honey, which does not support bacterial life, stored pollen will become rancid without proper care and has to be kept in honey cells.

Life of Worker Bee

- **Propolizing** - The walls of the hive will be covered with a thin coating of propolis, a resinous substance obtained from plants. In combination with enzymes added by the worker this will have antibacterial and antifungal properties. Propolis is also used to close off excessive ventilation and entrances.
- **Mortuary bees** - Dead bees and failed larvae must be removed from the hive to prevent disease and allow cells to be reused. They will be carried some distance from the hive by mortuary bees.
- **Fanning bees** - Worker bees fan the hive, cooling it with evaporated water brought by water carriers. They direct airflow into the hive or out of the hive depending on need.

Visit the life of the Worker Bee

(Days 18 - 21)

- **Guard Bees** - protect the entrance of the hive from enemies.
- **Soldier bees** - Soldiers hang around near the entrance and attack invaders. They work in concert with entrance guards.
- **Entrance guard bees** - These inspect incoming bees to ensure that they are bringing in food and have the correct hive odor. Other bees will be rejected or attacked with soldier bees.

Life of Worker Bee

- **Outside guard bees** - Outer guards may take short flights around the outside of the hive in response to disturbances.
- **Water carriers** - When the hive is in danger of overheating, these bees will obtain water, usually from within a short distance from the hive and bring it back to spread on the backs of fanning bees.

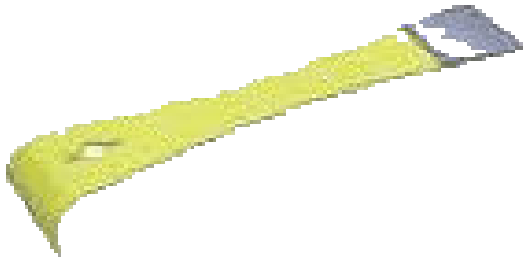
(Days 22 - 35)

- **Foraging bees** - The forager and scout bees travel (2 to 5 miles) to a nectar source, pollen source or to collect propolis.
- **Die in field** - The life span of worker bees depend on the time of year. Most worker bees live about 28 to 35 days. However, workers that are reared in September and October can live through the winter.

What do we need to harvest honey and how do we do it?

– Each stack is a hive. Each box is called a super. The supers are full of hanging frames. The frames are full of comb. The comb is full of honey, we hope!
The bees glue absolutely everything together with propolis, which resembles resin more than wax. You have to pry(interfere, meddle) everything apart.





So, we need bees with excess honey and a hive tool.

What do we need to harvest honey and how do we do it?

- There is only one correct way to smoke, but no two beekeepers agree what that is! My smoker is filled with pine straw. Smoke makes the bees unusually calm.



So, we need a smoker.

What do we need to harvest honey and how do we do it?

- Remove a super full of honey and get rid of the bees. I use a leaf blower without the snout. You can use a fume board and Bee-Gone but I do not use chemicals. Blow the bees out of the super in front of the hive, it's like taking them to Six Flags!



So, we need a leaf blower.

What do we need to harvest honey and how do we do it?

- Supers full of honey are heavy, shallow 37 pounds, medium 52 pounds and a deep 90 pounds so you need a device to carry from bee yard to honey house.



So, we need a wheel barrow, wagon or truck.

What do we need to harvest honey and how do we do it?

- You need a honey house (a place where the bees can not get into), a decapping tank (holds wax and a little bit of honey that dripped off the open frames).



So, we need a honey house and a decapping tank.

What do we need to harvest honey and how do we do it?

- Honey on a frame is somewhat like a bottle of soda, you can't taste it until it is open. Here we are using a decapping knife (gets hot like your mother's iron and has sharp edges on both sides).



So, we need a decapping knife.

What do we need to harvest honey and how do we do it?

– Sometime the bees do not build out the comb to where the decapping knife can cut it so we have to pick or scratch the capped comb open.



So, we need a pick.

What do we need to harvest honey and how do we do it?

- We must get the honey out of the frames of comb and we do this with the aid of an extractor. Years ago beekeepers would squeeze it out by hand.



So, we need an extractor.

What do we need to harvest honey and how do we do it?

- When the extractor gets full you need a bucket or buckets to store the honey in. This is also a good time to filter the honey and I use panty hose stretched across the top of the five gallon storage bucket.



So, we need five gallon storage bucket and a panty hose filter.

What do we need to harvest honey and how do we do it?

- Here the finished product, is a quart of honey. I usually let my honey sit for a few days so any particles that the panty hose did not filter out will rise to the top.



So, we need containers.

Bees can be dangerous, a beekeeper must take several safety precautions when working around a honey bee colony.

- A hat or veil is commonly used to keep the face and neck protected from stings.
- Gloves are another popular form of beekeeping protection, although many beekeepers complain that gloves restrict their movement.
- A hooded suit, typically made from a light colored fabric to help distinguish the beekeeper from the honey bee's natural predators, may also be used.







While working with a honey bee colony, a beekeeper uses a smoker to help calm the bees.

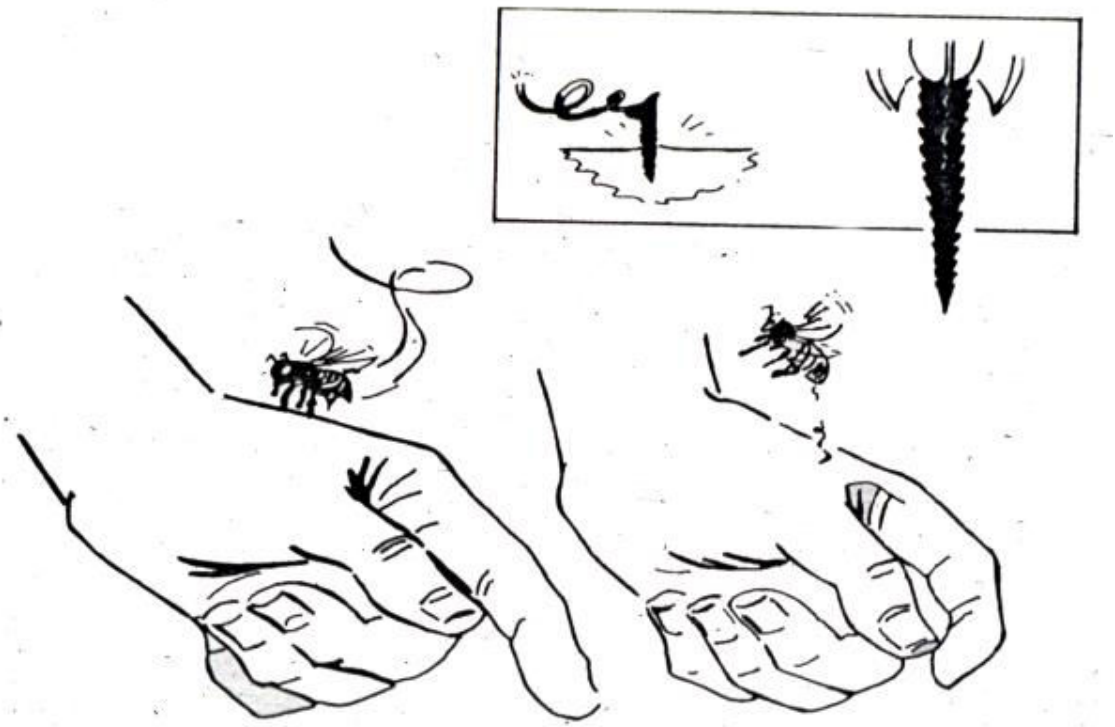
- Smoke is useful in beekeeping because it masks the guard bee's alarm pheromones and encourages the other bees to feed by tricking them into thinking they'll soon need to abandon their hive.
- The smoke gives the beekeeper enough time to inspect the colony and perform any needed maintenance.
- I use Pine needles for fuel in my bee smoker.





Bee Stings 101

- They HURT and....



we REACT

Bee Stings 201

- Normal Reaction

  - **Pain**
  - **Wheal(swelling) develops at puncture site**
  - **Redness develops around wheal**
  - **Swelling**
    - **a little at site**
    - **a lot at site (large local)**
  - **Itching**

Normal Reaction





Bee Stings 201

- Normal Reaction
- **Allergic Reaction**

  - **widespread, rapid swelling**
  - **itching of body**
    - **disorientation feeling**
    - **stomach upset**
    - **Loss of consciousness**

Bee Stings 201

- Normal Reaction
- Allergic Reaction
- Toxic Reaction
    - Too many stings at one time

Too Many Stings!!





Bee Stings 401

- What to do to avoid being stung.....
  -
  -
  -
  -
  -
  -
  -
  -
  -



And if you receive multiple stings ?

Bee Stings 601

A bee's front end is sweet and kind,
But never trust a bee's behind.
A bee can sting if it can sit,
So always stay in front of it!

Bee Stings 401

- What to do to avoid being stung…..
  - proper clothing
  - smoker
  - best environmental conditions
  - scrape sting out promptly
  - slow movements- no jerking or ‘windmills’
  - extra care around hive/flowers
  - don’t remove veil too soon
  - keep gentle bee stock



Poultry farming



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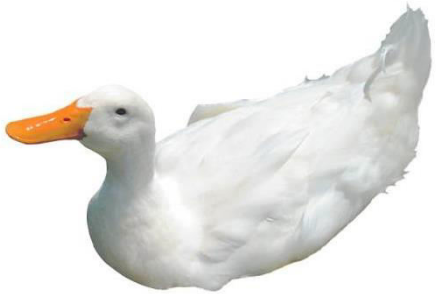
Introduction

Chicken, turkey, duck and goose are all types of birds called poultry.

They are reared for meat.









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Things to Know About Poultry Farming

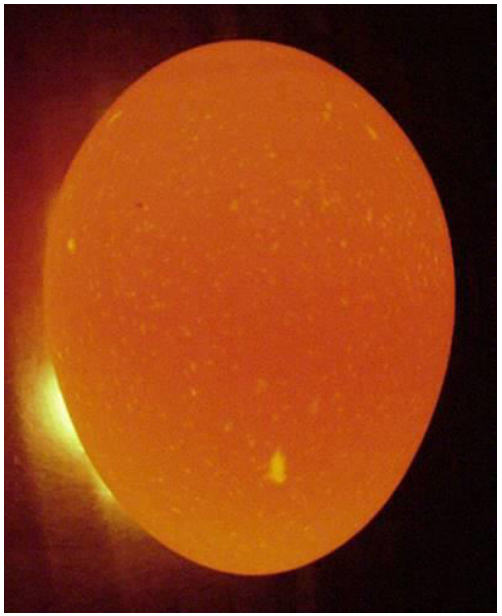
- Understanding poultry terms.
- Differentiate between a productive hen and a non- laying hen
- Identify parts of a chicken.
- Evaluate eggs.
- Where to get quality chicks from
- Where to sell the product
- Biosafety and
- Control of diseases

Basic Terms to Understand



- Scientific Name: Galine
- Chick: Newborn chicken
- Pullet: Young, immature female chicken less than 5-6 months of age
- Hen: Mature female chicken
- Rooster: Mature male chicken
- Roaster: male or female chicken 3-5 months of age and raised for the production of meat

More Terms



- Capon: Castrated male chicken
- Layer: Hen used for laying eggs
- Broiler/Flyer: Chickens grown for meat production
- Group Name: Flock
- Candling: examining a shell egg's content by holding it between one's eye and a light source
- Clutch: Nest of eggs

More Terms

- Egg: hard- shelled; reproductive body produced by a bird
- Green: description of chicks that have recently hatched
- Axial feather: short wing feather that separates primaries from secondaries
- Molt: To shed feathers periodically
- Oviposition: laying of an egg by a bird
- Plumage: Feathers of a bird

Introduction to chicken farming

Chickens farmed for meat are called broiler chickens.

A group of chickens is called a flock.



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Breeder farm

Eggs are laid by broiler hens (parent flock).

Male chickens are called cockerels and female chickens are called pullets or hens.

Male turkeys are called stags and

female turkeys are called hens.

The eggs are collected

and sent to the hatchery.



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Did you know?

At any one time there are approximately 7 million broiler chicken hens laying eggs for hatching in the UK.



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Hatchery

The eggs are incubated at the hatchery. They are kept warm, until the chicks start to hatch out of their shells.

Chickens hatch at around 20 days and turkeys hatch at around 27 days.

Baby chickens are called chicks.

Baby turkeys are called poults.

The hatched birds are then sorted and transported to Rearing farms.



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Do you know what a group of chicks is called?

A group of chicks (baby chickens)

is called a clutch or peep.



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About Chicks

When a chick hatches it can live healthily for up to two days without being given any food or water. This is because it still has nutrients in its stomach from when it was inside the egg.

This is why chicks do not need food or water when they are being moved to the rearing farm.



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Rearing farms

- The baby birds are reared in special large houses, which provide them with water and a special diet.
- These houses are cleaned before each new arrival of baby birds.
- Trained staff look after the birds in the houses every day.
- If the birds are ill, special medicines are given in their food or water.
- The birds are reared until they reach their required weight.



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Catching

- Once the birds reach their required weight they are transported to the processing plant.
- Trained staff called ‘catchers’ catch the birds and put them in special containers called modules.
- Catching is carried out quietly and with care to avoid unnecessary stress and to prevent injury to the birds.
- The modules are then loaded into lorries and taken to the processing plant.



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Egg- Type Hens

- Characteristics to look for when examining hens:
  - Bleaching of yellow pigment in the shanks, feet, and beak
  - Condition and capacity of the abdomen
  - Condition of plumage and rate of molt of the wing primaries
  - Vigor and vitality
  - Head characteristics

Bleaching of Body Pigment

- Order that the body pigment fades:
  - Vent
  - Eye Ring
  - Earlobe
  - Base of Beak
  - Tip of Beak
  - Bottom of Foot
  - Shank
  - Hock and Tip of Toe

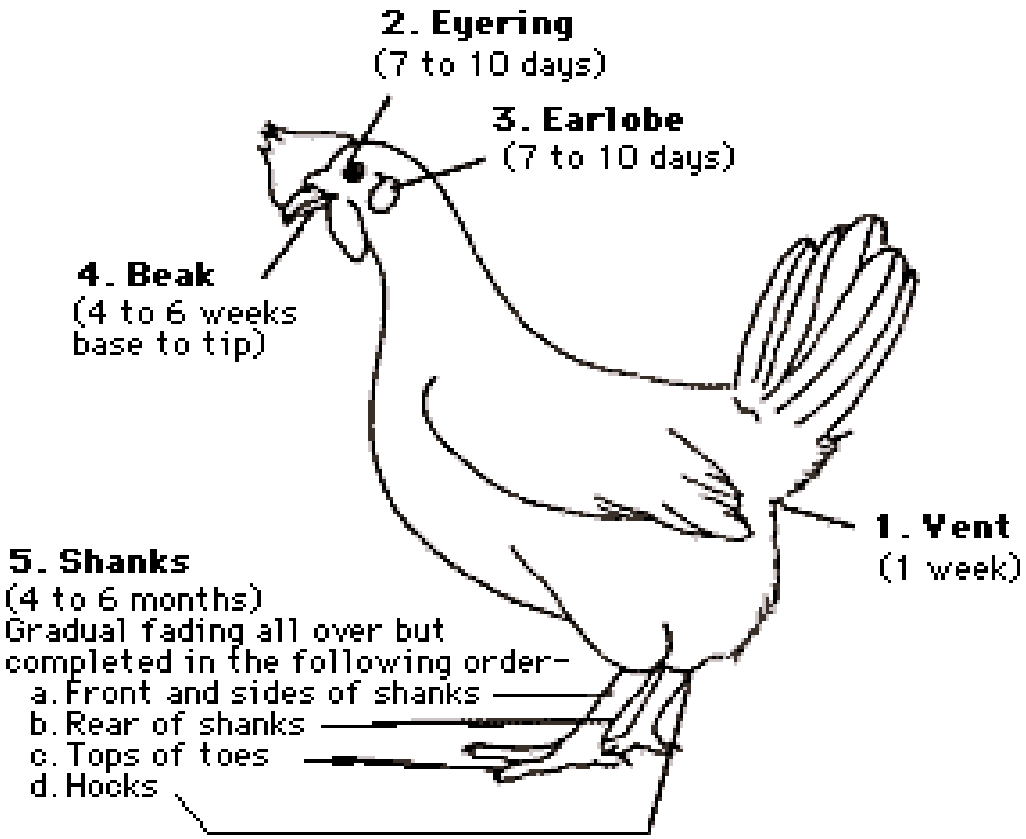


Figure 5. Photographs showing the loss of pigment from the front of the shanks and tops of toes



Hen has laid few eggs ▸ Hen has laid many eggs

Figure 4. Comparison of the yellow color in the eye ring, ear lobe and beak of a poor layer (photo on the left) and a good layer (photo on the right)

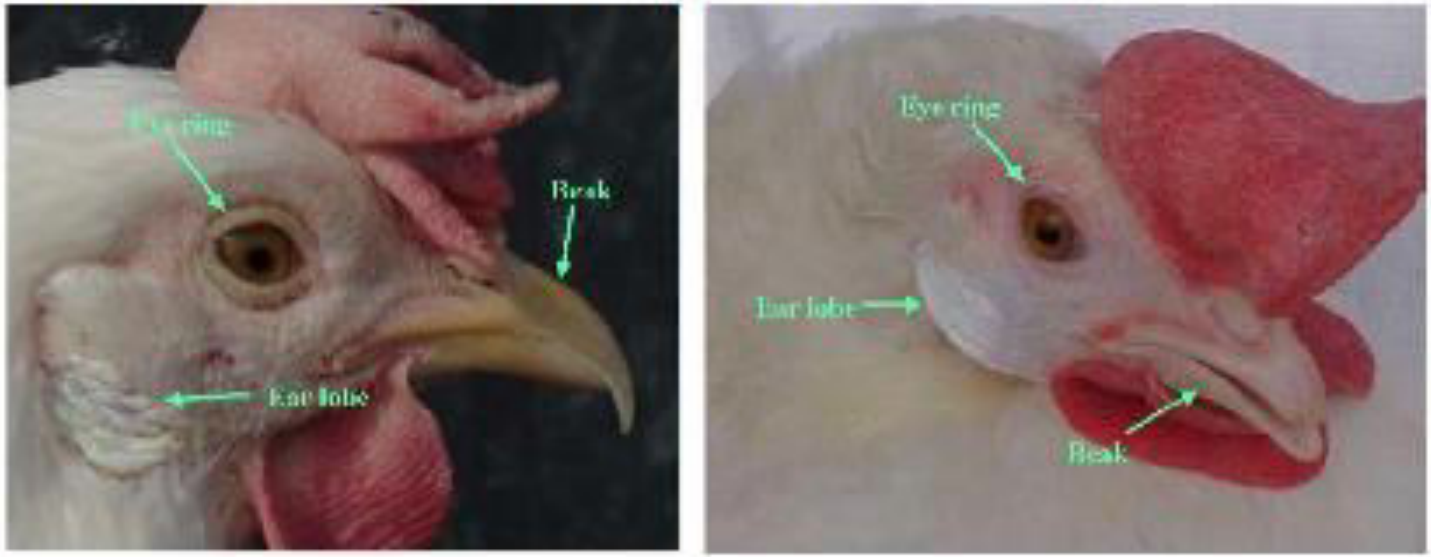


Figure 3. Comparison in the color of the skin around the vent of a poor layer (photo on the left) and a good layer (photo on the right)



Bleaching of Body Pigment

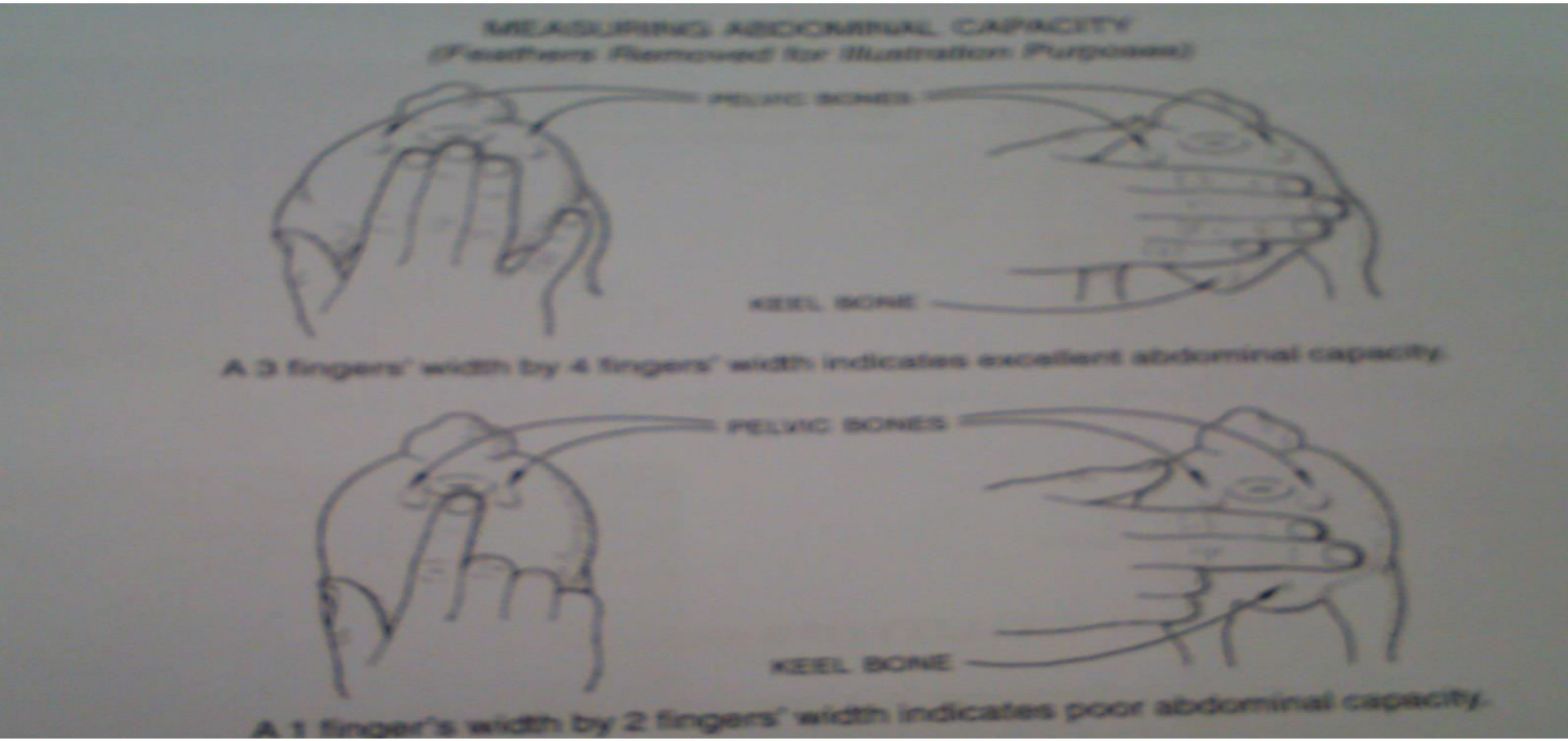
- Hens that show signs of returning pigment are decreasing in egg production
- Pigment returns to the body parts in the same order it faded
  - Returns 3 times quicker
  - Four factors:
    - Amount of pigment in feed
    - Health and vitality of hen
    - Whether the hen is confined or not
    - Size and coarseness of the hen

Condition and Capacity of Abdomen

- Good indicator of egg production
  - Abdomen of a layer is wide, soft (lacks fat), and expanded
  - Pelvic bones are thin and flexible
  - Vent is moist, large, and oblong in shape
- Non-layer
  - Narrow, hard (fatty), and contracted
  - Pelvic bones are thick and rigid
  - Vent has some moistness but is small and round in shape

Abdominal Capacity

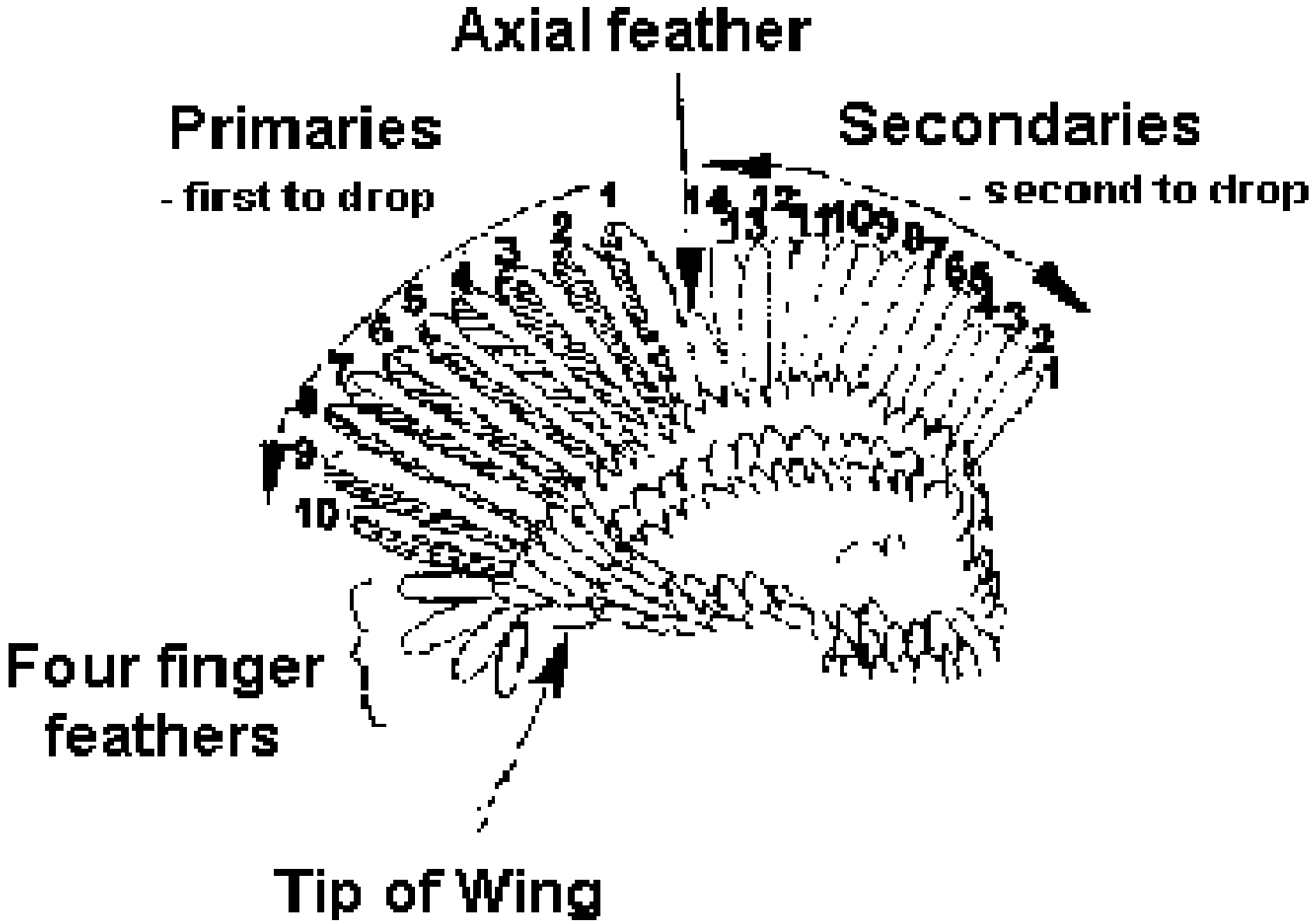
- Abdominal capacity of a hen is measured and expressed by ones fingers width.
  - Normal- 3 fingers width by 4 fingers width



Plumage and Rate of Molt

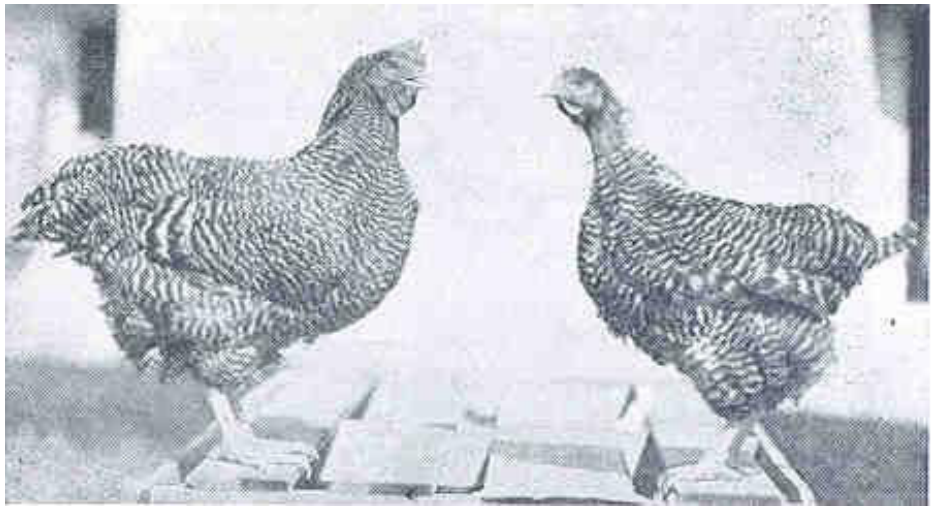
- Two factors considered in appraising the plumage of hens include condition (feather appearance) and molting rate (speed of shedding feathers)




Axial feather
Primaries
- first to drop
Secondaries
- second to drop
Four finger feathers
Tip of Wing

Vigor and Vitality

- A high producing hen appears vigorous, alert, and quick in movement
- Non-producing hen is sluggish

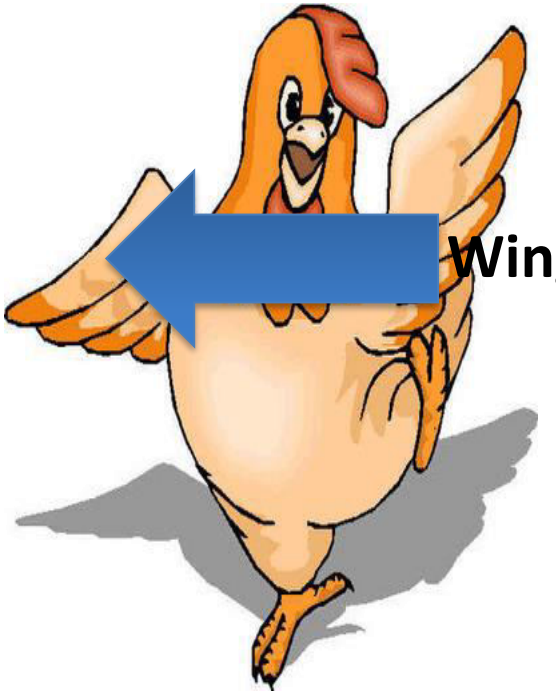


Head and Head Parts

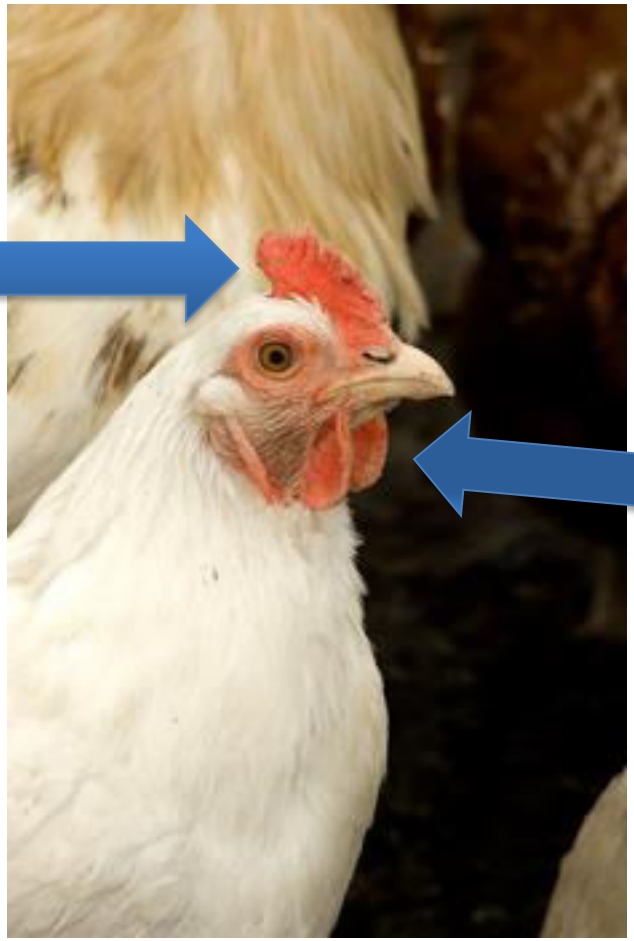
- If not trimmed- a productive hen's beak is short
  - The hens eyes are bright, alert and round.
  - Her skull is flat from side to side
  - Her comb and wattles are large, bright red, and glossy
  - They feel velvety soft and warm when touched.
- Non-producing hens beak is long
  - The hens eyes are dull, sleepy, and oblong.
  - Her skull is rounded from side to side
  - Her comb and wattles are shrunken and dull
  - They feel rough and cool when touched

Parts of A Chicken

Comb and Wattles



Comb

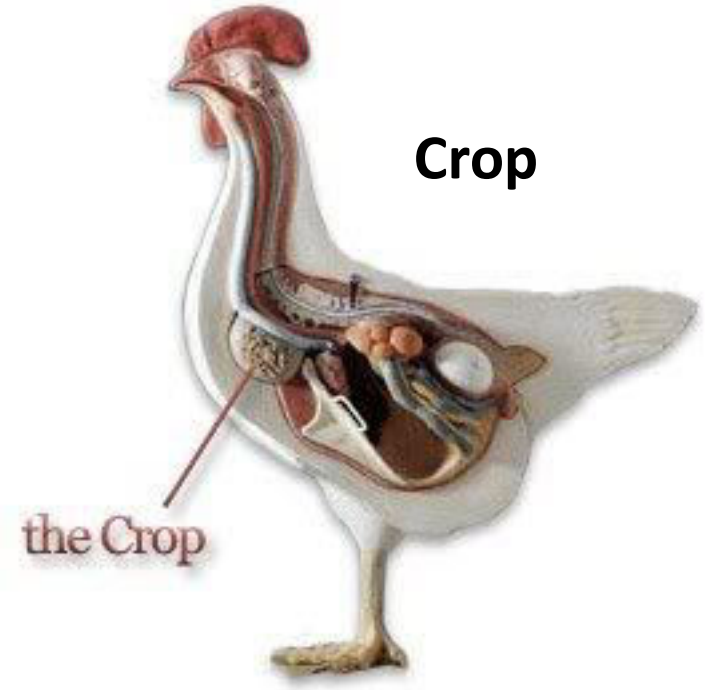


Wattles

Hock, Shank, And Feet

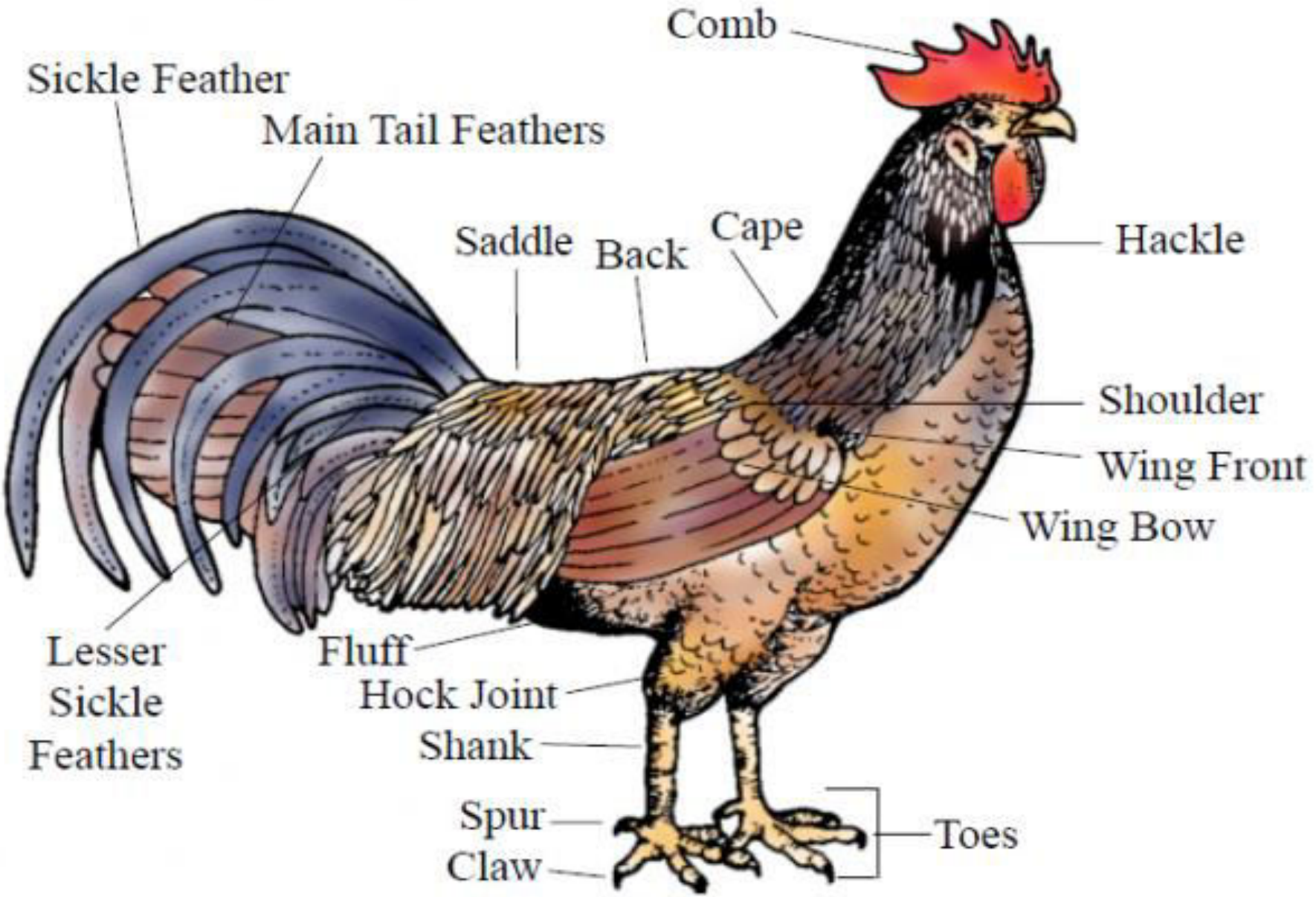


Crop and Vent

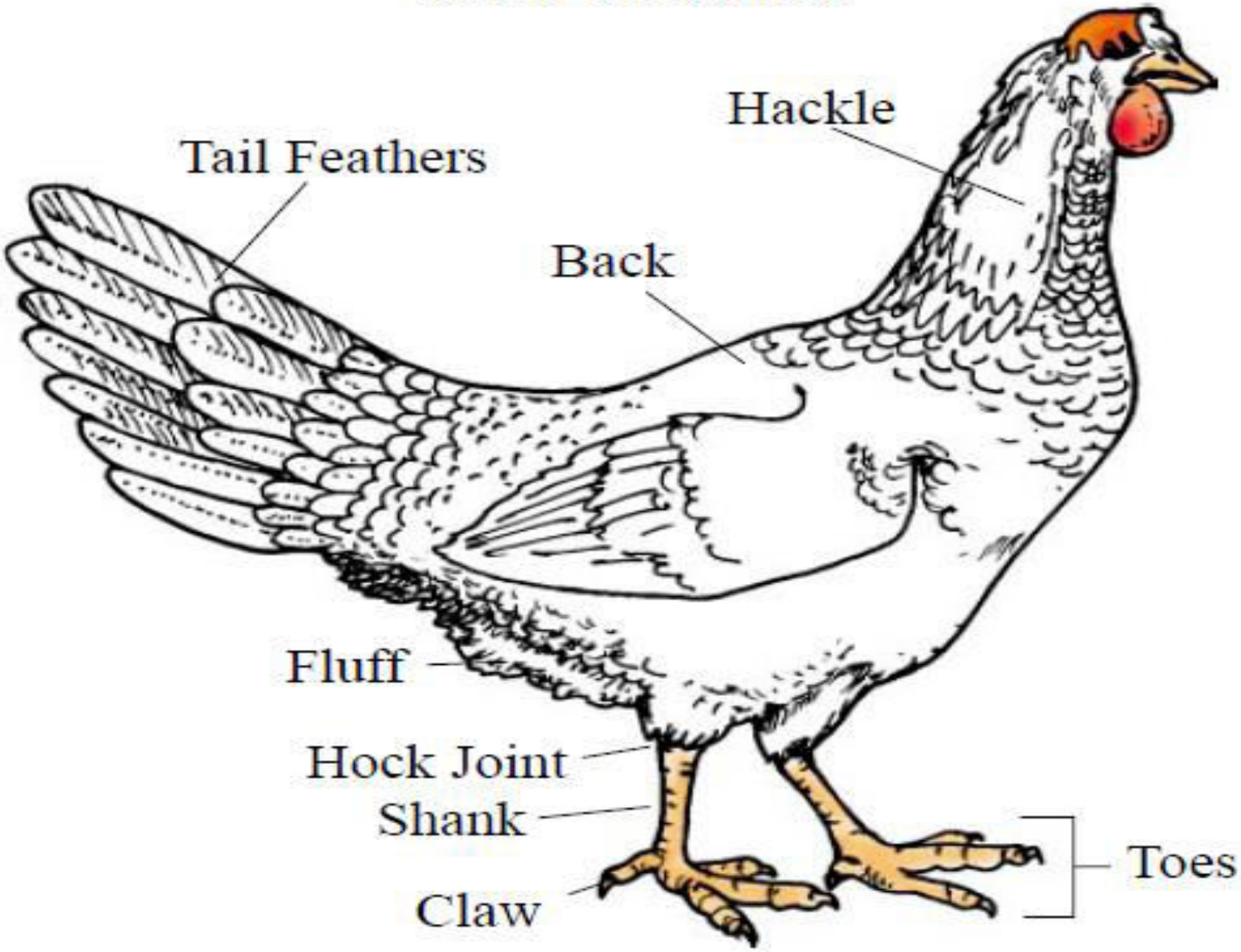




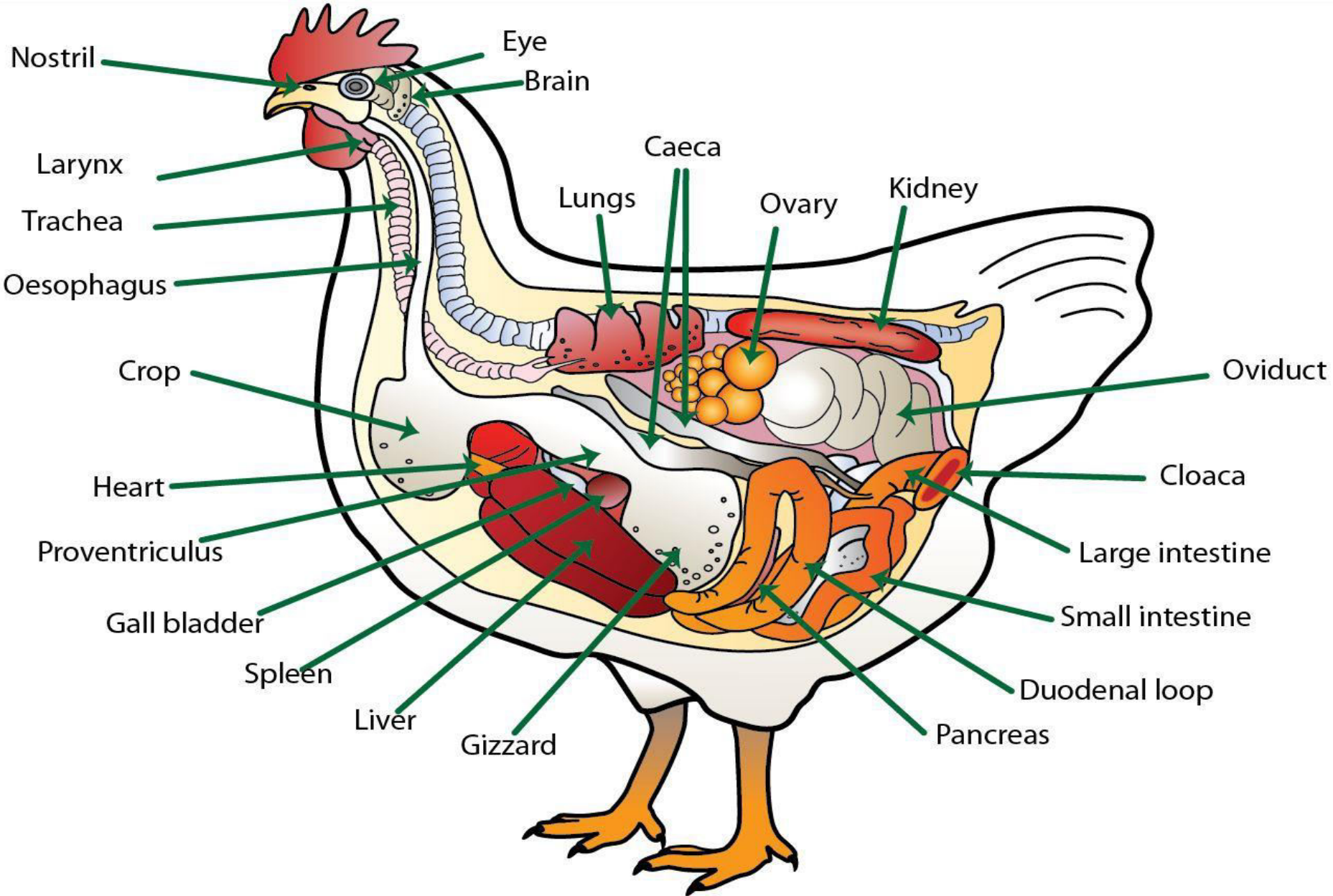
Parts of a Rooster



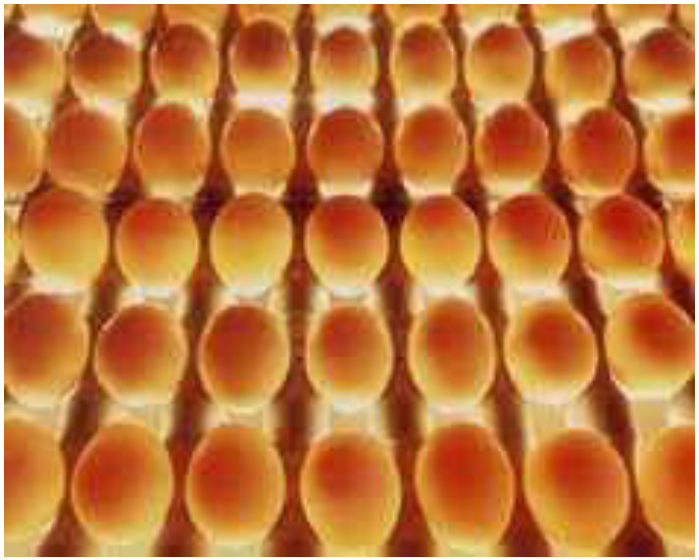
Parts of a Hen



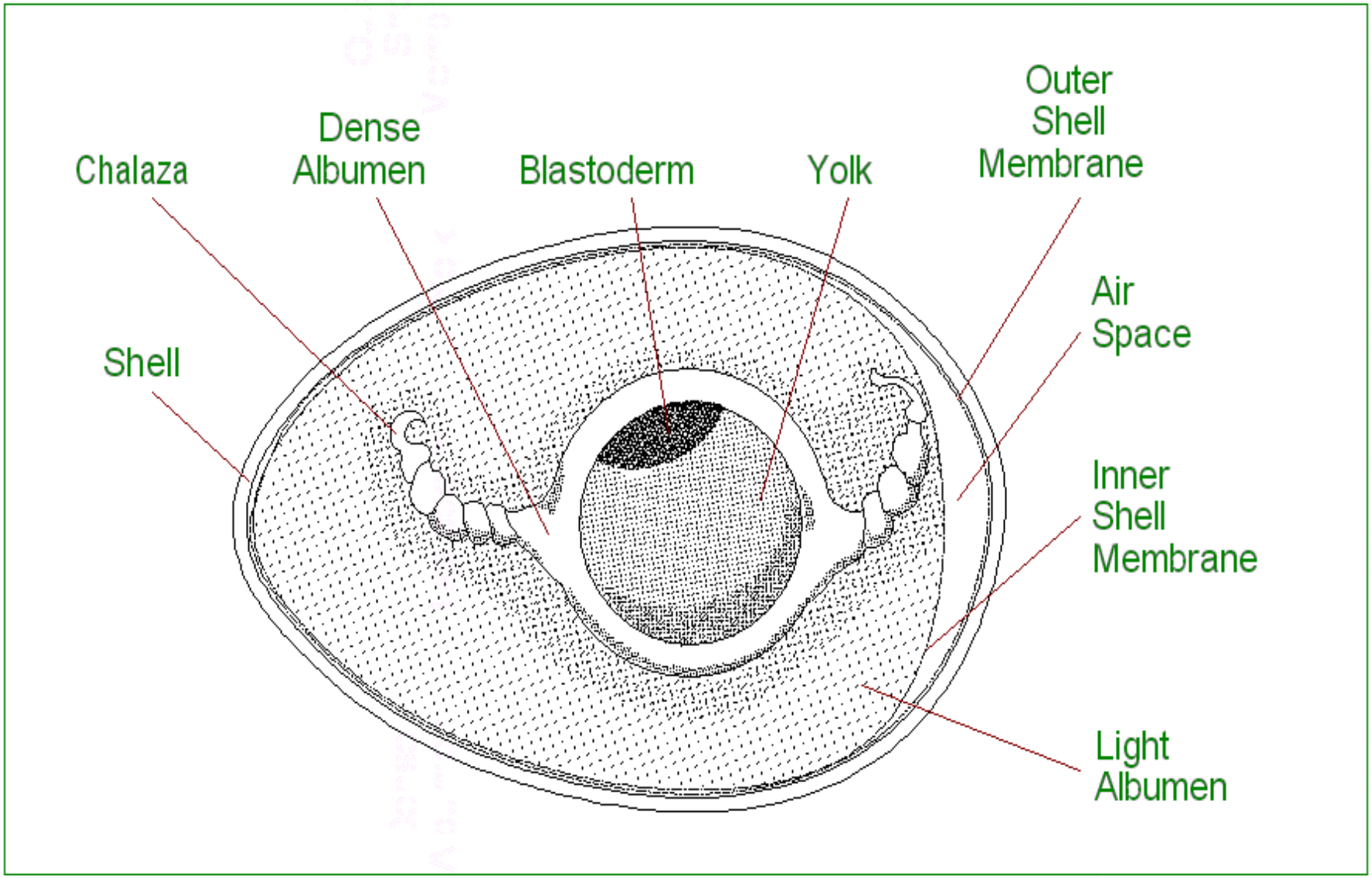
Body Parts of Chicken



Eggs



Egg and Its Parts



Candling Eggs

- Candling Demo
- What do we look for when we candle eggs?
  - Air Cell
  - Yolk
  - Cracks in shell
  - Blood spots and other foreign matter

Before We Candle Eggs

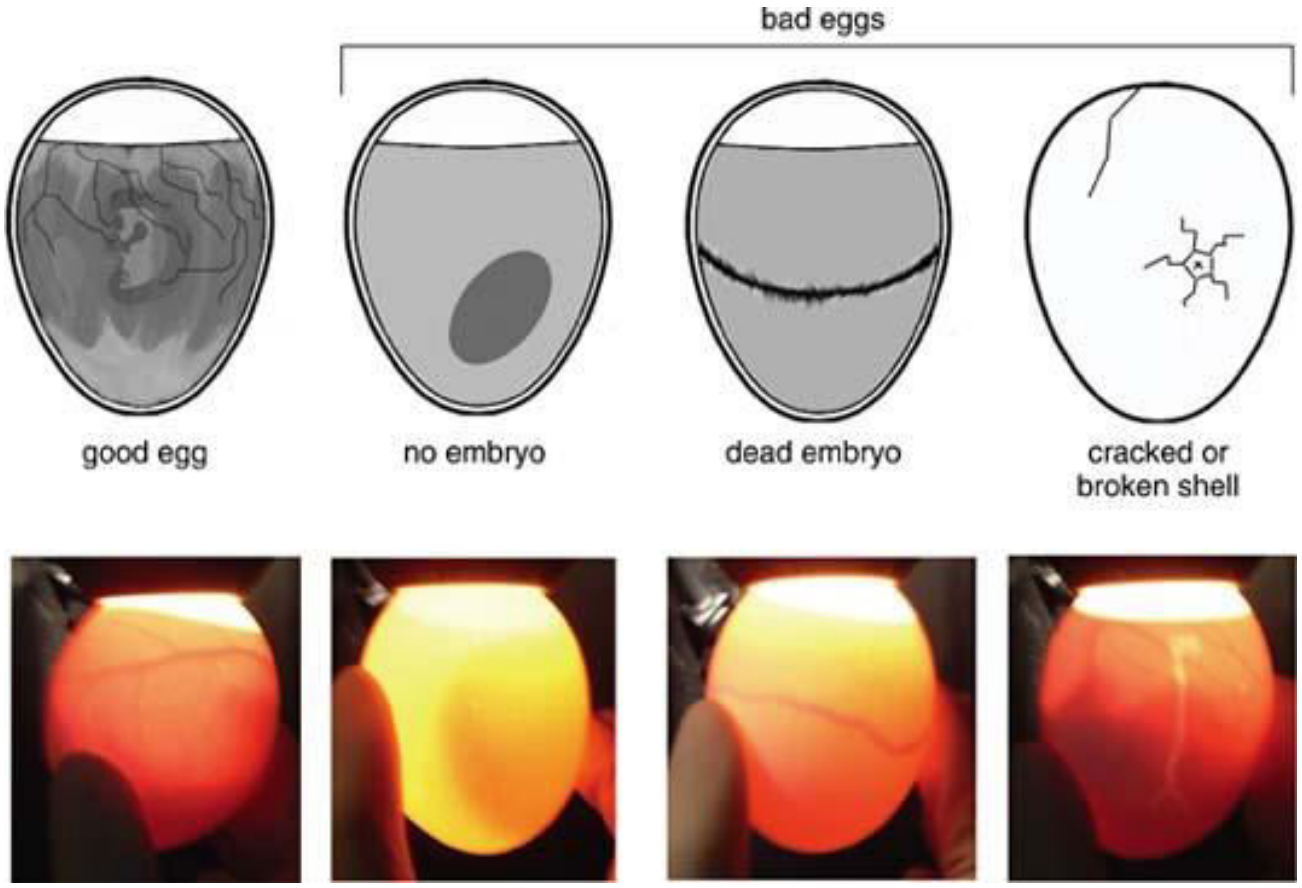
Air Cell

- Temperature- 105 degrees
- Is normally at the large end of the egg
- Quality Grades:
  - AA- up to 1/8”
  - A- 1/8”-3/16”
  - B- > 3/16”

Yolk

- Yolk size and shape
- Distinctness of yolk shadow outline
- Yolk defects and germ development

What Does Your Egg Look Like?



Processing Plant

- Here the birds are processed and packaged.
- After the birds have been processed they are weighed, and then either left whole or portioned.
- The meat is then packaged and labelled ready to be delivered to restaurants, shops and supermarkets.



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POULTRY FEEDING

Following facts should be considered when computing ration for poultry:



1-Feed must contain all essential nutrients in right amounts & proportion required.

2-Different standards per age should be followed.

3-Palatability of the ingredients which used.

POULTRY FEEDING

4-Unlike ruminants, poultry completely depend upon the dietary sources for all nutrients (essential AAs., vit.B groups & vit.K).

5-Include agro-industrial by-products to minimize cost of the ration,

6-Optimum level of ingredient inclusion as many of ingredients have a deleterious effect at higher levels.

7-Optimum Ca:P ratio for different purposes.



POULTRY FEEDING

Nutrients requirements of poultry:

1-Energy requirement:

- Ration for poultry calculated on the basis of ME.
- Poultry eat to satisfy their energy needs when fed free choice, thus must control the intake of all nutrients by including them in a definite proportion to available energy level..



POULTRY FEEDING

- High energy cereal grains are the principal energy sources.
- Fat may be added at levels of 3-8% to increase dietary energy concentrations.

Factors affecting feed intake:

1-Energy levels in the ration:

↑ energy level ⇒ ↓ feed intake

↓ energy level ⇒ ↑ feed intake



POULTRY FEEDING

2-Environmental temperature:(SET, 16-24C)

↑ Temp. ⇒ ↓ feed intake

↓ Temp. ⇒ ↑ feed intake

3-Health of the bird

4-Genetics

5-Form of the feed

6-Nutritive balance of the diet

7-Stress 8-Body size

9-Rate of growth & egg production



POULTRY FEEDING

2- Protein requirement:

- The amount of protein required is proportional to the energy level in the ration.
- Poultry required the 14 essential AAs.
- ↑ Temp. ⇒ ↓ feed intake ⇒ ↑ protein req.
  ↓ Temp. ⇒ ↑ feed intake ⇒ ↓ protein req.
- Some AAs can met by other AAs:

  Cystine ⇒ methionine, Tyrosine ⇒ phenylalanine

  Glysine ⇒ Serine

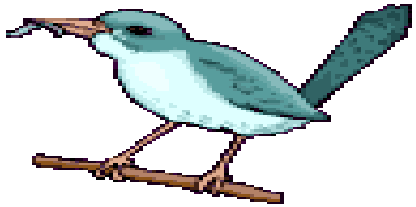


POULTRY FEEDING

- Overheating or under heating during processing can affect the availability of some amino acids.

3- Mineral requirements:

- The major minerals needed in poultry diets are Ca, P, Na & Cl.
- Trace minerals may be added if feeds grown on soil deficient in them.



POULTRY FEEDING

A-Calcium & Phosphorus:

- The recommended ratio P:Ca in diet of poultry is 1:1.2 (range 1:1 to 1:1.5)

  For laying hen 1:4 (Ca important for bone & shell formation)
- ↑ Ca in diet ⇒ ↓utilization of Mg, Mn & Zn.
- Inorganic P have a higher availability than organic P
- All P from animal origin & 40% from plant origin (wheat bran & rice bran) is available.

POULTRY FEEDING

B- Salt (NaCl):

- The amount added depend upon the feed ingredients.
- The recommended level in the ration 0.5-1% of the ration.
- Adult poultry can tolerate much higher inclusion but the water consumption increased.



POULTRY FEEDING

C- Manganese:

- Def. Of Mn cause perosis with slipped tendon.
- A free flowing Mn supplements should normally be included in all poultry feeds.
- Mn needed for egg production & hatchability.
- Mn carbonate, oxide, sulfate & commercial mineral mixture can be used.

POULTRY FEEDING

D- Iodine:

- Iodine included at rate of 0.5mg but when fish meal included at 5-10% no need iodine suppl.
- ↑ Ca & P in diet ⇒ ↑ iodine requirement

E- Magnesium:

- No Mg Suppl. Needed for poultry ration.
- ↑ Mg in diet ⇒laxation



POULTRY FEEDING

4-Vitamin requirements:

A- Vitamin A:

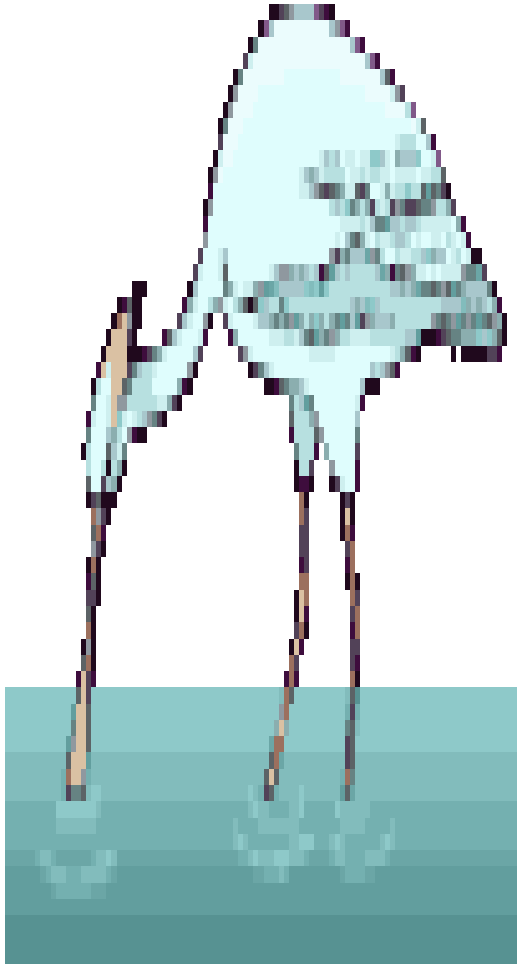
- Liberal supply of vit.A or carotene is needed for normal growth & health.
- Def. Symptoms: retardation of growth, emaciation, staggering gait & ruffled feathers, reduced immunity
- Sources: fish liver oils & other animal sources.



POULTRY FEEDING

B- Vitamin D:

- Vit.D required for bone formation, egg production, reproduction & prevention of rickets.
- Def. symptoms:poor growth, lameness & rickets.
- Poultry do not exposure to sunlight, ration must suppl. With vit.D.



POULTRY FEEDING

C- Vitamin E:

- Vit.E in vegetable is not readily available as in oil concentrates.
- Vit.E essential to prevent encyphalomalacia or crazy chick disease.

D- Vitamin K:

- Def. of vit. K $\Rightarrow$ delay clotting time of the blood & produce serious hemorrhage
- All mixtures should be suppl. With vit. K
- Treatment by sulfonamide $\Rightarrow \uparrow$ vit. K req.



POULTRY FEEDING

E- Riboflavin:

- Def. of vit.B2 ⇒curled-toe paralysis, dwarfism & degeneration of nerve trunks.
- Requirement: Broilers & breeder 4.4mg/kg

  Layers 2.5 mg/kg ration

F- Thiamin:

- Def. of thiamin ⇒ nerve deg., convulsion & heart abnormalities.



POULTRY FEEDING

G- Niacin:

- Def. of niacin $\Rightarrow$inflammation of tongue & mouth cavity (black tongue).
- Young chick required niacin more than adult due to less bacterial action synthesis.

H- Vit.B12:

- Animal proteins are good sources of vit.B12.
- Def. of vit.B12 $\Rightarrow$irritability, poor feathering & poor hatchability.

POULTRY FEEDING

Feeding space:

1 inch feeder space /chick for 2 weeks age & 2 inches after that.

Water:

- Bird drink about twice as much water by weight of feed consumed.
- Water consumption increase or decrease according to the environmental temperature.
- Some medications are administered in the drinking water.